MATH 3351

Advanced Calculus

Lecture Notes

Yoolkyu Park

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Chapter 1

Euclidean space and differentiation

The local geometry of a level set rests on linear algebra, limits, and linear approximation. We collect these foundations before using them for manifolds. The corresponding background is developed in Edwards [1, Chs. I–III]. We assume completeness of $\mathbb{R}$ and the elementary one-variable calculus, including the scalar mean value theorem.

1.1 Linear maps, orthogonality, and volume

Definition 1.1. For $x, y \in \mathbb{R}^n$, the Euclidean inner product and norm are

$$\langle x, y\rangle = \sum_{j=1}^{n} x_j y_j, \qquad \|x\| = \sqrt{\langle x, x\rangle}.$$

Vectors are orthogonal if their inner product is zero. For a subspace $V \subset \mathbb{R}^n$, its orthogonal complement is

$$V^{\perp} = \{w \in \mathbb{R}^n : \langle w, v\rangle = 0 \text{ for every } v \in V\}.$$

A map $A : \mathbb{R}^n \to \mathbb{R}^m$ is linear if $A(sx + ty) = sAx + tAy$. Its matrix in the standard bases has columns Ae_j. Its kernel, image, and rank are

$$\ker A = \{x : Ax = 0\}, \quad \operatorname{im} A = \{Ax : x \in \mathbb{R}^n\}, \quad \operatorname{rank} A = \dim \operatorname{im} A.$$

Its operator norm is $\|A\| = \sup_{\|v\|=1} \|Av\|$. The transpose A^T satisfies $\langle Av, w\rangle = \langle v, A^T w\rangle$.

Cauchy–Schwarz, $|\langle x, y\rangle| \le \|x\|\|y\|$, follows by minimizing $\|x - ty\|^2$ over t when $y \neq 0$; the case $y = 0$ is immediate. Expanding $\|x + y\|^2$ then gives the triangle inequality. In particular,

$$|x_j| \le \|x\| \le \sqrt{n} \max_j |x_j|.$$

These inequalities connect coordinate calculations to Euclidean estimates [1, Ch. I].

Exercise 1.1 (operator bounds and normal spaces)**.** Prove that every linear map has finite operator norm, that

$$\|Av\| \le \|A\|\|v\|, \qquad \|BA\| \le \|B\|\|A\|,$$

and that

$$\dim \ker A + \operatorname{rank} A = n, \qquad (\ker A)^{\perp} = \operatorname{im} A^T.$$

Describe orthogonal projection onto a subspace.

Solution. Writing $v = \sum_j v_j e_j$ gives

$$\|Av\| \le \sum_j |v_j| \|Ae_j\| \le \|v\| \left(\sum_j \|Ae_j\|^2 \right)^{1/2}.$$

Thus the supremum is finite. Homogeneity gives the first bound, including $v = 0$, and applying it twice gives the composition bound. The triangle inequality for operator norms follows in the same way.

Extend a basis of $\ker A$ to a basis of $\mathbb{R}^n$. The images of the added vectors form a basis of $\operatorname{im} A$: they span by expansion in the chosen basis, and a relation between their images would put the corresponding combination in the kernel. This proves rank–nullity.

Gram–Schmidt produces an orthonormal basis $u_1, \dots, u_k$ of any subspace V. Then

$$P_V x = \sum_{j=1}^{k} \langle x, u_j \rangle u_j, \qquad x - P_V x \in V^\perp.$$

This decomposition is unique, since $V \cap V^\perp = \{0\}$. Hence $\mathbb{R}^n = V \oplus V^\perp$ and $\dim V^\perp = n - \dim V$. Finally,

$$v \in (\operatorname{im} A^T)^\perp \iff \langle Av, w \rangle = 0 \text{ for every } w \in \mathbb{R}^m \iff Av = 0.$$

Taking orthogonal complements proves the asserted normal-space identity. In matrix language, the rows of A span $(\ker A)^\perp$.

Definition 1.2. For an $n \times n$ matrix $A = (a_{ij})$, define

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} a_{\sigma(j), j},$$

where S_n is the set of permutations of $\{1, \dots, n\}$. Equivalently, determinant is the alternating multilinear function of the columns which takes value 1 on $(e_1, \dots, e_n)$. It is a polynomial in the entries, hence continuous. We use the elementary identities

$$\det(AB) = \det A \det B, \qquad \det A^T = \det A, \qquad A \text{ invertible} \iff \det A \neq 0.$$

For independent vectors $v_1, \dots, v_k \in \mathbb{R}^n$, define their parallelepiped volume as the product of the successive perpendicular heights obtained by Gram–Schmidt. For dependent vectors the k-dimensional volume is zero.

Exercise 1.2 (the determinant measures parallelepiped volume)**.** If V is the $n \times k$ matrix with columns $v_1, \dots, v_k$, prove

$$\operatorname{Vol}_k(v_1, \dots, v_k) = \sqrt{\det(V^T V)}.$$

In particular, for $k = n$ the volume is $|\det V|$.

Solution. For independent columns, Gram–Schmidt gives $V = QR$, where $Q^T Q = I_k$ and R is upper triangular with positive diagonal entries equal to the successive heights. Therefore $V^T V = R^T R$ and

$$\sqrt{\det(V^T V)} = |\det R| = \prod_{j=1}^{k} R_{jj}.$$

For dependent columns, $V^T V$ has a nonzero kernel, so its determinant and the defined volume both vanish. If $k = n$, $\det(V^T V) = (\det V)^2$. Thus $|\det A|$ is the volume of the image of the unit cube under a square linear map A; its determinant also records orientation. Volume scaling for general regions will be proved with change of variables.

1.2 Topology and completeness of Euclidean space

Definition 1.3. The open ball with center a and radius $r > 0$ is $B(a, r) = \{x : \|x - a\| < r\}$; its closed ball is $\overline{B}(a, r) = \{x : \|x - a\| \le r\}$. A set $U \subset \mathbb{R}^n$ is open if every $a \in U$ has a ball $B(a, r) \subset U$. A set is closed if its complement is open. A neighborhood of a is a set containing an open ball about a.

For $E \subset \mathbb{R}^n$, its interior E° consists of points having a ball contained in E; its closure $\overline{E}$ consists of points whose every ball meets E. Its boundary is

$$\partial E = \overline{E} \cap \overline{\mathbb{R}^n \setminus E} = \overline{E} \setminus E^\circ.$$

A subset of E is relatively open if it equals $E \cap U$ for some open $U \subset \mathbb{R}^n$. The set E is compact if every cover of E by open subsets of $\mathbb{R}^n$ has a finite subcover. It is bounded if it lies in some ball.

Exercise 1.3 (completeness and Heine–Borel)**.** A sequence (x_j) is Cauchy if for every $\varepsilon > 0$ there is N such that $\|x_j - x_k\| < \varepsilon$ for $j, k \ge N$. Prove the following facts.

(i) A set is closed exactly when it contains the limits of all its convergent sequences.

(ii) Every Cauchy sequence in $\mathbb{R}^n$ converges. Every closed subset, with the Euclidean distance, is complete.

(iii) A subset $K \subset \mathbb{R}^n$ is compact exactly when it is closed and bounded, and exactly when every sequence in K has a subsequence converging to a point of K.

Solution. If E is closed and $x_j \in E$ tends to $x \notin E$, a ball about x in the complement contradicts convergence. Conversely, if E is not closed, some $x \in \overline{E} \setminus E$ admits points $x_j \in E \cap B(x, 1/j)$, which converge to x. This proves (i).

Each coordinate of a Cauchy sequence is Cauchy in $\mathbb{R}$, hence converges. The coordinate inequalities in Section 1.1 give convergence in norm. Part (i) keeps the limit in a closed subset, proving (ii).

Every bounded real sequence has a convergent subsequence: repeatedly bisect a bounded closed interval, retain a half containing infinitely many terms, and select increasing indices in the nested halves. Completeness of $\mathbb{R}$ gives their common limiting point. Applying this successively to the n coordinates proves that every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence. If K is closed, the limit belongs to K.

Suppose now that every sequence in K has a subsequence converging in K. The set is bounded, since otherwise one could choose $x_j \in K$ with $\|x_j\| > j$, and is closed by (i). Given an open cover, there is $\delta > 0$ such that for every $x \in K$, the ball $B(x, \delta)$ is contained in some member of the cover. Otherwise choose $x_j \in K$ for which $B(x_j, 1/j)$ is contained in no member. A subsequence tends to $x \in K$; an open cover member containing x contains $B(x, r)$ for some $r > 0$. For large subsequence indices, $B(x_j, 1/j) \subset B(x, r)$, a contradiction. Boundedness allows a finite grid of cells of diameter less than δ covering K. Choose a point of K in each occupied cell. The associated δ-balls cover K, and choosing one cover member for each gives a finite subcover.

Finally, compactness implies boundedness by the cover $\{B(0, j) : j \ge 1\}$. To prove closedness, fix $p \notin K$. For nonempty K, the balls $B(x, \|x - p\|/3)$, $x \in K$, have a finite subcover with centers $x_1, \dots, x_s$. Put $r = \min_i \|x_i - p\|/3 > 0$. If $z \in K$ lies in the ith ball, then $\|z - p\| > 2\|x_i - p\|/3 \ge 2r$. Thus $B(p, r)$ misses K. The empty set is closed as well. This proves all equivalences.

Definition 1.4. A set E is connected if it cannot be written as the union of two disjoint nonempty relatively open sets. It is path connected if any $a, b \in E$ can be joined by a continuous map

$c : [0,1] \to E$ with $c(0) = a$ and $c(1) = b$. It is convex if $(1-t)a + tb \in E$ whenever $a, b \in E$ and $0 \le t \le 1$.

Here and below, continuity on a subset uses distances within that subset. Explicit limit definitions appear in the next section.

Exercise 1.4 (connectedness and paths)**.** Prove that continuous images of connected sets are connected, that path connected sets are connected, and that an open connected subset of $\mathbb{R}^n$ is polygonally path connected. Explain why a convex set is path connected.

Solution. A separation of a continuous image pulls back to a separation of its domain, because preimages of relatively open sets are relatively open. Hence continuous images preserve connectedness. The interval $[0,1]$ is connected by the one-variable intermediate value property. If a path connected set had a separation, a path between its two parts would pull it back to a separation of $[0,1]$.

Let U be open and connected and fix $a \in U$. Let A consist of points joinable to a by finitely many line segments in U. For $x \in U$, choose a ball $B(x,r) \subset U$. Any two points of the ball are joined by a segment within it. Thus if $x \in A$, the ball lies in A; if $x \notin A$, the ball misses A, since a path to any point of the ball could be extended to x. Both A and $U \setminus A$ are relatively open. Since $a \in A$, connectedness gives $A = U$. For convex sets the single path $c(t) = (1-t)a + tb$ suffices.

1.3 Limits and continuous maps

Definition 1.5. Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ and let a be an accumulation point of E, meaning that every $B(a,r)$ meets $E \setminus \{a\}$. We write $\lim_{x \to a,\, x \in E} f(x) = L$ if for each $\varepsilon > 0$ there is $\delta > 0$ such that

$$x \in E, \quad 0 < \|x - a\| < \delta \quad \Longrightarrow \quad \|f(x) - L\| < \varepsilon.$$

The map is continuous at $a \in E$ if for each $\varepsilon > 0$ there is $\delta > 0$ such that $x \in E$ and $\|x - a\| < \delta$ imply $\|f(x) - f(a)\| < \varepsilon$. It is continuous on E if this holds at every point. It is uniformly continuous if one δ works for all pairs $x, y \in E$:

$$\|x - y\| < \delta \quad \Longrightarrow \quad \|f(x) - f(y)\| < \varepsilon.$$

It is Lipschitz with constant $L \ge 0$ if $\|f(x) - f(y)\| \le L\|x - y\|$ for every $x, y \in E$.

Limits and continuity of vector-valued maps are equivalent to their coordinate versions, by the coordinate inequalities. Sums, scalar products, and compositions of continuous maps are continuous; quotients of scalar functions are continuous wherever the denominator is nonzero. These facts follow directly from the limit definitions [1, Ch. I].

Exercise 1.5 (sequences and compactness under continuous maps)**.** Prove the sequential criterion for limits and continuity. Then prove that a continuous map $f : K \to \mathbb{R}^m$ on a compact set has compact image and is uniformly continuous. Deduce that a real-valued continuous function on a nonempty compact set attains its maximum and minimum.

Solution. The ε–δ definition of a limit implies $f(x_j) \to L$ whenever $x_j \in E \setminus \{a\}$ tends to a. If the definition fails, some $\varepsilon_0 > 0$ allows a choice $0 < \|x_j - a\| < 1/j$ with $\|f(x_j) - L\| \ge \varepsilon_0$, contradicting that sequential property. The continuity criterion is identical, allowing $x_j = a$ and taking $L = f(a)$.

This criterion also shows that preimages of relatively open sets under a continuous map are relatively open: at a point mapping into an open ball, continuity supplies a sufficiently small domain ball.

For a sequence $y_j = f(x_j)$ in $f(K)$, compactness gives $x_{j_\ell} \to x \in K$. Continuity gives $y_{j_\ell} \to f(x)$. The sequential characterization of compactness proves that $f(K)$ is compact. If uniform continuity failed, there would be $\varepsilon_0 > 0$ and $x_j, y_j \in K$ with

$$\|x_j - y_j\| < 1/j, \qquad \|f(x_j) - f(y_j)\| \geq \varepsilon_0.$$

Pass to a subsequence $x_{j_\ell} \to x \in K$. Then $y_{j_\ell} \to x$ also, and continuity contradicts the second inequality.

For scalar f, the nonempty set $f(K) \subset \mathbb{R}$ is closed and bounded. Its supremum and infimum exist by completeness of $\mathbb{R}$ and are limits of sequences in $f(K)$; closedness puts both in $f(K)$. Their preimages are the required extrema.

1.4 The derivative as a linear approximation

Definition 1.6. Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$. The map is differentiable at $a \in U$ if there is a linear map $Df(a) : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$f(a+h) = f(a) + Df(a)h + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \longrightarrow 0 \quad (h \to 0,\ h \neq 0).$$

The remainder is denoted $o(\|h\|)$. The notations $\mathrm{d}f_a$ and $f'(a)$ also denote this total derivative. For $v \in \mathbb{R}^n$, the directional derivative, when it exists, is

$$D_v f(a) = \lim_{t \to 0} \frac{f(a+tv) - f(a)}{t}.$$

The partial derivative $D_j f(a)$ is $D_{e_j} f(a)$. The Jacobian matrix has entries $\partial f_i / \partial x_j$, with m rows and n columns. The map is C^1 on U if it is differentiable there and $x \mapsto Df(x)$ is continuous in operator norm.

For a curve $c : I \to \mathbb{R}^n$ on an open interval, $Dc(t)s = sc'(t)$, where $c'(t)$ is the vector of coordinate derivatives. Thus velocity is a total derivative with one-dimensional domain [1, Ch. II].

Exercise 1.6 (uniqueness and consequences of differentiability)**.** Prove that the total derivative is unique, that differentiability implies continuity, and that $D_v f(a) = Df(a)v$. Deduce the Jacobian formula for $Df(a)$.

Solution. If A and B are two derivatives, then $(A - B)h = o(\|h\|)$. Put $h = tv$ for fixed $v \neq 0$, divide norms by $|t|$, and let $t \to 0$ to get $(A - B)v = 0$. Thus $A = B$.

The expansion gives $\|f(a+h) - f(a)\| \leq \|Df(a)\|\|h\| + o(\|h\|) \to 0$. For fixed $v \neq 0$, dividing the expansion at $h = tv$ by t gives $D_v f(a) = Df(a)v$; for $v = 0$ both sides vanish. In particular $Df(a)e_j = D_j f(a)$, so the partial derivatives are its columns. The component partials are therefore its matrix entries.

Exercise 1.7 (directional derivatives need not give a total derivative)**.** Set

$$f(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Show that f is continuous at the origin and all directional derivatives there exist, but f is not differentiable there.

Solution. Since $|f(x,y)| \le |x|$, continuity holds. For $v = (u, w) \neq 0$,

$$D_v f(0,0) = \frac{u^3}{u^2 + w^2}.$$

In particular, $D_1 f(0,0) = 1$ and $D_2 f(0,0) = 0$, so a total derivative would have to be $A(h,k) = h$. Along (t,t),

$$\frac{|f(t,t) - A(t,t)|}{\|(t,t)\|} = \frac{|t/2 - t|}{\sqrt{2}|t|} = \frac{1}{2\sqrt{2}},$$

which does not tend to zero. Existence of partial derivatives, or even all directional derivatives, is insufficient.

Exercise 1.8 (continuous partial derivatives suffice)**.** Suppose all partial derivatives of $f : U \to \mathbb{R}^m$ exist on a neighborhood of a and are continuous at a. Prove that f is differentiable at a. Consequently, f is C^1 on an open set exactly when all its component partial derivatives exist and are continuous there.

Solution. Choose a ball about a inside that neighborhood. For small h, pass from a to $a + h$ one coordinate at a time, writing

$$p_j = a + \sum_{\ell=1}^{j} h_\ell e_\ell, \qquad p_0 = a.$$

All intervening segments lie in the ball. For each component f_i and each $h_j \neq 0$, the scalar mean value theorem gives a point z_{ij} on the segment from p_{j-1} to p_j such that

$$f_i(p_j) - f_i(p_{j-1}) = h_j D_j f_i(z_{ij}).$$

When $h_j = 0$ this difference is zero. All $z_{ij} \to a$ as $h \to 0$. Let A have entries $A_{ij} = D_j f_i(a)$. For any $\eta > 0$, continuity makes all entry differences less than η for sufficiently small h. Telescoping then yields

$$|f_i(a+h) - f_i(a) - (Ah)_i| \le \eta \sum_j |h_j| \le \eta\sqrt{n}\|h\|.$$

The vector remainder has norm at most $\eta\sqrt{mn}\|h\|$, proving differentiability. Finally, for a matrix M, $\max_{i,j} |M_{ij}| \le \|M\| \le \sqrt{mn}\max_{i,j} |M_{ij}|$. Thus continuity of the entries and operator-norm continuity are equivalent, proving the C^1 assertion.

1.5 The chain rule and the mean value inequality

Exercise 1.9 (chain rule with dimensions)**.** Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open, let $f : U \to \mathbb{R}^m$ satisfy $f(U) \subset V$, and let $g : V \to \mathbb{R}^p$. If f is differentiable at a and g at $f(a)$, prove

$$D(g \circ f)(a) = Dg(f(a))Df(a).$$

The matrix product has dimensions $(p \times m)(m \times n) = p \times n$.

Solution. Put $A = Df(a)$, $B = Dg(f(a))$, and $k = f(a+h)-f(a) = Ah+r(h)$, where $r(h) = o(\|h\|)$. Hence $\|k\| \le C\|h\|$ for all sufficiently small h, for some finite C. Write

$$g(f(a)+k) - g(f(a)) = Bk + s(k), \qquad s(k) = o(\|k\|),$$

with $s(0) = 0$. Substitution gives the remainder $Br(h) + s(k)$. Its first term is $o(\|h\|)$. Given $\eta > 0$, the second has norm at most $\eta\|k\| \le \eta C\|h\|$ for sufficiently small h, including when $k = 0$. Thus it too is $o(\|h\|)$, proving the formula. Continuous matrix multiplication also shows that a composition of C^1 maps is C^1.

Definition 1.7. For scalar $f : U \subset \mathbb{R}^n \to \mathbb{R}$, the gradient is the column vector

$$\nabla f(a) = (D_1 f(a), \dots, D_n f(a))^T.$$

If f is differentiable, then $Df(a)h = \langle \nabla f(a), h\rangle$. In particular, for a differentiable curve c,

$$(f \circ c)'(t) = \langle \nabla f(c(t)), c'(t)\rangle.$$

Exercise 1.10 (the squared norm and its gradient)**.** Let $F : U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable and set $q(x) = \frac{1}{2}\|F(x)\|^2$. Compute ∇q. Explain the direction of steepest increase of a scalar differentiable function.

Solution. For $s(y) = \frac{1}{2}\|y\|^2$, expansion gives

$$s(y+k) - s(y) = \langle y, k\rangle + \tfrac{1}{2}\|k\|^2, \qquad Ds(y)k = \langle y, k\rangle.$$

The chain rule therefore gives

$$Dq(x)h = \langle F(x), DF(x)h\rangle, \qquad \nabla q(x) = DF(x)^T F(x).$$

For a unit vector v, $D_v f(a) = \langle \nabla f(a), v\rangle \le \|\nabla f(a)\|$ by Cauchy–Schwarz. If the gradient is nonzero, equality is attained at $v = \nabla f(a)/\|\nabla f(a)\|$. If it is zero, every directional derivative is zero.

Exercise 1.11 (multivariable mean value inequality)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable on an open set containing the segment $[x, y]$. If $\|Df(z)\| \le M$ for every z on that segment, prove

$$\|f(y) - f(x)\| \le M\|y - x\|.$$

Deduce that a derivative bound on a convex open set makes f Lipschitz there, and that $Df = 0$ on a connected open set makes f constant. Continuity of Df is not required for this inequality.

Solution. If $d = f(y) - f(x) = 0$, the inequality holds. Otherwise put $w = d/\|d\|$ and apply the scalar mean value theorem to

$$\phi(t) = \langle w, f(x + t(y - x))\rangle, \qquad 0 \le t \le 1.$$

The chain rule and that theorem give, for some $0 < t_* < 1$,

$$\|d\| = \phi(1) - \phi(0) = \langle w, Df(x + t_*(y - x))(y - x)\rangle \le M\|y - x\|.$$

Convexity puts every connecting segment in the domain, proving the Lipschitz statement. In a connected open set, join two points by a polygonal path as above and apply the inequality with $M = 0$ to each segment. The values at the endpoints are equal.

Exercise 1.12 (why there is no vector mean value equality)**.** For $c(t) = (\cos t, \sin t)$ on $[0, 2\pi]$, show that there is no $\xi \in (0, 2\pi)$ such that $c(2\pi) - c(0) = 2\pi c'(\xi)$.

Solution. The left side is zero, whereas $c'(t) = (-\sin t, \cos t)$ has norm 1 at every t. Thus the right side has norm 2π. Applying the scalar mean value theorem separately to components need not produce a common intermediate point. The norm inequality is the valid general statement for vector-valued maps [1, Chs. II–III].

1.6 Local inverses and implicit differentiation

Exercise 1.13 (stability of an invertible linear map)**.** Suppose $A : \mathbb{R}^n \to \mathbb{R}^n$ is invertible and $\|A^{-1}(B - A)\| \le q < 1$. Prove that B is invertible, that

$$\|B^{-1}\| \le \frac{\|A^{-1}\|}{1 - q},$$

and that inversion is continuous on the set of invertible matrices.

Solution. Put $E = A^{-1}(B - A)$, so $B = A(I + E)$. The estimate

$$\|(I + E)v\| \ge (1 - q)\|v\|$$

shows that $I + E$ is injective. Rank–nullity makes it surjective as well, and the same estimate bounds its inverse by $(1 - q)^{-1}$. This proves the inverse bound. If $B \to A$, the hypothesis holds with, for example, $q = 1/2$ for all sufficiently close B, and

$$B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1}$$

tends to zero in operator norm by that bound. Thus nearby matrices remain invertible and their inverses vary continuously.

Exercise 1.14 (contractions on a closed ball)**.** Let $K = \overline{B}(a, r) \subset \mathbb{R}^n$ and let $T : K \to K$ satisfy $\|T(x) - T(y)\| \le q\|x - y\|$ for all $x, y \in K$, where $0 \le q < 1$. Prove that T has a unique fixed point.

Solution. This is the closed-ball form of the contraction argument in Chapter 4; it uses only completeness, so it is independent of the inverse function theorem. Choose $x_0 \in K$ and put $x_{j+1} = T(x_j)$. If $0 < q < 1$, then

$$\|x_{j+1} - x_j\| \le q^j\|x_1 - x_0\|, \qquad \|x_k - x_j\| \le \frac{q^j}{1 - q}\|x_1 - x_0\| \quad (k > j).$$

Thus (x_j) is Cauchy and converges to $x_* \in K$, since the closed ball is complete. The Lipschitz estimate makes T continuous, so $T(x_*) = x_*$. Two fixed points would have distance at most q times their distance and hence coincide. If $q = 0$, T is constant and its value is the unique fixed point.

Theorem 1.8 (Inverse function theorem)**.** *Let $U \subset \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}^n$ be C^1, and suppose $A = Df(a)$ is invertible at $a \in U$. There are open neighborhoods $V \subset U$ of a and W of $f(a)$ such that $f : V \to W$ is a bijection with a C^1 inverse $g : W \to V$. At every $y \in W$,*

$$Dg(y) = \big[Df(g(y))\big]^{-1}.$$

Exercise 1.15 (prove the inverse function theorem)**.** Prove the theorem using the mean value inequality and the closed-ball contraction result. Specify the neighborhoods and prove surjectivity, injectivity, an inverse Lipschitz bound, differentiability of the inverse, and continuity of its derivative.

Solution. Normalize by setting

$$F(x) = A^{-1}(f(x) - f(a)), \qquad H(x) = F(x) - (x - a).$$

Then $F(a) = 0$, $DF(a) = I$, and $H(a) = 0$. Fix $0 < q < 1$. By continuity choose $r > 0$ such that $K = \overline{B}(a, r) \subset U$ and

$$\|DH(x)\| = \|A^{-1}Df(x) - I\| \le q \qquad (x \in K).$$

The mean value inequality along segments in K gives

$$\|H(x) - H(x')\| \le q\|x - x'\|, \qquad \|F(x) - F(x')\| \ge (1 - q)\|x - x'\|. \tag{*}$$

In particular, F and f are injective on K.

Put $\rho = (1 - q)r$. For $z \in B(0, \rho)$, define

$$T_z(x) = a + z - H(x) \qquad (x \in K).$$

It has Lipschitz constant at most q, and

$$\|T_z(x) - a\| \le \|z\| + q\|x - a\| \le \|z\| + qr < r.$$

Thus it maps K into itself. Its unique fixed point x solves $F(x) = z$. Moreover $\|x - a\| \le \|z\|/(1 - q) < r$, so the solution is in the open ball. Define

$$W = f(a) + AB(0, \rho), \qquad V = B(a, r) \cap f^{-1}(W).$$

The invertible linear map A is a homeomorphism, since A and A^{-1} satisfy operator bounds. Hence W is open; continuity of f makes V open. They contain $f(a)$ and a, respectively. The fixed-point construction proves surjectivity of $f : V \to W$, and (*) proves injectivity. If g denotes its inverse, (*) also gives

$$\|g(y) - g(y')\| \le \frac{\|A^{-1}\|}{1 - q}\|y - y'\| \qquad (y, y' \in W). \tag{**}$$

Thus g is Lipschitz and continuous.

For $x \in V$, the bound $\|A^{-1}Df(x) - I\| \le q$ and the linear stability exercise show that $Df(x)$ is invertible. Fix $y \in W$, write $x = g(y)$ and $B = Df(x)$, and let $k \to 0$ with $y + k \in W$. Set $h = g(y + k) - g(y)$. Estimate (**) gives $\|h\| \le C\|k\|$, and differentiability of f at x gives

$$k = Bh + r_x(h), \qquad r_x(h) = o(\|h\|).$$

Consequently

$$h = B^{-1}k - B^{-1}r_x(h) = B^{-1}k + o(\|k\|).$$

This proves differentiability and the asserted formula for $Dg(y)$. Finally g and Df are continuous, and matrix inversion is continuous at every invertible matrix by the stability exercise. Hence $y \mapsto [Df(g(y))]^{-1}$ is continuous. Thus g is C^1.

Theorem 1.9 (Implicit function theorem)**.** *Let $\Omega \subset \mathbb{R}^k \times \mathbb{R}^m$ be open and $G : \Omega \to \mathbb{R}^m$ be C^1. Suppose $G(a,b) = 0$ and the $m \times m$ matrix $D_v G(a,b)$ is invertible. There are open neighborhoods P of a and Q of b, with $P \times Q \subset \Omega$, and a C^1 map $h : P \to Q$ such that $h(a) = b$ and*

$$G(u,v) = 0 \text{ in } P \times Q \quad \Longleftrightarrow \quad v = h(u).$$

For every $u \in P$, $D_v G(u, h(u))$ is invertible and

$$Dh(u) = -\big[D_v G(u,h(u))\big]^{-1} D_u G(u,h(u)).$$

Here Dh and $D_u G$ are $m \times k$ matrices.

Exercise 1.16 (implicit differentiation in several dependent variables)**.** Explain how the inverse function theorem gives the stated neighborhoods and derive the derivative formula. For $m = 1$, express the result using partial derivatives.

Solution. The map $\Phi(u,v) = (u, G(u,v))$ has block derivative

$$D\Phi(a,b) = \begin{pmatrix} I_k & 0 \\ D_u G(a,b) & D_v G(a,b) \end{pmatrix},$$

which is invertible. Its local C^1 inverse has the form $(u,w) \mapsto (u, \psi(u,w))$, since Φ preserves u. This is the local graph construction used for regular level sets in Chapter 2. For precision, let O and O' be the open source and target neighborhoods of this inverse. Choose a product $P_0 \times J \subset O'$ about $(a,0)$ and set $h(u) = \psi(u,0)$ on P_0. Shrink O so that $D_v G$ remains invertible, which is possible by continuity and linear stability, and restrict the inverse neighborhoods accordingly before choosing the product. Choose a product $P_1 \times Q \subset O \cap \Omega$ about (a,b). Continuity of h allows an open $P \subset P_0 \cap P_1$ about a with $h(P) \subset Q$. If $(u,v) \in P \times Q$ is a zero of G, injectivity of Φ on O forces $v = h(u)$; the converse follows from its definition. This gives precisely the claimed product neighborhoods.

Differentiate $G(u, h(u)) = 0$ by the chain rule:

$$D_u G(u,h(u)) + D_v G(u,h(u)) Dh(u) = 0.$$

Multiplication on the left by the inverse of $D_v G$ gives the formula, with its stated dimensions. For one scalar dependent variable v it reads

$$D_j h(u) = -\frac{D_{u_j} G(u,h(u))}{D_v G(u,h(u))} \qquad (1 \le j \le k).$$

The invertibility hypothesis permits the local solution and justifies this division [1, Ch. III].

Chapter 2

Patches, manifolds, and constrained extrema

2.1 Patches and tangent spaces

2.1.1 A manifold is locally a graph

The guiding picture is a set which, near each point, looks like a graph over a coordinate plane. A curve has one free coordinate; a surface in three-dimensional space has two. The remaining coordinates are differentiable functions of the free coordinates [1].

For $1 \le i \le n$, let $\pi_i : \mathbb{R}^n \to \mathbb{R}^{n-1}$ denote the projection which deletes the ith coordinate:

$$\pi_i(x_1, \dots, x_n) = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n).$$

The hat notation $(x_1, \dots, \widehat{x_i}, \dots, x_n)$ means the same thing.

Definition 2.1. An $(n-1)$-dimensional *patch* in $\mathbb{R}^n$ is a set of the form

$$P = \{x \in \mathbb{R}^n : \pi_i(x) \in U,\ x_i = h(\pi_i(x))\},$$

where $U \subset \mathbb{R}^{n-1}$ is open and $h : U \to \mathbb{R}$ is C^1. More generally, after permuting coordinates, a k-dimensional patch is

$$P = \{(u, h(u)) : u \in U\}, \qquad U \subset \mathbb{R}^k \text{ open}, \quad h : U \to \mathbb{R}^{n-k} \text{ of class } C^1.$$

Thus $n-k$ coordinates are functions of the other k coordinates.

Definition 2.2. A subset $M \subset \mathbb{R}^n$ is a k-dimensional *manifold* if for every $a \in M$ there is an open set $V \subset \mathbb{R}^n$ containing a such that $M \cap V$ is a k-dimensional patch.

Note 2.3. Throughout this chapter, patches are C^1 and manifolds are embedded manifolds without boundary. The local graph condition implies the existence of a tangent space; the existence of tangent directions alone does not establish the local graph condition.

Example 2.4 (The unit circle)**.** The circle $S^1 = \{(x, y) : x^2 + y^2 = 1\}$ is covered by four open semicircles:

$$\begin{aligned}
y > 0: &\quad y = \sqrt{1 - x^2}, \quad -1 < x < 1,\\
y < 0: &\quad y = -\sqrt{1 - x^2}, \quad -1 < x < 1,\\
x > 0: &\quad x = \sqrt{1 - y^2}, \quad -1 < y < 1,\\
x < 0: &\quad x = -\sqrt{1 - y^2}, \quad -1 < y < 1.
\end{aligned}$$

Each is a one-dimensional patch. Every point is in at least one of them, so S^1 is a one-dimensional manifold. Two patches using only $y = \pm\sqrt{1-x^2}$ would miss the points $(\pm 1, 0)$; those points require the other choice of free coordinate.

Example 2.5 (One-dimensional patches in three dimensions)**.** A curve in three dimensions can have two dependent coordinates. For example,

$$(x, \sqrt{1-x^2}, 1), \quad -1 < x < 1, \qquad \text{and} \qquad (x, 1-x^2, 0), \quad x \in \mathbb{R},$$

are one-dimensional patches in $\mathbb{R}^3$. In the first case $y = \sqrt{1-x^2}$ and $z = 1$; in the second $y = 1 - x^2$ and $z = 0$.

2.1.2 Velocities of curves give the tangent space

Definition 2.6. For a manifold M and $a \in M$, define

$$T_a M = \{c'(0) : c : (-\varepsilon, \varepsilon) \to M \text{ is } C^1,\ c(0) = a,\ \varepsilon > 0\}.$$

This is the tangent space at a. It is a linear subspace of $\mathbb{R}^n$; the affine tangent plane is $a + T_a M$.

Proposition 2.7 (Tangent space of a graph)**.** *Suppose $F(u) = (u, h(u))$ parametrizes a k-patch and $a = F(b)$. Then*

$$T_a M = \{(v, Dh(b)v) : v \in \mathbb{R}^k\} = \operatorname{im} DF(b).$$

In particular, $\dim T_a M = k$.

Proof. Suppose $c(t) \in M$, $c(0) = a$, and let $u(t)$ be the free coordinates of $c(t)$. Locally $c(t) = F(u(t))$, with $u(0) = b$. The chain rule gives

$$c'(0) = DF(b)u'(0) = (u'(0), Dh(b)u'(0)).$$

Conversely, given $v \in \mathbb{R}^k$, openness of U ensures that $b + tv \in U$ for sufficiently small $|t|$. The curve $c(t) = F(b + tv)$ satisfies $c(0) = a$ and $c'(0) = (v, Dh(b)v)$. Thus the displayed set is exactly the set of velocities. The first k coordinates of $DF(b)v$ equal v, so $DF(b)$ is injective and its image has dimension k.

In the surface case $F(x, y) = (x, y, f(x, y))$, we obtain

$$c'(0) = x'(0)(1, 0, D_1 f(b)) + y'(0)(0, 1, D_2 f(b)).$$

Thus a basis for the tangent space is

$$(1, 0, D_1 f(b)), \qquad (0, 1, D_2 f(b)).$$

For $a = (b_1, b_2, f(b))$, the affine tangent plane has equation

$$z - f(b) = D_1 f(b)(x - b_1) + D_2 f(b)(y - b_2).$$

Example 2.8 (The cone at its vertex)**.** The graph $z = \sqrt{x^2 + y^2}$ is smooth away from the origin, where

$$\nabla z = \left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}} \right).$$

This formula is undefined at the origin. The vertex is not a manifold point.

Exercise 2.1 (why another graph cannot repair the cone)**.** Prove that the upper cone $C = \{(x, y, z) : z = \sqrt{x^2 + y^2}\}$ is not a two-dimensional C^1 manifold at the origin.

Solution. Let $c(t) = (x(t), y(t), z(t))$ be a differentiable curve in C with $c(0) = 0$, defined for both positive and negative t. Since $z(t) \geq 0 = z(0)$, we have $z'(0) = 0$. Also

$$|x(t)| \leq z(t), \qquad |y(t)| \leq z(t).$$

Dividing by $|t|$ and taking limits gives $x'(0) = y'(0) = 0$. Therefore every such curve has velocity zero. At a two-dimensional patch point the velocities form a two-dimensional space, by the proposition. This contradiction rules out every choice of graph coordinates, not just the displayed square-root graph.

2.2 Regular level sets

2.2.1 The implicit function theorem and normal vectors

Theorem 2.9 (Implicit function theorem: one equation)**.** *Let $G : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be C^1, with Ω open. Suppose $G(a) = 0$ and $D_i G(a) \neq 0$. Then near a the equation $G(x) = 0$ can be solved uniquely for x_i as a C^1 function of the other $n - 1$ coordinates. More precisely, there are neighborhoods U of $\pi_i(a)$ and V of a and a C^1 function $h : U \to \mathbb{R}$ such that*

$$G^{-1}(0) \cap V = \{x \in V : \pi_i(x) \in U,\ x_i = h(\pi_i(x))\}.$$

Exercise 2.2 (obtain the local graph from the inverse function theorem)**.** Recall the inverse function theorem: a C^1 map between open subsets of $\mathbb{R}^n$ with invertible derivative at a has a C^1 local inverse near a. Use it to prove the preceding implicit function statement.

Solution. Permute coordinates so that $i = n$, and write $x = (u, v) \in \mathbb{R}^{n-1} \times \mathbb{R}$. Set

$$H(u, v) = (u, G(u, v)).$$

Its derivative has block form

$$DH(a) = \begin{pmatrix} I & 0 \\ D_u G(a) & D_v G(a) \end{pmatrix}, \qquad \det DH(a) = D_v G(a) \neq 0.$$

The inverse function theorem gives open neighborhoods W of a and W' of $(\pi_n(a), 0)$ on which H is a C^1 bijection with C^1 inverse. Restrict W' to a product $U \times J$ containing $(\pi_n(a), 0)$. Because the first coordinates of H are u, the inverse necessarily has the form

$$H^{-1}(u, w) = (u, \psi(u, w)).$$

Set $h(u) = \psi(u, 0)$ and take $V = H^{-1}(U \times J)$. Then $G(u, v) = 0$ in V exactly when $v = h(u)$, and h is C^1. Uniqueness follows from injectivity of H on this neighborhood.

Theorem 2.10 (Regular level sets)**.** *Let $\Omega \subset \mathbb{R}^n$ be open and let $G : \Omega \to \mathbb{R}^m$ be C^1, where $1 \leq m \leq n$. Write $G = (G_1, \ldots, G_m)$, and let*

$$M = \{x \in \Omega : G(x) = 0,\ \operatorname{rank} DG(x) = m\}.$$

Then M is an $(n-m)$-dimensional manifold, and at every $a \in M$,

$$T_aM = \ker DG(a), \qquad (T_aM)^\perp = \operatorname{span}\{\nabla G_1(a), \dots, \nabla G_m(a)\}.$$

In particular, the gradients of the constraints are linearly independent normal vectors.

Exercise 2.3 (prove the regular level set description)**.** Establish the theorem, including equality of the tangent space and the kernel.

Solution. Since $DG(a)$ has rank m, some m columns form an invertible square submatrix. After permuting coordinates write $x = (u, v)$ with $u \in \mathbb{R}^{n-m}$, $v \in \mathbb{R}^m$, and $D_vG(a)$ invertible. Apply the inverse function theorem to

$$H(u, v) = (u, G(u, v)).$$

Its block triangular derivative is invertible. Exactly as in the preceding solution, the local inverse on a product neighborhood gives a C^1 graph $v = h(u)$ for $G = 0$. An $m \times m$ minor nonzero at a stays nonzero nearby by continuity, so all points of this local zero set are regular. Thus the graph also describes M locally.

For any curve c in M, differentiating $G(c(t)) = 0$ gives $DG(a)c'(0) = 0$. Hence $T_aM \subset \ker DG(a)$. The graph calculation gives $\dim T_aM = n - m$, while rank–nullity gives $\dim \ker DG(a) = n - m$. Inclusion and equal dimensions prove equality. Each row $\nabla G_j(a)^T$ annihilates the kernel. Their span has dimension m, the dimension of its orthogonal complement, proving the normal-space identity.

Note 2.11. For one equation, differentiation along a curve gives the identity:

$$G(c(t)) = 0 \quad \Longrightarrow \quad \nabla G(c(t)) \cdot c'(t) = 0 \quad \Longrightarrow \quad \nabla G(a) \cdot c'(0) = 0.$$

The nonzero-gradient hypothesis is essential for concluding that the entire level set is locally a hypersurface. A zero gradient alone does not decide whether the set is a manifold.

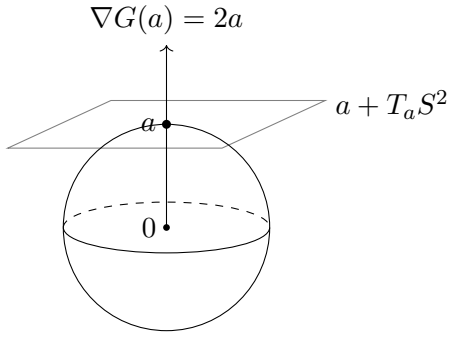
Example 2.12 (The unit sphere)**.** Let $G(x, y, z) = x^2 + y^2 + z^2 - 1$. Then $G^{-1}(0) = S^2$ and

$$\nabla G(x, y, z) = (2x, 2y, 2z), \qquad \nabla G(a) = 2a \neq 0 \quad (a \in S^2).$$

Thus S^2 is a two-dimensional manifold. At $a \in S^2$,

$$T_aS^2 = \{v : a \cdot v = 0\}, \qquad a + T_aS^2 = \{x : a \cdot x = 1\}.$$

For example, at $(0, 0, 1)$ the tangent space is $v_3 = 0$ and the affine tangent plane is $z = 1$.



The radius is normal to the tangent plane.

2.2.2 The torus homework and its radii

Note 2.13. For a ring torus obtained by rotating a circle of radius r whose center is distance R from the axis, assume $R > r > 0$. When the circle crosses the axis, the resulting surface can have singular points.

Exercise 2.4 (The ring torus)**.** Suppose $R > r > 0$. Rotate $(y - R)^2 + z^2 = r^2$ about the z-axis. Prove that the resulting set is a two-dimensional manifold.

Solution. Writing $\rho = \sqrt{x^2 + y^2}$, the rotated set is

$$T = \{(x, y, z) : (\rho - R)^2 + z^2 = r^2\}.$$

On T, $R - r \le \rho \le R + r$, so $\rho > 0$. On the open set $\rho > 0$ the function

$$G(x, y, z) = (\rho - R)^2 + z^2 - r^2$$

is C^1, and

$$\nabla G = \left(2(\rho - R)\frac{x}{\rho}, 2(\rho - R)\frac{y}{\rho}, 2z\right).$$

Its squared length on T is $4((\rho - R)^2 + z^2) = 4r^2 > 0$. Thus T is a regular level set and hence a two-manifold. A normal at a point is the displayed gradient, and its tangent space is the kernel of the corresponding dot product.

Exercise 2.5 (A spindle torus and its singularities)**.** Show why rotating $(y - 1)^2 + z^2 = 4$ does not give a two-dimensional C^1 manifold everywhere.

Solution. A parametrization of the rotated circle, with a signed radial coordinate, is

$$F(\theta, \phi) = ((1 + 2\cos\theta)\cos\phi, (1 + 2\cos\theta)\sin\phi, 2\sin\theta).$$

At $\theta_0 = 2\pi/3$, all values of ϕ give $a = (0, 0, \sqrt{3})$. Holding ϕ fixed and varying θ gives curves through a whose velocities are

$$v_\phi = (-\sqrt{3}\cos\phi, -\sqrt{3}\sin\phi, -1).$$

If the set were a two-manifold at a, its two-dimensional tangent space would contain all these vectors. But $v_0, v_\pi, v_{\pi/2}$ span $\mathbb{R}^3$: the sum of the first two gives a nonzero vertical vector, their difference gives a nonzero x vector, and the third then gives a nonzero y vector. This is impossible in a two-dimensional tangent space. The analogous point $(0, 0, -\sqrt{3})$ is singular as well.

2.3 Lagrange multipliers

2.3.1 One constraint and several constraints

Theorem 2.14 (Lagrange multipliers)**.** *Let $\Omega \subset \mathbb{R}^n$ be open and let $G : \Omega \to \mathbb{R}^m$ be C^1. Let $G(a) = 0$ and suppose* $\operatorname{rank} DG(a) = m$*. If $f : \Omega \to \mathbb{R}$ is differentiable and f restricted to $G^{-1}(0)$ has a local maximum or minimum at a, then*

$$\nabla f(a) = \lambda_1 \nabla G_1(a) + \cdots + \lambda_m \nabla G_m(a)$$

for some $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$. For one constraint $g = 0$, this is $\nabla f(a) = \lambda \nabla g(a)$.

Exercise 2.6 (the geometric multiplier proof)**.** Explain why the gradient of the objective is in the normal space at a constrained extremum.

Solution. Given any tangent vector $v \in T_aM$, choose a C^1 curve c in M with $c(0) = a$ and $c'(0) = v$. The one-variable function $f \circ c$ has an extremum at 0, so

$$0 = (f \circ c)'(0) = \nabla f(a) \cdot v.$$

Thus $\nabla f(a) \in (T_aM)^\perp$. The regular level set theorem identifies this orthogonal complement with the span of the independent constraint gradients, giving the asserted coefficients. This argument is local and gives a necessary condition; it does not decide whether a candidate is an extremum or whether any global extremum exists.

Note 2.15. Two independent constraint equations in $\mathbb{R}^4$ give a two-dimensional manifold. A point on a circle has one degree of freedom, and a point on a line has another.

2.3.2 A box of volume 1000

Example 2.16 (Least surface area)**.** Find the positive dimensions of a rectangular box of volume 1000 with least surface area. Write

$$xyz = 1000, \qquad S(x, y, z) = 2xy + 2yz + 2xz, \qquad g(x, y, z) = xyz - 1000.$$

For $x, y, z > 0$, $\nabla g = (yz, xz, xy) \neq 0$. The multiplier equations are

$$\begin{aligned} 2(y + z) &= \lambda yz, \\ 2(x + z) &= \lambda xz, \\ 2(x + y) &= \lambda xy. \end{aligned}$$

Dividing by the positive products gives

$$2\left(\frac{1}{y} + \frac{1}{z}\right) = 2\left(\frac{1}{x} + \frac{1}{z}\right) = 2\left(\frac{1}{x} + \frac{1}{y}\right) = \lambda.$$

Hence $x = y = z$, and the volume condition yields $x = y = z = 10$. The corresponding area is $S = 600$.

Exercise 2.7 (prove that the box candidate is the global minimum)**.** The multiplier calculation finds a candidate. Show that no other positive box of volume 1000 has smaller area.

Solution. Apply the arithmetic–geometric mean inequality to the three positive numbers xy, yz, xz:

$$xy + yz + xz \geq 3\sqrt[3]{(xy)(yz)(xz)} = 3(xyz)^{2/3} = 300.$$

Therefore $S \geq 600$. Equality holds precisely when $xy = yz = xz$, which for positive side lengths means $x = y = z = 10$. The candidate attains the bound, so existence and global minimality both follow. No compactness assertion about the unbounded constraint set is needed.

2.3.3 Distance between a circle and a line

Example 2.17 (The setup in four variables)**.** Find the minimum distance between $x^2 + y^2 = 1$ and $u + v = 4$. Minimize squared distance,

$$f(x, y, u, v) = (x - u)^2 + (y - v)^2,$$

subject to

$$g_1 = x^2 + y^2 - 1 = 0, \qquad g_2 = u + v - 4 = 0.$$

The gradients are

$$\nabla f = (2(x - u), 2(y - v), -2(x - u), -2(y - v)),$$
$$\nabla g_1 = (2x, 2y, 0, 0), \qquad \nabla g_2 = (0, 0, 1, 1).$$

They give the multiplier equations

$$2(x - u) = 2\lambda_1 x, \qquad 2(y - v) = 2\lambda_1 y,$$
$$-2(x - u) = \lambda_2, \qquad -2(y - v) = \lambda_2,$$

together with the two constraints. The constraint gradients are independent because $(x, y) \neq (0, 0)$ on the circle and their nonzero coordinates occupy disjoint blocks.

Exercise 2.8 (finish the distance calculation and justify the minimum)**.** Solve the preceding system and identify the global minimum distance.

Solution. The last two equations give $x - u = y - v =: d$. The first two then give $d = \lambda_1 x = \lambda_1 y$. If $\lambda_1 = 0$, then $d = 0$ and the two points coincide, forcing a point on the unit circle to have coordinate sum 4, impossible since $|x + y| \leq \sqrt{2}$. Therefore $x = y = \pm 1/\sqrt{2}$. From $u = x - d$, $v = y - d$, and $u + v = 4$, we obtain $d = x - 2$ and $u = v = 2$. These yield distances $2\sqrt{2} - 1$ and $2\sqrt{2} + 1$.

For a direct global check, fix (x, y) on the circle. Put $n = (1, 1)/\sqrt{2}$. Every point q on the line has $n \cdot q = 2\sqrt{2}$. Cauchy–Schwarz gives

$$\|q - (x, y)\| \geq |n \cdot (q - (x, y))| = \frac{4 - x - y}{\sqrt{2}} \geq \frac{4 - \sqrt{2}}{\sqrt{2}} = 2\sqrt{2} - 1.$$

At $(x, y) = (1/\sqrt{2}, 1/\sqrt{2})$ and $(u, v) = (2, 2)$ both inequalities are equalities. Thus the minimum is $2\sqrt{2} - 1$. The other multiplier point is the farthest circle point from the line after perpendicular projection; it is not a global maximum of the distance over all pairs, since the line is unbounded.

Chapter 3

Taylor approximation and critical points

3.1 Taylor's theorem in one variable

3.1.1 Polynomial approximation and the remainder

Recall that differentiability at a means

$$f(a+h) = f(a) + f'(a)h + R_1(h), \qquad \lim_{h\to 0} \frac{R_1(h)}{h} = 0.$$

The polynomial $f(a) + f'(a)h$ is the first-order approximation. Taylor's formula gives polynomial approximations of higher degree and measures the error [1].

Definition 3.1. If f has derivatives through order k at a, its kth Taylor polynomial, written in the increment $h = x - a$, is

$$P_k(h) = \sum_{j=0}^{k} \frac{f^{(j)}(a)}{j!} h^j.$$

We define the remainder by

$$R_k(h) = f(a+h) - P_k(h).$$

Here $f^{(0)} = f$. The polynomial as a function of x is $P_k(x-a)$; the distinction between x and the increment h prevents an extra shift of the center.

Theorem 3.2 (Taylor's theorem with Lagrange remainder)**.** *Let f be C^{k+1} on an open interval containing the closed interval between a and x. For $x \neq a$ there is a point ξ strictly between a and x such that*

$$f(x) = \sum_{j=0}^{k} \frac{f^{(j)}(a)}{j!}(x-a)^j + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x-a)^{k+1}.$$

For $k = 0$, this is the mean value theorem.

Exercise 3.1 (prove the Lagrange remainder formula)**.** Prove the theorem by repeated applications of Rolle's theorem.

Solution. Fix $x \neq a$ and set

$$C = \frac{f(x) - P_k(x-a)}{(x-a)^{k+1}}, \qquad F(t) = f(t) - P_k(t-a) - C(t-a)^{k+1}.$$

Then $F(x) = 0$ and $F^{(j)}(a) = 0$ for $0 \leq j \leq k$. Suppose first $a < x$. Applying Rolle's theorem to F on $[a, x]$ gives $t_1 \in (a, x)$ with $F'(t_1) = 0$. Since $F'(a) = 0$ when $k \geq 1$, applying Rolle to F' on $[a, t_1]$ gives $t_2 \in (a, t_1)$ with $F''(t_2) = 0$. Continue this procedure. At the last step, $F^{(k)}$ vanishes at a and at a point to its right, giving $F^{(k+1)}(\xi) = 0$. The case $x < a$ uses the intervals with reversed endpoints in precisely the same Rolle argument. For $k = 0$, only the first step is needed. Since the $(k+1)$st derivative of P_k is zero,

$$0 = f^{(k+1)}(\xi) - (k+1)!C.$$

Substituting C yields the formula.

Corollary 3.3 (A usable error bound)**.** *If $|f^{(k+1)}(t)| \leq M$ on the interval between a and $a+h$, then*

$$|R_k(h)| \leq \frac{M}{(k+1)!}|h|^{k+1}.$$

If this bound holds in a fixed neighborhood of a, then

$$\lim_{h \to 0} \frac{R_k(h)}{|h|^k} = 0.$$

Proof. Take absolute values in the Lagrange formula. For $h \neq 0$, division by $|h|^k$ leaves a bound $M|h|/(k+1)!$, which tends to zero.

Note 3.4. The last limit concerns $h \to 0$ with k fixed. Convergence of an infinite Taylor series concerns $k \to \infty$ with x fixed. The two limits answer different questions and require different arguments.

3.1.2 The Peano form of the remainder

Exercise 3.2 (Taylor approximation using only C^k regularity)**.** Suppose f is C^k near a. Prove the Peano form

$$f(a+h) = P_k(h) + o(|h|^k) \qquad (h \to 0).$$

For $k = 0$, interpret this as continuity.

Solution. For $k \geq 1$, repeated use of the fundamental theorem of calculus gives

$$f(a+h) = \sum_{j=0}^{k-1} \frac{f^{(j)}(a)}{j!}h^j + \frac{1}{(k-1)!}\int_0^h (h-t)^{k-1} f^{(k)}(a+t)\, \mathrm{d}t.$$

For completeness, the identity for $k = 1$ is the fundamental theorem itself. To pass from k to $k+1$, substitute $f^{(k)}(a+t) = f^{(k)}(a) + \int_0^t f^{(k+1)}(a+s)\, \mathrm{d}s$ into the formula, or integrate by parts once; this gives the next term $f^{(k)}(a)h^k/k!$ and the next integral remainder. Integration by parts also verifies the identity for negative h, using oriented integrals.

Subtracting the constant $f^{(k)}(a)$ inside the integral yields

$$R_k(h) = \frac{1}{(k-1)!}\int_0^h (h-t)^{k-1}(f^{(k)}(a+t) - f^{(k)}(a))\,\mathrm{d}t.$$

Given $\varepsilon > 0$, continuity gives $\delta > 0$ such that the difference in parentheses has absolute value at most ε whenever $|t| < \delta$. For $0 < |h| < \delta$, taking absolute values of the integral, in either orientation, gives

$$|R_k(h)| \le \frac{\varepsilon}{(k-1)!}\frac{|h|^k}{k} = \frac{\varepsilon}{k!}|h|^k.$$

This proves the limit. For $k = 0$, $R_0(h) = f(a+h) - f(a) \to 0$ by continuity.

3.1.3 The first nonzero derivative test

Theorem 3.5. *Suppose f is C^k near a, $k \ge 1$, and*

$$f'(a) = \cdots = f^{(k-1)}(a) = 0, \qquad f^{(k)}(a) \neq 0.$$

Then:

(a) If k is even and $f^{(k)}(a) > 0$, then a is a strict local minimum.
(b) If k is even and $f^{(k)}(a) < 0$, then a is a strict local maximum.
(c) If k is odd, then a is neither a local minimum nor a local maximum.

Proof. Translate a to zero. Taylor's formula gives

$$f(a+h) - f(a) = h^k\left(\frac{f^{(k)}(a)}{k!} + \frac{R_k(h)}{h^k}\right).$$

The expression in parentheses tends to the nonzero number $f^{(k)}(a)/k!$, so for sufficiently small nonzero h it has that number's sign. If k is even, $h^k > 0$ on both sides of zero, giving the stated strict extremum. If k is odd, h^k changes sign across zero, giving values both above and below $f(a)$ arbitrarily close to a.

Example 3.6. Suppose $f'(a) = f''(a) = 0$ and $f'''(a) \neq 0$. Then

$$f(a+h) - f(a) = \frac{f'''(a)}{6}h^3 + o(|h|^3),$$

so a is neither a maximum nor a minimum. A vanishing first derivative alone does not imply an extremum.

3.2 Original applications of one-variable Taylor theory

3.2.1 The exponential and its series

Example 3.7. For $f(x) = e^x$ and $a = 0$, every derivative at zero is 1, so

$$P_k(x) = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^k}{k!}, \qquad R_k(x) = \frac{e^\xi}{(k+1)!}x^{k+1}$$

for some ξ between 0 and x. If $x > 0$, then

$$0 < R_k(x) \le \frac{e^x x^{k+1}}{(k+1)!};$$

if $x < 0$, then

$$|R_k(x)| \le \frac{|x|^{k+1}}{(k+1)!}.$$

Equivalently, for every real x,

$$|R_k(x)| \le e^{\max(0,x)} \frac{|x|^{k+1}}{(k+1)!}.$$

Exercise 3.3 (complete the factorial-limit argument)**.** Show that for each fixed $x \in \mathbb{R}$, $|x|^k/k! \to 0$, and deduce $e^x = \sum_{j=0}^{\infty} x^j/j!$ for every real x.

Solution. Choose an integer K such that $|x|/(K+1) \le 1/2$. For every $k \ge K$, the ratio between consecutive nonnegative terms $|x|^k/k!$ is at most $1/2$. Consequently

$$\frac{|x|^{K+j}}{(K+j)!} \le \frac{|x|^K}{K!} 2^{-j} \to 0.$$

The remainder bound above therefore tends to zero as $k \to \infty$. Passing to the limit in $e^x = P_k(x) + R_k(x)$ proves the series identity.

3.2.2 Taylor's theorem proves that e is irrational

Example 3.8. Suppose for contradiction that $e = p/q$, where $p, q \in \mathbb{Z}$ and $q > 0$. Multiplying numerator and denominator if necessary, take $q \ge 3$. Taylor's theorem at 0, evaluated at 1, gives

$$\frac{p}{q} = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{q!} + R_q(1), \qquad R_q(1) = \frac{e^\xi}{(q+1)!}, \quad 0 < \xi < 1.$$

We have $e < 3$: from the exponential series, $n! \ge 2^{n-1}$ for $n \ge 2$, strictly for $n \ge 3$, so

$$e = 2 + \sum_{n=2}^{\infty} \frac{1}{n!} < 2 + \sum_{n=2}^{\infty} \frac{1}{2^{n-1}} = 3.$$

Multiplying the Taylor identity by $q!$ yields

$$(q-1)!p = \sum_{j=0}^{q} \frac{q!}{j!} + q! R_q(1).$$

The left side and the finite sum are integers. However,

$$0 < q! R_q(1) = \frac{e^\xi}{q+1} < \frac{3}{q+1} < 1.$$

Their difference cannot be this noninteger number. Thus e is irrational.

3.2.3 Approximating $\sqrt{2}$ near 1.96

Example 3.9. Use the fact that $\sqrt{1.96} = 1.4$. Let $f(x) = x^{1/2}$, $a = 1.96$, and $h = 0.04$. The derivatives are

$$f'(x) = \frac{1}{2}x^{-1/2}, \quad f''(x) = -\frac{1}{4}x^{-3/2}, \quad f'''(x) = \frac{3}{8}x^{-5/2}, \quad f^{(4)}(x) = -\frac{15}{16}x^{-7/2}.$$

It is convenient to put $s = 1.4$, so $a = s^2$. The cubic approximation is

$$\begin{aligned} P_3(0.04) &= s + \frac{0.04}{2s} - \frac{(0.04)^2}{8s^3} + \frac{(0.04)^3}{16s^5} \\ &= 1.4142135717260665\ldots. \end{aligned}$$

For $\xi \in (1.96, 2)$, the error satisfies

$$R_3(0.04) = -\frac{15}{16 \cdot 4!}\xi^{-7/2}(0.04)^4,$$

so

$$|R_3(0.04)| \le \frac{15}{16 \cdot 24}(1.96)^{-7/2}(0.04)^4 < 9.487 \times 10^{-9}.$$

In fact $R_3 < 0$, so the cubic approximation is an upper bound for $\sqrt{2}$.

3.2.4 A smooth function need not equal its Taylor series

Example 3.10. Consider

$$f(x) = \begin{cases} 0, & x \le 0, \\ e^{-1/x^2}, & x > 0. \end{cases}$$

It is infinitely differentiable, and $f^{(k)}(0) = 0$ for every $k \ge 0$. Its Taylor series at zero is therefore identically zero, although $f(x) > 0$ for every $x > 0$. Thus smoothness does not imply equality with a Taylor series in any neighborhood of the center.

Exercise 3.4 (prove smoothness and vanishing of all derivatives)**.** Supply the argument for the claims in the preceding example.

Solution. First, for every nonnegative integer m,

$$\lim_{x \downarrow 0} x^{-m} e^{-1/x^2} = 0.$$

Indeed, set $u = 1/x^2$. For an integer $N > m/2$, the exponential series gives $e^u \ge u^N/N!$ for $u > 0$, whence

$$0 \le u^{m/2} e^{-u} \le N! u^{m/2 - N} \to 0.$$

For $x > 0$, repeated differentiation gives

$$f^{(k)}(x) = P_k(1/x)e^{-1/x^2}$$

for a polynomial P_k. Start with $P_0 = 1$. If the formula holds for k, the chain and product rules give

$$P_{k+1}(t) = -t^2 P_k'(t) + 2t^3 P_k(t).$$

The limit already established implies that these expressions, and each divided by x, tend to zero as $x \downarrow 0$.

Now proceed inductively across zero. Continuity of f follows from the case $m = 0$. Suppose the globally defined kth derivative equals the displayed expression for $x > 0$ and zero for $x \leq 0$. Its derivative at zero is

$$\lim_{h \to 0} \frac{f^{(k)}(h) - 0}{h} = 0,$$

using the polynomial decay estimate from the right and the zero value from the left. Its derivative away from zero is the next displayed polynomial expression, which tends to zero at zero. Thus the $(k+1)$st derivative exists everywhere and is continuous at zero. Induction proves $f \in C^\infty$ and $f^{(k)}(0) = 0$ for all k.

3.3 Taylor's theorem in several variables

3.3.1 Partial derivatives and multinomial notation

Suppose $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is C^k. Partial derivatives of total order at most k exist and are continuous; their order of differentiation may be interchanged. For example, $D_1 D_2 f = D_2 D_1 f$ when $f \in C^2$.

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_n) \in (\mathbb{N} \cup \{0\})^n$, write

$$|\alpha| = \alpha_1 + \cdots + \alpha_n, \quad \alpha! = \alpha_1! \cdots \alpha_n!, \quad h^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}, \quad D^\alpha f = D_1^{\alpha_1} \cdots D_n^{\alpha_n} f.$$

Recall the binomial and multinomial formulas:

$$(a+b)^r = \sum_{j=0}^{r} \frac{r!}{j!(r-j)!} a^j b^{r-j}, \qquad (z_1 + \cdots + z_n)^r = \sum_{|\alpha|=r} \frac{r!}{\alpha!} z^\alpha.$$

For a fixed vector $h = (h_1, \ldots, h_n)$ we define

$$D_h f = \sum_{i=1}^{n} h_i D_i f.$$

Repeated directional differentiation gives

$$D_h^r f = (h_1 D_1 + \cdots + h_n D_n)^r f = \sum_{|\alpha|=r} \frac{r!}{\alpha!} h^\alpha D^\alpha f.$$

The equality follows by expanding the operator product: the coefficient counts the different orders in which the same partial derivatives occur, and C^r regularity allows those derivatives to commute. Here $D_h^0 f = f$. The vector h need not have length one.

For $n = 2$, this becomes

$$D_h^r f = \sum_{j=0}^{r} \frac{r!}{j!(r-j)!} h_1^j h_2^{r-j} D_1^j D_2^{r-j} f;$$

in particular,

$$D_h^2 f = h_1^2 D_1^2 f + 2 h_1 h_2 D_1 D_2 f + h_2^2 D_2^2 f.$$

3.3.2 Reduce the calculation to a line segment

Definition 3.11. The kth Taylor polynomial of f at a, in the increment h, is

$$P_k(h) = \sum_{r=0}^{k} \frac{D_h^r f(a)}{r!} = \sum_{|\alpha| \le k} \frac{D^\alpha f(a)}{\alpha!} h^\alpha.$$

The remainder is $R_k(h) = f(a+h) - P_k(h)$.

Theorem 3.12 (Multivariable Lagrange remainder)**.** *Suppose $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is C^{k+1} on an open set containing the line segment from a to $a+h$. If $h \neq 0$, there is $\theta \in (0,1)$, and hence $\xi = a + \theta h$, such that*

$$f(a+h) = \sum_{r=0}^{k} \frac{D_h^r f(a)}{r!} + \frac{D_h^{k+1} f(\xi)}{(k+1)!}.$$

Exercise 3.5 (prove the line-segment formula and its error estimate)**.** Derive the theorem from one-variable Taylor theory. Show that if all partial derivatives of order $k+1$ have absolute value at most M on that segment, then

$$|R_k(h)| \le \frac{M}{(k+1)!}(|h_1| + \cdots + |h_n|)^{k+1} \le \frac{M n^{(k+1)/2}}{(k+1)!} \|h\|^{k+1}.$$

Solution. Define $g(t) = f(a+th)$ for t in an open interval containing $[0,1]$. The chain rule and induction give $g^{(r)}(t) = D_h^r f(a+th)$. Apply one-variable Taylor's theorem at $t = 0$ with endpoint 1:

$$g(1) = \sum_{r=0}^{k} \frac{g^{(r)}(0)}{r!} + \frac{g^{(k+1)}(\theta)}{(k+1)!}.$$

This is precisely the claimed formula. Expanding D_h^{k+1} and taking absolute values gives

$$|D_h^{k+1} f(\xi)| \le M \sum_{|\alpha|=k+1} \frac{(k+1)!}{\alpha!} |h|^\alpha = M(|h_1| + \cdots + |h_n|)^{k+1}.$$

Here $|h|^\alpha$ means $|h_1|^{\alpha_1} \cdots |h_n|^{\alpha_n}$. Finally Cauchy–Schwarz gives $\sum_i |h_i| \le \sqrt{n}\|h\|$, proving the estimate.

Corollary 3.13. *Under C^{k+1} regularity near a,*

$$\lim_{h \to 0} \frac{R_k(h)}{\|h\|^k} = 0.$$

Proof. Choose a closed ball about a contained in the domain. The finitely many continuous derivatives of order $k+1$ are bounded on that ball by compactness. Apply the preceding estimate to each sufficiently short segment and divide by $\|h\|^k$.

Exercise 3.6 (the multivariable Peano remainder under C^k regularity)**.** Prove that $f \in C^k$ near a already implies $R_k(h) = o(\|h\|^k)$.

Solution. For $k \geq 1$, apply the integral formula in the one-variable Peano argument to $g(t) = f(a+th)$ between 0 and 1. Subtract the constant part of $g^{(k)}$ to obtain

$$R_k(h) = \frac{1}{(k-1)!} \int_0^1 (1-t)^{k-1} \big(D_h^k f(a+th) - D_h^k f(a)\big)\, \mathrm{d}t.$$

There are finitely many derivatives $D^\alpha f$ with $|\alpha| = k$. Their continuity at a implies that, for $\|h\| < \delta$, all differences $|D^\alpha f(a+th) - D^\alpha f(a)|$ are at most ε uniformly for $0 \leq t \leq 1$. The multinomial estimate then gives

$$|R_k(h)| \leq \frac{\varepsilon}{k!}(|h_1| + \cdots + |h_n|)^k \leq \frac{\varepsilon n^{k/2}}{k!}\|h\|^k.$$

Since ε is arbitrary, the quotient tends to zero. For $k = 0$, the assertion is continuity of f at a.

Note 3.14. The point ξ depends on a, h, k and is somewhere on the segment, not necessarily at its midpoint. This single-point Lagrange formula is stated for real-valued f. A vector-valued map need not have one common ξ for all components. The segment must lie in the domain; an arbitrary open domain need not contain the segment joining two of its points.

3.3.3 Examples of Taylor polynomials

Example 3.15 (Exponential of the sum of coordinates)**.** Let $f(x) = e^{x_1+\cdots+x_n}$ and $a = 0$. For every multi-index, $D^\alpha f(0) = 1$, so

$$D_h^r f(0) = \sum_{|\alpha|=r} \frac{r!}{\alpha!} h^\alpha = (h_1 + \cdots + h_n)^r.$$

Consequently

$$P_k(h) = \sum_{r=0}^{k} \frac{(h_1 + \cdots + h_n)^r}{r!}.$$

Writing $s = h_1 + \cdots + h_n$, the line-segment remainder is

$$R_k(h) = \frac{e^{\theta s} s^{k+1}}{(k+1)!} \quad \text{for some } \theta \in (0,1),$$

with $R_k(h) = 0$ if $s = 0$.

Example 3.16 (The product xy at $(1,1)$)**.** Let $f(x,y) = xy$, $h_1 = x-1$, $h_2 = y-1$. At $(1,1)$,

$$f = 1, \quad D_1 f = D_2 f = 1, \quad D_1^2 f = D_2^2 f = 0, \quad D_1 D_2 f = 1.$$

Thus

$$\begin{aligned} P_2(h) &= 1 + h_1 + h_2 + \tfrac{1}{2}(0 + 2h_1h_2 + 0) \\ &= 1 + h_1 + h_2 + h_1h_2 = (1+h_1)(1+h_2) = xy. \end{aligned}$$

The second-degree remainder vanishes identically. The factor 2 in the mixed derivative cancels the 2! in Taylor's formula.

Example 3.17 (Multiply Taylor approximations: $e^x \sin x$)**.** At zero, write

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + r(x), \qquad \sin x = x - \frac{x^3}{6} + s(x),$$

where $r(x) = o(|x|^3)$ and $s(x) = o(|x|^3)$. Multiplication of the polynomial parts gives

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\left(x - \frac{x^3}{6}\right) = x + x^2 + \frac{x^3}{3} - \frac{x^5}{12} - \frac{x^6}{36}.$$

The x^4 terms cancel. The remaining terms involving r or s are

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)s(x) + \left(x - \frac{x^3}{6}\right)r(x) + r(x)s(x),$$

all $o(|x|^3)$, since the polynomial factors are bounded near zero and both remainders tend to zero. The terms of degree five and six are also $o(|x|^3)$. Hence

$$e^x \sin x = x + x^2 + \frac{x^3}{3} + o(|x|^3), \qquad P_3(x) = x + x^2 + \frac{x^3}{3}.$$

The remainder estimates justify discarding every term of order higher than three.

3.4 The Hessian and the second derivative test

3.4.1 Quadratic forms at a critical point

Suppose a is a critical point of $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, meaning $Df(a) = 0$, and f is C^2 near a. Its Hessian is the symmetric matrix

$$H_f(a) = \left(D_i D_j f(a)\right)_{1 \le i,j \le n}.$$

The second-order Taylor formula is

$$f(a + h) - f(a) = q(h) + R_2(h), \qquad q(h) = \frac{1}{2} h^T H_f(a) h, \qquad R_2(h) = o(\|h\|^2).$$

In particular, for $h \neq 0$,

$$q(h) = \|h\|^2 q\left(\frac{h}{\|h\|}\right).$$

Thus the sign of the quadratic part is determined by its values on the unit sphere.

Definition 3.18. For a real symmetric matrix A, the quadratic form $q_A(v) = v^T A v$ is

- positive definite if $q_A(v) > 0$ for every $v \neq 0$;
- negative definite if $q_A(v) < 0$ for every $v \neq 0$;
- indefinite if it is positive for some vector and negative for another.

It is positive semidefinite if $q_A(v) \ge 0$ for all v, and negative semidefinite if $q_A(v) \le 0$ for all v. The positive factor $1/2$ in the Taylor quadratic form changes none of these sign properties.

Theorem 3.19 (Second derivative test)**.** *At a critical point a of a C^2 real-valued function:*

(a) A positive definite Hessian gives a strict local minimum.

(b) A negative definite Hessian gives a strict local maximum.

(c) An indefinite Hessian gives values both above and below $f(a)$ arbitrarily near a, so a is a saddle point.

A semidefinite Hessian alone is inconclusive.

Exercise 3.7 (prove the second derivative test)**.** Explain why positivity in every direction controls the remainder uniformly, and why indefiniteness prevents an extremum.

Solution. If q is positive definite, continuity and compactness of the unit sphere give

$$m = \min_{\|u\|=1} q(u) > 0.$$

The strict inequality follows because the minimum is attained and every unit vector has positive value. Hence $q(h) \geq m\|h\|^2$. For sufficiently small h, $|R_2(h)| \leq (m/2)\|h\|^2$, so

$$f(a+h) - f(a) \geq (m/2)\|h\|^2 > 0 \quad (h \neq 0).$$

This proves the minimum assertion. Apply it to $-f$ to prove the maximum assertion.

If q is indefinite, choose unit vectors u, v with $q(u) > 0$ and $q(v) < 0$. Along either fixed direction,

$$\frac{f(a+tu) - f(a)}{t^2} = q(u) + \frac{R_2(tu)}{t^2} \longrightarrow q(u)$$

(and the same for v). Thus for arbitrarily small nonzero t the first difference is positive and the second is negative. An extremum is impossible.

3.4.2 Sylvester's criterion

Let A be a real symmetric $n \times n$ matrix. For $1 \leq k \leq n$, let Δ_k be the determinant of its upper-left $k \times k$ submatrix. These are the *leading principal minors*.

Theorem 3.20 (Sylvester's criterion)**.**

$$\begin{aligned} A \text{ is positive definite} &\iff \Delta_k > 0 \quad (1 \leq k \leq n), \\ A \text{ is negative definite} &\iff (-1)^k \Delta_k > 0 \quad (1 \leq k \leq n). \end{aligned}$$

If $\det A \neq 0$ *and neither definiteness condition holds, then A is indefinite.*

Exercise 3.8 (justify the determinant test)**.** Prove the criterion using completion of squares, and justify its last assertion without assuming that a zero determinant means an inconclusive Hessian in every case.

Solution. We prove the positive definite criterion by induction on the size. The one-dimensional case is immediate. Write

$$A = \begin{pmatrix} a & b^T \\ b & C \end{pmatrix}, \qquad S = C - \frac{bb^T}{a} \quad (a > 0).$$

For $v = (t, z)$, completion of the square gives

$$v^T A v = a\left(t + \frac{b^T z}{a}\right)^2 + z^T S z.$$

Thus A is positive definite exactly when $a > 0$ and S is positive definite. One direction follows from the sum of the two positive terms, noting that $z = 0$, $t \neq 0$ gives a positive first term. Conversely, if A is positive definite then $a > 0$, and choosing $t = -b^T z/a$ proves $z^T S z > 0$ for every $z \neq 0$.

Eliminating the first column in the leading blocks gives

$$\Delta_{j+1}(A) = a\Delta_j(S) \qquad (1 \leq j \leq n-1).$$

Indeed, subtracting b_i/a times the first row from each subsequent row leaves a block upper triangular matrix with diagonal blocks a and the appropriate leading block of S, without changing its determinant. Since $\Delta_1(A) = a$, positivity of all leading minors of A is equivalent to $a > 0$ and positivity of all leading minors of S. Induction proves the first equivalence. Apply it to $-A$, whose kth leading minor is $(-1)^k\Delta_k(A)$, to prove the second.

For the last assertion, if a quadratic form is not indefinite, it has a constant weak sign, so it is positive or negative semidefinite. Suppose A is positive semidefinite and $v^T A v = 0$. For every w and every real t,

$$0 \leq (v + tw)^T A(v + tw) = 2t\, v^T A w + t^2 w^T A w.$$

If $v^T A w \neq 0$, sufficiently small t of the opposite sign would make the expression negative. Hence $v^T A w = 0$ for all w, so $Av = 0$. If A is invertible, this forces $v = 0$, showing that positive semidefiniteness is strict. The negative case follows by replacing A with $-A$. Therefore an invertible symmetric matrix which is neither positive nor negative definite must be indefinite.

Note 3.21. The nonzero-determinant condition in the last assertion is sufficient, not necessary for indefiniteness. For example $\mathrm{diag}(1, -1, 0)$ is indefinite although singular. Also, leading minors alone with weak inequalities are not the corresponding test for positive semidefiniteness: for example $\mathrm{diag}(0, -1)$ has both leading minors zero but takes negative values.

Exercise 3.9 (the two-variable test and its limits)**.** Derive the usual Hessian discriminant test for two variables and give examples explaining why a semidefinite quadratic part does not settle the extremum question.

Solution. Write

$$H_f(a) = \begin{pmatrix} A & B \\ B & C \end{pmatrix}, \qquad D = AC - B^2.$$

Sylvester's criterion gives a strict local minimum when $A > 0$, $D > 0$, and a strict local maximum when $A < 0$, $D > 0$. If $D < 0$, the matrix is invertible and neither definiteness criterion holds, so it is indefinite and a is a saddle. If $D = 0$, further terms may matter. At the origin the functions

$$x^2 + y^4, \qquad x^2 - y^4$$

share the same positive semidefinite Hessian $\mathrm{diag}(2, 0)$. The first is strictly positive away from the origin, giving a strict minimum; the second is positive on the x-axis and negative on the y-axis, giving a saddle. A completely zero Hessian is also inconclusive: $x^4 + y^4$ has a strict minimum, $-x^4 - y^4$ a strict maximum, and $x^4 - y^4$ a saddle.

Chapter 4

Fixed points and Newton's method

4.1 From a tangent line to an iteration

4.1.1 Newton's geometric construction

Suppose we want to approximate a solution x_* of $f(x) = 0$. Consider a differentiable function whose derivative is nonzero on an interval and whose endpoint values have opposite signs. Continuity gives a root, and a derivative of constant nonzero sign gives uniqueness. Choosing a good initial point and proving convergence remain separate tasks.

Given x_0 near x_*, linearize at x_0:

$$f(x_*) - f(x_0) \approx f'(x_0)(x_* - x_0).$$

Since $f(x_*) = 0$, this suggests

$$x_* \approx x_0 - \frac{f(x_0)}{f'(x_0)} =: x_1.$$

Geometrically, the tangent line at $(x_0, f(x_0))$ is

$$y = f(x_0) + f'(x_0)(x - x_0).$$

Its intersection with the x-axis has coordinate x_1. Repeating the construction gives Newton's method [1]:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

The denominator must be nonzero at every iterate. If the tangent intercept lies to the left of x_0, its slope satisfies

$$\frac{f(x_0)}{x_0 - x_1} = f'(x_0).$$

It is this line intercept, not a general equality with the actual root, that gives the formula.

4.2 Contraction mappings

4.2.1 The contraction mapping principle on an interval

Definition 4.1. A map $\varphi : [a, b] \to [a, b]$ is a contraction with constant k, where $0 \le k < 1$, if

$$|\varphi(x) - \varphi(y)| \le k|x - y| \qquad (x, y \in [a, b]).$$

A point x_* is a fixed point if $\varphi(x_*) = x_*$.

The requirement that the interval maps into itself is part of the definition being used here. A slope bound alone does not provide it.

Theorem 4.2 (Contraction mapping principle)**.** *Let $\varphi : [a, b] \to [a, b]$ be a contraction with constant $k < 1$. There is exactly one fixed point $x_* \in [a, b]$. For any $x_0 \in [a, b]$, the iterates*

$$x_{n+1} = \varphi(x_n)$$

converge to x_*. For $0 < k < 1$ and $n \geq 0$,

$$|x_n - x_*| \leq \frac{k^n}{1-k}|x_1 - x_0|, \qquad |x_n - x_*| \leq k^n|x_0 - x_*|.$$

If $k = 0$, the map is constant and $x_1 = x_*$.

Proof. The self-map condition ensures that every iterate lies in $[a, b]$. Repeated use of the contraction inequality gives

$$|x_{n+1} - x_n| \leq k|x_n - x_{n-1}| \leq k^n|x_1 - x_0|.$$

For integers $m > n \geq 0$, telescoping and the triangle inequality give

$$\begin{aligned} |x_m - x_n| &\leq \sum_{j=n}^{m-1} |x_{j+1} - x_j| \\ &\leq \sum_{j=n}^{m-1} k^j|x_1 - x_0| \leq \frac{k^n}{1-k}|x_1 - x_0|. \end{aligned}$$

The last bound tends to zero with n, so (x_n) is Cauchy. Completeness of $\mathbb{R}$ gives a limit x_*, and closedness of $[a, b]$ gives $x_* \in [a, b]$. A contraction is continuous because it is Lipschitz. Consequently

$$\varphi(x_*) = \varphi(\lim x_n) = \lim \varphi(x_n) = \lim x_{n+1} = x_*.$$

If y_* is another fixed point, then

$$|x_* - y_*| = |\varphi(x_*) - \varphi(y_*)| \leq k|x_* - y_*|.$$

Since $1 - k > 0$, this forces $x_* = y_*$. Letting $m \to \infty$ in the telescoping bound proves the first error estimate. Comparing x_n directly with the fixed point and iterating the contraction inequality proves the second. The case $k = 0$ follows directly from the definition.

Exercise 4.1 (an error bound from the most recent step)**.** Show that for $n \geq 1$,

$$|x_n - x_*| \leq \frac{k}{1-k}|x_n - x_{n-1}|.$$

Explain how to use this as a stopping rule.

Solution. Starting from the nth step, contraction gives $|x_{n+j+1} - x_{n+j}| \leq k^{j+1}|x_n - x_{n-1}|$ for $j \geq 0$. Sum the tail and pass to its limit:

$$|x_* - x_n| \leq \sum_{j=0}^{\infty} k^{j+1}|x_n - x_{n-1}| = \frac{k}{1-k}|x_n - x_{n-1}|.$$

Thus an error at most ε is certified once the right-hand side is at most ε, provided the map and its contraction constant have already been justified. A small difference between successive iterates without a known contraction estimate is not by itself an error certificate. For $k = 0$, all iterates from x_1 onward already equal the fixed point.

4.3 A fixed-slope method with interval convergence

4.3.1 Interval convergence

Theorem 4.3. *Let f be continuous on $[a,b]$ and differentiable on (a,b), with*

$$f(a) < 0 < f(b), \qquad 0 < m \le f'(x) \le M \quad (a < x < b).$$

Then f has a unique zero $x_* \in (a,b)$. For every $x_0 \in [a,b]$, the sequence

$$x_{n+1} = x_n - \frac{f(x_n)}{M}$$

stays in $[a,b]$ and converges to x_*. If $m < M$, then for every $n \ge 0$,

$$|x_n - x_*| \le \frac{|f(x_0)|}{m}\left(1 - \frac{m}{M}\right)^n.$$

If $m = M$, the first iteration already gives the root.

Proof. Define $\varphi(x) = x - f(x)/M$. Its derivative satisfies

$$0 \le \varphi'(x) = 1 - \frac{f'(x)}{M} \le 1 - \frac{m}{M} =: k < 1.$$

By the mean value theorem φ is nondecreasing and satisfies $|\varphi(x) - \varphi(y)| \le k|x - y|$. Moreover,

$$a < \varphi(a) = a - \frac{f(a)}{M} \le \varphi(x) \le \varphi(b) = b - \frac{f(b)}{M} < b.$$

Thus φ is a contraction of $[a,b]$ into itself. Its unique fixed point satisfies

$$x_* = x_* - f(x_*)/M, \qquad \text{hence } f(x_*) = 0.$$

This proves existence, uniqueness, and convergence. For $m < M$, use the contraction error bound and $|x_1 - x_0| = |f(x_0)|/M$:

$$|x_n - x_*| \le \frac{k^n}{1-k}\frac{|f(x_0)|}{M} = \frac{|f(x_0)|}{m}\left(1 - \frac{m}{M}\right)^n.$$

If $m = M$, the mean value theorem makes φ constant, so its first image is its fixed point.

Note 4.4. The number M need only be an upper bound for f', not necessarily an attained maximum. The strictly positive lower bound m controls how close the contraction constant is to 1. If $f' < 0$ instead, apply the theorem to $-f$; with $0 < m \le -f' \le M$ the iteration becomes $x_{n+1} = x_n + f(x_n)/M$.

Exercise 4.2 (relate the error to the residual)**.** Under the preceding hypotheses, show that for any $x \in [a,b]$,

$$m|x - x_*| \le |f(x)| \le M|x - x_*|.$$

Solution. If $x = x_*$ both sides vanish. Otherwise, the mean value theorem on the interval joining x to x_* gives

$$f(x) - f(x_*) = f'(c)(x - x_*)$$

for some interior point c. Take absolute values and use $f(x_*) = 0$ and $m \le f'(c) \le M$. In particular, a residual $|f(x)|$ implies an error at most $|f(x)|/m$. Combining this at x_0 with the contraction comparison $|x_n - x_*| \le k^n|x_0 - x_*|$ also proves the theorem's error estimate.

4.4 When Newton's method converges

4.4.1 A local convergence theorem

Exercise 4.3 (local quadratic convergence at a simple root)**.** Suppose f is C^2 near α, with $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. Prove that every sufficiently close initial point produces well-defined Newton iterates converging to α, and prove an estimate

$$|x_{n+1} - \alpha| \leq C|x_n - \alpha|^2$$

with a constant independent of n.

Solution. Choose a closed interval $I_0 = [\alpha - r_0, \alpha + r_0]$ in the domain such that $|f'| \geq m_0 > 0$ throughout I_0. This is possible by continuity and the simple-root assumption. On this compact interval, $|f''| \leq L$ for some finite L. Define the Newton map

$$N(x) = x - \frac{f(x)}{f'(x)}, \qquad N(\alpha) = \alpha.$$

It is C^1 on I_0, with

$$N'(x) = \frac{f(x)f''(x)}{(f'(x))^2}, \qquad N'(\alpha) = 0.$$

By continuity shrink to $I = [\alpha - r, \alpha + r]$ so that $|N'(x)| \leq q < 1$ there. The mean value theorem gives

$$|N(x) - \alpha| = |N(x) - N(\alpha)| \leq q|x - \alpha| \leq qr < r.$$

Thus N maps I into itself and is a contraction. The contraction theorem proves that every $x_0 \in I$ produces iterates staying in I and converging to the unique fixed point α.

For the stronger estimate, apply Taylor's theorem at x_n, evaluated at α. For some ξ_n between them,

$$0 = f(\alpha) = f(x_n) + f'(x_n)(\alpha - x_n) + \tfrac{1}{2}f''(\xi_n)(\alpha - x_n)^2.$$

Rearranging gives the exact error identity

$$x_{n+1} - \alpha = \frac{f''(\xi_n)}{2f'(x_n)}(x_n - \alpha)^2.$$

Therefore $|x_{n+1} - \alpha| \leq (L/(2m_0))|x_n - \alpha|^2$. The bound remains valid if an iterate equals the root, since all subsequent iterates then equal the root as well. This is quadratic convergence in the stated estimate sense; it may be even faster when the second derivative vanishes.

4.4.2 A worked comparison using $\sqrt{2}$

Exercise 4.4 (compare Taylor, fixed slope, and Newton for the same root)**.** Apply the two iterative methods to $f(x) = x^2 - 2$ on $[1, 2]$. Give a guaranteed error bound for the fixed-slope method, and show directly that Newton's method starting at 1 converges to $\sqrt{2}$.

Solution. Here $f(1) = -1$, $f(2) = 2$, and $2 \leq f'(x) = 2x \leq 4$. The fixed-slope theorem applies with $m = 2$ and $M = 4$:

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{4}.$$

Starting at $x_0 = 1$ gives

$$x_1 = \frac{5}{4}, \qquad x_2 = \frac{87}{64}, \qquad |x_n - \sqrt{2}| \le \frac{1}{2}\left(\frac{1}{2}\right)^n.$$

The derivative of its iteration map is $1 - x/2$, between 0 and $1/2$ on $[1, 2]$, in agreement with the theorem.

Newton's method is

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

Starting at $x_0 = 1$ gives

$$x_1 = \frac{3}{2}, \qquad x_2 = \frac{17}{12}, \qquad x_3 = \frac{577}{408}.$$

Let $\alpha = \sqrt{2}$. For every $x > 0$,

$$\frac{1}{2}\left(x + \frac{2}{x}\right) - \alpha = \frac{(x-\alpha)^2}{2x} \ge 0.$$

Thus every iterate from x_1 onward is at least α. If $x \ge \alpha$, then $2/x \le x$, so its Newton image is at most x. The sequence from x_1 is decreasing and bounded below by α, hence converges to a positive limit ℓ. Passing to the limit in the iteration gives $\ell = (\ell + 2/\ell)/2$, so $\ell^2 = 2$ and $\ell = \alpha$. The displayed identity also gives

$$|x_{n+1} - \alpha| = \frac{|x_n - \alpha|^2}{2x_n} \le \frac{|x_n - \alpha|^2}{2\alpha} \qquad (n \ge 1).$$

The cubic Taylor approximation uses the nearby value $\sqrt{1.96} = 1.4$ once, with a fourth-derivative error bound. Both approaches give controlled approximations, but the iteration repeatedly improves its current point.

4.4.3 Why convergence hypotheses matter

Exercise 4.5 (a Newton cycle despite a root)**.** Show that Newton's method for $f(x) = x^3 - 2x + 2$, starting at $x_0 = 0$, never converges to a root. Check also that the equation does have a real root.

Solution. We have $f(0) = 2$, $f'(0) = -2$, so $N(0) = 0 - 2/(-2) = 1$. Also $f(1) = 1$, $f'(1) = 1$, so $N(1) = 1 - 1 = 0$. Therefore the iterates alternate between 0 and 1, with nonzero derivatives at every step. They do not converge. Yet $f(-2) = -2$ and $f(-1) = 3$, so the intermediate value theorem supplies a real root in $(-2, -1)$. Existence of a root and nonzero denominators along the computed orbit do not suffice; a convergence argument must control the initial point and the iteration map.

Chapter 5

Area, signed integrals, and the fundamental theorem

5.1 Defining area by rectangles

5.1.1 Inner and outer approximation

We define the area $a(R)$ of a rectangle R to be its length times its width. We take rectangles to be closed. A finite family of rectangles is called nonoverlapping when their interiors are pairwise disjoint; common edges are permitted.

Definition 5.1. A bounded set $S \subset \mathbb{R}^2$ has area α if, given $\varepsilon > 0$, there exist both:

(i) nonoverlapping rectangles $R_1, \dots, R_k \subset S$ with $\sum_i a(R_i) > \alpha - \varepsilon$;
(ii) rectangles $R'_1, \dots, R'_\ell$ covering S with $\sum_j a(R'_j) < \alpha + \varepsilon$.

We write $a(S) = \alpha$. This is the finite-rectangle, or Jordan, notion of area.

Let $\mathcal{A}$ be the collection of bounded sets having area. The following properties hold:

(A) If $S, T \in \mathcal{A}$ and $S \subset T$, then $a(S) \le a(T)$.
(B) If $S, T \in \mathcal{A}$ and $S \cap T = \varnothing$, then $S \cup T \in \mathcal{A}$ and $a(S \cup T) = a(S) + a(T)$.
(C) Congruent sets in $\mathcal{A}$ have the same area.
(D) A rectangle has area equal to base times height.

For example, the set of points in $[0,1]^2$ whose two coordinates are rational does not have area in this sense: every nondegenerate inner rectangle fails to lie in the set, whereas a finite closed-rectangle cover must also cover its closure, the whole square. Thus its inner area is 0 and its outer area is 1.

5.1.2 Continuity and the region under a graph

Recall: f is continuous at a if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$, for x in its domain. We say f is uniformly continuous on D if one δ works for every pair $x, y \in D$:

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

Recall also that a continuous function on a compact set is uniformly continuous.

Suppose $f : [a, b] \to [0, \infty)$ is continuous. The set

$$Q_a^b(f) = \{(x, y) : a \le x \le b,\ 0 \le y \le f(x)\}$$

has area. We define

$$\int_a^b f(x)\ \mathrm{d}x = a(Q_a^b(f)).$$

Note: if $0 \le m \le f(x) \le M$, then monotonicity and the rectangle formula give

$$m(b-a) \le \int_a^b f \le M(b-a).$$

Exercise 5.1 (Why a continuous graph encloses an area)**.** Supply the rectangle argument underlying this definition.

Solution. Partition $[a, b]$ into intervals I_i of length less than a uniform-continuity scale for f. Let $m_i = \min_{I_i} f$ and $M_i = \max_{I_i} f$. Rectangles of heights m_i lie below the graph, and those of heights M_i cover the ordinate set. Given $\eta > 0$, choose the partition so that $M_i - m_i < \eta$. The difference between the two total areas is at most

$$\sum_i (M_i - m_i)|I_i| \le \eta(b-a).$$

The supremum of the inner rectangle areas and the infimum of the outer rectangle areas therefore agree. Their common value gives the required α, with arbitrarily accurate inner and outer approximations. Degenerate intervals or zero heights cause no difficulty.

5.2 Signed integrals and elementary rules

5.2.1 Positive and negative parts

For continuous $f : [a, b] \to \mathbb{R}$, we define

$$f^+(x) = \max\{f(x), 0\}, \qquad f^-(x) = \max\{-f(x), 0\}.$$

Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$. Both parts are nonnegative and continuous. We define the signed integral by

$$\int_a^b f = \int_a^b f^+ - \int_a^b f^-.$$

Thus area below the horizontal axis contributes with a minus sign. We use the conventions $\int_a^a f = 0$ and $\int_b^a f = -\int_a^b f$.

Lemma 5.2 (Additivity in the interval)**.** *If $a < c < b$, then*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof. For a nonnegative function, split the ordinate set at $x = c$. The pieces share only a vertical line segment of area zero, so their areas add. Apply this separately to f^+ and f^- and subtract.

Lemma 5.3 (Bounds for a signed integral)**.** *If $m \le f(x) \le M$ on $[a,b]$, then*

$$m(b-a) \le \int_a^b f \le M(b-a).$$

Proof. For the upper bound, if $M > 0$, then $f^+ \le M$, so $\int f = \int f^+ - \int f^- \le \int f^+ \le M(b-a)$. If $M \le 0$, then $f^+ = 0$ and $f^- = -f \ge -M \ge 0$, giving $\int f = -\int f^- \le M(b-a)$. Apply the upper bound to $-f \le -m$ to obtain the lower bound.

5.3 The fundamental theorem of calculus

5.3.1 An integral as an antiderivative

Theorem 5.4 (Fundamental theorem, Part 1)**.** *Suppose $f : [a,b] \to \mathbb{R}$ is continuous and*

$$F(x) = \int_a^x f(t)\ \mathrm{d}t.$$

Then F is C^1 and $F' = f$ on (a,b), with the corresponding one-sided derivatives at the endpoints.

Proof. Fix $x \in (a,b)$. For $h > 0$ small enough, let $m(h)$ and $M(h)$ be the minimum and maximum of f on $[x, x+h]$. By additivity and the preceding bounds,

$$m(h) \le \frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\ \mathrm{d}t \le M(h).$$

Continuity gives $m(h), M(h) \to f(x)$. For $h < 0$, write the same quotient as the average of f over $[x+h, x]$; the same bounds apply. The squeeze theorem gives $F'(x) = f(x)$. Since f is continuous, so is F'.

5.3.2 Evaluation by an antiderivative

Corollary 5.5 (Fundamental theorem, Part 2)**.** *If f is continuous and $G' = f$, then*

$$\int_a^b f(x)\ \mathrm{d}x = G(b) - G(a).$$

Proof. Let $F(x) = \int_a^x f$. Then $F' = G'$, so the mean value theorem implies that $F - G$ is constant. Since $F(a) = 0$, this constant is $-G(a)$. Evaluate at b.

Theorem 5.6 (Linearity)**.** *For continuous f, g and $c \in \mathbb{R}$,*

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g, \qquad \int_a^b cf = c\int_a^b f.$$

Proof. Choose antiderivatives F, G using Part 1. Then $(F+G)' = f+g$ and $(cF)' = cf$. Part 2 gives both identities. For instance,

$$\int_a^b (f+g) = (F+G)(b) - (F+G)(a) = [F(b) - F(a)] + [G(b) - G(a)].$$

5.3.3 Substitution and integration by parts

Theorem 5.7 (One-variable substitution)**.** *Let $u : [a, b] \to \mathbb{R}$ be C^1, and let g be continuous on an interval containing $u([a, b])$. Then*

$$\int_a^b g(u(x))u'(x)\ \mathrm{d}x = \int_{u(a)}^{u(b)} g(t)\ \mathrm{d}t.$$

Proof. Choose G with $G' = g$. The chain rule gives $(G \circ u)' = (g \circ u)u'$, and therefore

$$\int_a^b g(u(x))u'(x)\ \mathrm{d}x = G(u(b)) - G(u(a)) = \int_{u(a)}^{u(b)} g(t)\ \mathrm{d}t.$$

Note that this signed formula does not require u to be one-to-one. Reversed limits already account for the sign of an orientation-reversing substitution.

For C^1 functions f, g, integration by parts gives

$$\int_a^b f(x)g'(x)\ \mathrm{d}x = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x)\ \mathrm{d}x.$$

Exercise 5.2 (Deriving integration by parts)**.** Recover this formula from the product rule.

Solution. The product rule says $(fg)' = f'g + fg'$. Integrating and using the fundamental theorem yields $[fg]_a^b = \int_a^b f'g + \int_a^b fg'$. Subtract the first integral on the right.

Chapter 6

Content, step functions, and multiple integrals

6.1 Content and negligible sets

6.1.1 Boxes in $\mathbb{R}^n$

A closed interval in $\mathbb{R}^n$ means a box

$$I = \prod_{j=1}^{n} [a_j, b_j], \qquad V(I) = \prod_{j=1}^{n} (b_j - a_j).$$

It need not be a one-dimensional interval. The definition of area extends by replacing rectangles with boxes and area with volume [1].

Definition 6.1. A bounded set $A \subset \mathbb{R}^n$ is *contented* with volume $V(A)$ if, given $\varepsilon > 0$, it admits a finite inner approximation by nonoverlapping closed boxes of total volume greater than $V(A) - \varepsilon$, and a finite outer box cover of total volume less than $V(A) + \varepsilon$. It is *negligible* if it is contented and $V(A) = 0$.

In particular, a bounded set is negligible exactly when it has finite box covers of arbitrarily small total volume. Any subset of a negligible set is negligible.

Theorem 6.2 (Boundary criterion)**.** *A bounded set A is contented if and only if its boundary*

$$\partial A = \overline{A} \cap \overline{\mathbb{R}^n \setminus A}$$

is negligible.

Exercise 6.1 (The boundary criterion)**.** Complete the geometric argument behind this criterion.

Solution. Suppose first that ∂A is negligible. Cover it by finitely many open boxes whose total volume is less than ε; this is possible by slightly enlarging a closed-box cover. Since ∂A is compact, a sufficiently fine rectangular grid has every cell meeting ∂A contained in the union of this cover. The sum of the volumes of those cells is at most the volume of the cover, hence less than ε. Every remaining grid cell is wholly on one side of the boundary. Cells contained in A give an inner

approximation, and adjoining the boundary cells gives an outer approximation, with volume gap less than ε.

Conversely, choose finitely many inner boxes $I_i \subset A$ and outer boxes J_j covering A, with total outer volume minus total inner volume less than ε. Their union $O = \bigcup_j J_j$ is closed and contains $\overline{A}$. Put $H = \bigcup_i \operatorname{int} I_i$. Then $H \subset \operatorname{int} A$ and $\partial A \subset O \setminus H$. Refine all the boxes by one common grid. Away from the finitely many grid faces, the boxes of $O \setminus H$ have total volume at most $\sum_j V(J_j) - \sum_i V(I_i) < \varepsilon$. The grid faces are negligible: cover each face by a slab of arbitrarily small thickness. This produces a finite cover of ∂A with total volume arbitrarily close to ε. Since ε was arbitrary, ∂A is negligible.

Corollary 6.3. *If A, B are contented, then $A \cup B$, $A \cap B$, and $A \setminus B$ are contented.*

Proof. Each of these boundaries is contained in $\partial A \cup \partial B$. A finite union of negligible sets is negligible, by combining covers, so the boundary criterion applies.

Note 6.4. Here negligible means that finite box covers have arbitrarily small total volume. This differs from Lebesgue measure zero: the rational points of a box are countable but are not Jordan negligible, since they are dense in the box.

6.2 The geometric definition of an integral

6.2.1 Bounded support and ordinate sets

We say $f : \mathbb{R}^n \to \mathbb{R}$ has bounded support if there exists $R > 0$ such that $\|x\| > R$ implies $f(x) = 0$. For nonnegative f, let

$$O_f = \{(x, t) \in \mathbb{R}^{n+1} : 0 < t \leq f(x)\}.$$

Suppose f is bounded and has bounded support. We say f is Riemann integrable if both O_{f^+} and O_{f^-} are contented, and then define

$$\int f = V(O_{f^+}) - V(O_{f^-}).$$

For a set A, its characteristic function is

$$\chi_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

We define $\int_A f = \int \chi_A f$ whenever the latter exists. A function initially defined only on A is understood to be extended by zero when this notation is used.

The integral has four basic properties: integrable functions form a vector space; integration is linear; $f \geq 0$ implies $\int f \geq 0$; and $\int \chi_A = V(A)$ for contented A.

6.2.2 Admissible functions

Definition 6.5. A function $f : \mathbb{R}^n \to \mathbb{R}$ is *admissible* if it is bounded, has bounded support, and is continuous except on a negligible set.

Theorem 6.6. *Every admissible function is integrable.*

To prove this, bound $|f|$ by M; cover the discontinuities by boxes of small total volume; use uniform continuity on what remains; then use boxes of small diameter.

Exercise 6.2 (Completing the admissibility argument)**.** Turn this outline into an upper and lower approximation argument.

Solution. Take a box Q containing the support in its interior, and let E be a negligible set containing all discontinuities. Its closure is negligible, since a finite closed-box cover of E also covers $\overline{E}$. Choose a finite union G of open boxes covering $\overline{E} \cap Q$, with total volume less than $\varepsilon/(4M)$ when $M > 0$. The function is continuous on the compact set $K = Q \setminus G$, hence uniformly continuous there.

Take a sufficiently fine grid of Q. The total volume of grid cells that meet G can be made less than $\varepsilon/(2M)$: only a thin neighborhood of the finitely many box faces is added to G. On each remaining cell, all points lie in K, and uniform continuity makes the oscillation of f less than $\varepsilon/(2V(Q))$. The upper minus lower sums are consequently bounded by

$$2M\frac{\varepsilon}{2M} + \frac{\varepsilon}{2V(Q)}V(Q) = \frac{3}{2}\varepsilon.$$

Starting with $2\varepsilon/3$ in place of ε gives the desired arbitrarily small gap. The case $M = 0$ is immediate. These upper and lower step approximations imply integrability, as formulated below.

Proposition 6.7. *If A is contented and $f : A \to \mathbb{R}$ is bounded and continuous, its zero extension is admissible. In particular $\int_A f$ exists. If f is admissible, then $f\chi_A$ is admissible.*

Proof. For the first assertion the zero extension is continuous on $\operatorname{int} A$ and on $\mathbb{R}^n \setminus \overline{A}$; possible discontinuities lie in ∂A. For the second, the discontinuities of the product lie in the union of those of f and ∂A. This union is negligible.

6.2.3 Graph volume and elementary estimates

A bounded continuous function on a contented $A \subset \mathbb{R}^n$ has negligible graph

$$G_f = \{(x, f(x)) : x \in A\} \subset \mathbb{R}^{n+1}.$$

Here graph volume is $(n+1)$-dimensional volume.

Exercise 6.3 (Justifying the graph assertion)**.** Explain why the graph has zero volume even when A is not closed.

Solution. Extend f by zero to a bounded admissible function h on a box Q. For any $\eta > 0$, choose a partition for which $\sum_I \operatorname{osc}_I(h)V(I) < \eta$. Over each cell I, the graph is covered by the slab between $\inf_I h$ and $\sup_I h$, enlarged vertically by δ at both ends. The total covering volume is at most $\eta + 2\delta V(Q)$. Cell faces can be covered by additional slabs of arbitrarily small total volume, or included in the closed covering boxes. Let $\delta \to 0$ and then $\eta \to 0$. This covers the graph of $h|_Q$, hence G_f, by arbitrarily small finite box covers.

For integrable functions on contented sets, the following properties are used repeatedly:

$$f \le g \implies \int_A f \le \int_A g,$$
$$|f| \le M \implies |\int_A f| \le MV(A),$$
$$\int_{A\cup B} f = \int_A f + \int_B f - \int_{A\cap B} f.$$

If $A \cap B$ is negligible, its integral term is zero. If bounded integrable f, g agree except on a negligible set, then $\int_A f = \int_A g$. To see the last statement, cover the exceptional set by boxes of total volume less than ε; if $|f - g| \le C$, their integral difference has absolute value at most $C\varepsilon$, and then let $\varepsilon \to 0$.

6.3 Step functions and Riemann sums

6.3.1 Step approximation

A step function is a finite linear combination of characteristic functions of bounded boxes:

$$h = \sum_{i=1}^{p} a_i \chi_{I_i}, \qquad a_i \in \mathbb{R}.$$

The boxes may be refined into a family with disjoint interiors. Then

$$\int h = \sum_{i=1}^{p} a_i V(I_i).$$

Finite sums of step functions and scalar multiples are step functions, with

$$\int (h + k) = \int h + \int k, \qquad \int ch = c \int h.$$

Note 6.8. Closed partition boxes can overlap on their faces, where their characteristic functions may sum to more than one. Faces have zero volume. For pointwise upper and lower bounds, use disjoint half-open cells and assign the faces separately; their indicators are step functions with integral zero.

Theorem 6.9 (Step-function criterion)**.** *Let f be bounded with bounded support. Then f is integrable if and only if for every $\varepsilon > 0$ there exist step functions h, k with*

$$h \le f \le k, \qquad \int (k - h) < \varepsilon.$$

In that case $\int h \le \int f \le \int k$.

The geometric content definition and this criterion describe the same integral: rectangles approximating the ordinate sets give upper and lower step approximations, and conversely the ordinate volumes trapped between such steps have arbitrarily small gap.

To prove closure under addition, suppose

$$h_1 \le f \le k_1, \quad h_2 \le g \le k_2, \qquad \int (k_i - h_i) < \varepsilon/2.$$

Then

$$h_1 + h_2 \le f + g \le k_1 + k_2, \qquad \int \big((k_1 + k_2) - (h_1 + h_2)\big) < \varepsilon.$$

Thus $f + g$ is integrable. Its integral is $\int f + \int g$, since both this number and $\int (f + g)$ are trapped between the same sums of lower and upper integrals, with arbitrarily small gap.

6.3.2 Tagged partitions

A partition $P = \{Q_1, \dots, Q_k\}$ of a box Q consists of closed subboxes with disjoint interiors and union Q. A selection of tags is $\xi = \{\xi_1, \dots, \xi_k\}$ with $\xi_j \in Q_j$. Define

$$R(f, P, \xi) = \sum_{j=1}^{k} f(\xi_j) V(Q_j), \qquad \operatorname{mesh}(P) = \max_j \operatorname{diam} Q_j.$$

Theorem 6.10 (Riemann-sum criterion)**.** *Suppose f is bounded and zero outside Q. It is integrable with integral I if and only if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$|I - R(f, P, \xi)| < \varepsilon$$

for every tagged partition with $\operatorname{mesh}(P) < \delta$.

Exercise 6.4 (Why the tags do not matter in the limit)**.** Prove this criterion using step approximations.

Solution. For a fixed step function, all cells not meeting one of its finitely many box faces have the exact constant value required by its integral. Cells meeting these faces lie in thin slabs, whose total volume tends to zero with the mesh. Boundedness therefore implies that its Riemann sums converge to its integral, independently of tags. For integrable f, choose $h \le f \le k$ with $\int (k - h) < \varepsilon$. The inequalities $R(h, P, \xi) \le R(f, P, \xi) \le R(k, P, \xi)$ and the convergence just proved squeeze the sums of f to $\int f$.

Conversely, suppose every sufficiently fine tagged sum is within ε of I. Fix such a partition. In every cell choose tags with values arbitrarily close to its infimum, and separately tags with values arbitrarily close to its supremum. Finiteness of the partition permits the total approximation error to be arbitrarily small. Its lower and upper sums are therefore within ε of I, up to an arbitrarily small error. Their difference is at most 2ε. Since ε is arbitrary, the step-function criterion proves integrability and identifies the integral as I.

6.4 Limits and integrals

6.4.1 Pointwise convergence is insufficient

Suppose f_n is integrable for each n and $f_n(x) \to f(x)$ at every point. It need not follow that $\int f_n \to \int f$. Consider

$$f_n(x) = n \chi_{(0,1/n)}(x).$$

For each x, eventually $f_n(x) = 0$, including $x = 0$, so $f_n \to 0$ pointwise. But

$$\int_0^1 f_n(x) \, \mathrm{d}x = n \frac{1}{n} = 1$$

for every n. The integral of the limiting function is 0.

Exercise 6.5 (The uniform-convergence repair)**.** Prove that uniform convergence on a fixed finite box permits passage to the limit under the integral.

Solution. Suppose f_n is integrable on Q and $\|f_n - f\|_\infty \to 0$. Given $\varepsilon > 0$, choose n with $\|f_n - f\|_\infty < \varepsilon/(4V(Q))$. Choose lower and upper steps for f_n with gap integral less than $\varepsilon/2$. Subtracting this uniform error from the lower step on Q and adding it to the upper step gives steps bounding f with gap less than ε. Thus f is integrable. The bound

$$|\int_Q f_n - \int_Q f| \leq V(Q)\|f_n - f\|_\infty \longrightarrow 0$$

then proves the limit identity. A zero-volume box is immediate.

Chapter 7

Iterated integrals and parameter integrals

7.1 Fubini's theorem and domains between graphs

7.1.1 The section hypothesis

Theorem 7.1 (Fubini, in the bounded Riemann setting)**.** *Let $A \subset \mathbb{R}^m$ and $B \subset \mathbb{R}^n$ be contented. Suppose f is bounded and Riemann integrable on $A \times B$, and for every $x \in A$ the section $y \mapsto f(x, y)$ is Riemann integrable on B. Then*

$$F(x) = \int_B f(x, y)\ \mathrm{d}y$$

is integrable on A, and

$$\int_{A\times B} f(x, y)\ \mathrm{d}(x, y) = \int_A \left(\int_B f(x, y)\ \mathrm{d}y \right) \mathrm{d}x.$$

If the sections in the other direction are also integrable, x and y may be interchanged.

Note 7.2. A Riemann-integrable function of several variables need not have integrable sections at every parameter. Interchanging the two orders of integration requires the section hypotheses in both directions.

Exercise 7.1 (An exceptional section)**.** Give an integrable function on $[0, 1]^2$ whose section at $x = 0$ is not Riemann integrable.

Solution. Set $f(0, y) = 1$ for rational $y \in [0, 1]$, and set $f(x, y) = 0$ otherwise. The function is bounded and vanishes outside the negligible vertical segment $\{0\} \times [0, 1]$. Upper step approximations supported in a strip of width δ and lower approximation 0 show that its double integral is 0. At $x = 0$, however, the section is the rational-indicator function: on every nondegenerate subinterval its infimum is 0 and supremum is 1. Its lower and upper integrals are 0 and 1, so that section is not Riemann integrable.

Exercise 7.2 (Fubini by upper and lower steps)**.** Explain why integrable sections give the displayed equality.

Solution. Extend f by zero to a product of containing boxes. Choose step functions $h \le f \le k$ with arbitrarily small integral gap, refined into product boxes $I_i \times J_i$. For a product indicator, integration in y gives $V(J_i)\chi_{I_i}(x)$. Thus the functions

$$H(x) = \int h(x,y)\ \mathrm{d}y, \qquad K(x) = \int k(x,y)\ \mathrm{d}y$$

are steps and satisfy $H \le F \le K$. Direct computation on product boxes gives

$$\int (K - H) = \int (k - h), \quad \int H = \int h, \quad \int K = \int k.$$

The step criterion proves integrability of F. Both $\int F$ and $\int f$ lie between $\int h$ and $\int k$, whose gap is arbitrarily small; hence they agree. Face values can be handled by the zero-volume face convention already described.

7.1.2 Regions with variable limits

Suppose $A \subset \mathbb{R}^n$ is contented, $f_1, f_2 : A \to \mathbb{R}$ are bounded continuous functions, and $f_1 \le f_2$. Set

$$C = \{(x,y) : x \in A,\ f_1(x) \le y \le f_2(x)\}.$$

Then C is contented. If $g : C \to \mathbb{R}$ is bounded and continuous, then

$$\int_C g = \int_A \left(\int_{f_1(x)}^{f_2(x)} g(x,y)\ \mathrm{d}y \right) \mathrm{d}x.$$

Note 7.3. Continuity on a bounded set that is not closed does not imply boundedness. On compact domains, continuous functions are automatically bounded.

Exercise 7.3 (Why such regions are contented)**.** Verify the domain assertion and explain the integration formula.

Solution. Take M bounding $|f_1|$ and $|f_2|$. A boundary point of C lies either over ∂A, or on one of the two graphs. Indeed, over an interior point of A, a point strictly between the two continuous graphs is interior to C, while a point strictly outside them is exterior. The set $\partial A \times [-M, M]$ is negligible: multiply a small-volume finite cover of ∂A by the bounded interval. The two graphs are negligible by the preceding graph result. Thus ∂C is negligible and C is contented. Extend g by zero to a containing product box. This extension is integrable because it is bounded and continuous except possibly on ∂C. Every vertical section is continuous between its two endpoints and zero outside, with only endpoint jumps. Fubini therefore gives the asserted formula.

The unit ball $B \subset \mathbb{R}^3$ can be written as

$$B = \{(x,y,z) : x^2 + y^2 \le 1,\ -\sqrt{1 - x^2 - y^2} \le z \le \sqrt{1 - x^2 - y^2}\}.$$

Consequently a bounded continuous g can first be integrated in z, and then over the disk in the (x,y) plane. This is the same variable-limit construction with A equal to that disk.

7.2 Changing the order of integration

7.2.1 A triangle, two ways

Let T have vertices $(0,0),(1,0),(1,1)$. We want

$$I = \int_T xy^2 \, \mathrm{d}(x,y).$$

The triangle is described by $0 \le x \le 1$, $0 \le y \le x$, and sets up integration in y first.

Exercise 7.4 (Complete the two triangle calculations)**.** Evaluate I in the indicated order and then in the reverse order.

Solution. Integrating in y first,

$$I = \int_0^1 \int_0^x xy^2 \, \mathrm{d}y \, \mathrm{d}x = \int_0^1 \frac{x^4}{3} \, \mathrm{d}x = \frac{1}{15}.$$

For the other order, $0 \le y \le 1$ and $y \le x \le 1$:

$$I = \int_0^1 \int_y^1 xy^2 \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{2} \int_0^1 (y^2 - y^4) \, \mathrm{d}y = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{1}{15}.$$

7.2.2 The exponential boundary

Draw a picture before changing the order. Consider

$$I = \int_0^{\log 2} \int_{e^x - 2}^0 \frac{\cos\big((\log(y+2))^2\big)}{y+2} \, \mathrm{d}y \, \mathrm{d}x.$$

The region is bounded by $x = 0$, $y = 0$, and $y = e^x - 2$. Solving for x gives $x = \log(y+2)$, so the same region is

$$-1 \le y \le 0, \qquad 0 \le x \le \log(y+2).$$

Hence

$$\begin{aligned} I &= \int_{-1}^0 \int_0^{\log(y+2)} \frac{\cos\big((\log(y+2))^2\big)}{y+2} \, \mathrm{d}x \, \mathrm{d}y \\ &= \int_{-1}^0 \frac{\log(y+2)}{y+2} \cos\big((\log(y+2))^2\big) \, \mathrm{d}y. \end{aligned}$$

Set $u = (\log(y+2))^2$. Then $\mathrm{d}u = 2\log(y+2) \, \mathrm{d}y/(y+2)$ and the bounds are 0 and $(\log 2)^2$. Therefore

$$I = \frac{1}{2} \int_0^{(\log 2)^2} \cos u \, \mathrm{d}u = \frac{1}{2} \sin\big((\log 2)^2\big).$$

7.2.3 A parabolic boundary

Consider the integral

$$I = \int_0^1 \int_{x^2}^1 x\sqrt{1-y^2}\ \mathrm{d}y\ \mathrm{d}x.$$

Its region is $0 \le x \le 1$, $x^2 \le y \le 1$, equivalently $0 \le y \le 1$, $0 \le x \le \sqrt{y}$. Reversing the order gives

$$\begin{aligned} I &= \int_0^1 \int_0^{\sqrt{y}} x\sqrt{1-y^2}\ \mathrm{d}x\ \mathrm{d}y = \frac{1}{2}\int_0^1 y\sqrt{1-y^2}\ \mathrm{d}y \\ &= \frac{1}{4}\int_0^1 u^{1/2}\ \mathrm{d}u = \frac{1}{6}, \qquad u = 1-y^2. \end{aligned}$$

7.2.4 Mass in an octant of a ball

Suppose a body has density $\rho(x,y,z) = xyz$ and occupies

$$B = \{(x,y,z) : x,y,z \ge 0,\ x^2+y^2+z^2 \le 1\}.$$

This is one eighth of the unit ball. Its mass is

$$\begin{aligned} \int_B \rho &= \int_0^1 \int_0^{\sqrt{1-z^2}} \int_0^{\sqrt{1-y^2-z^2}} xyz\ \mathrm{d}x\ \mathrm{d}y\ \mathrm{d}z \\ &= \frac{1}{2}\int_0^1 z \int_0^{\sqrt{1-z^2}} y(1-y^2-z^2)\ \mathrm{d}y\ \mathrm{d}z \\ &= \frac{1}{8}\int_0^1 z(1-z^2)^2\ \mathrm{d}z = \frac{1}{16}\int_0^1 u^2\ \mathrm{d}u = \frac{1}{48}. \end{aligned}$$

Here $u = 1-z^2$ in the last step. The raw cube bounds $0 \le x,y,z \le 1$ would include points outside the ball; the curved upper limits are essential.

7.3 Differentiating an integral with respect to a parameter

7.3.1 A sufficient hypothesis

Uniform convergence gives a useful criterion for differentiating an integral with respect to a parameter.

Exercise 7.5 (Differentiation under a finite integral)**.** Suppose $H(t,x)$ and $\partial H/\partial t$ are continuous on $J \times [a,b]$, where J is an open interval. Prove

$$G(t) = \int_a^b H(t,x)\ \mathrm{d}x \quad \Longrightarrow \quad G'(t) = \int_a^b \frac{\partial H}{\partial t}(t,x)\ \mathrm{d}x.$$

Solution. Fix $t \in J$ and a closed parameter interval about t contained in J. The derivative is uniformly continuous on the product of this interval with $[a,b]$. By the fundamental theorem in the parameter variable,

$$\frac{H(t+h,x)-H(t,x)}{h} = \int_0^1 \frac{\partial H}{\partial t}(t+sh,x)\ \mathrm{d}s.$$

The difference between this expression and $\partial H/\partial t(t,x)$ tends uniformly to zero for $x \in [a,b]$. Integrating this difference gives an error bounded by $(b-a)$ times its uniform norm. The error tends to zero, proving the formula.

7.3.2 A cosine integral

Suppose

$$g(t) = \int_1^2 \frac{\cos(\pi t x^2)}{x}\ \mathrm{d}x.$$

Differentiating gives

$$g'(t) = -\pi \int_1^2 x \sin(\pi t x^2)\ \mathrm{d}x.$$

Use the substitution $u = \pi t x^2$ to evaluate the derivative.

Exercise 7.6 (finish the parameter derivative)**.** Complete the calculation for $t \neq 0$, and determine the derivative at $t = 0$ without dividing by t.

Solution. For $t \neq 0$, put $u = \pi t x^2$, $\mathrm{d}u = 2\pi t x\ \mathrm{d}x$. Thus

$$g'(t) = -\frac{1}{2t} \int_{\pi t}^{4\pi t} \sin u\ \mathrm{d}u = \frac{\cos(4\pi t) - \cos(\pi t)}{2t}.$$

At $t = 0$, use the derivative integral: $g'(0) = -\pi \int_1^2 x \sin 0\ \mathrm{d}x = 0$. This is also the limit of the quotient formula, since $\cos(4\pi t) - \cos(\pi t) = O(t^2)$. The apparent singularity is removable. Also $g(0) = \int_1^2 \mathrm{d}x/x = \log 2$.

7.3.3 A logarithmic parameter integral

Suppose $\alpha \neq \pm 1$. We want to evaluate

$$J(\alpha) = \int_0^\pi \log(1 - 2\alpha \cos x + \alpha^2)\ \mathrm{d}x.$$

The logarithm's argument is positive on $[0, \pi]$ for such α. On a closed parameter interval avoiding ± 1 it has a positive lower bound, so differentiation is justified. For $\alpha \neq 0$,

$$\begin{aligned} J'(\alpha) &= \int_0^\pi \frac{2\alpha - 2\cos x}{1 - 2\alpha\cos x + \alpha^2}\ \mathrm{d}x \\ &= \frac{1}{\alpha}\left[\pi + (\alpha^2 - 1)\int_0^\pi \frac{\mathrm{d}x}{1 - 2\alpha\cos x + \alpha^2}\right]. \end{aligned}$$

Use the tangent half-angle substitution $u = \tan(x/2)$:

$$\cos x = \frac{1 - u^2}{1 + u^2}, \qquad \mathrm{d}x = \frac{2\ \mathrm{d}u}{1 + u^2}.$$

First apply the substitution on $[0, b]$ with $b < \pi$, then let $b \uparrow \pi$. The original integrand is continuous on $[0, \pi]$ for $\alpha \neq \pm 1$, and the transformed integrand has an $O(u^{-2})$ tail, so this endpoint limit is justified. The remaining integral becomes

$$\begin{aligned} \int_0^\pi \frac{\mathrm{d}x}{1 - 2\alpha\cos x + \alpha^2} &= 2\int_0^\infty \frac{\mathrm{d}u}{(1-\alpha)^2 + (1+\alpha)^2 u^2} \\ &= \frac{2}{|1-\alpha||1+\alpha|}\left[\arctan\left(\frac{|1+\alpha|}{|1-\alpha|}u\right)\right]_0^\infty \\ &= \frac{\pi}{|1 - \alpha^2|}. \end{aligned}$$

Consequently $J'(\alpha) = 0$ for $0 < |\alpha| < 1$, and $J'(\alpha) = 2\pi/\alpha$ for $|\alpha| > 1$. Continuity at 0 and $J(0) = 0$ imply $J = 0$ throughout $(-1, 1)$.

For the exterior range, put $\beta = 1/\alpha \in (-1, 1)$:

$$\begin{aligned} J(\alpha) &= \int_0^{\pi} \log\bigl(\alpha^2(1 - 2\beta\cos x + \beta^2)\bigr)\, \mathrm{d}x \\ &= J(\beta) + \pi \log(\alpha^2) = 2\pi \log|\alpha|. \end{aligned}$$

Therefore

$$J(\alpha) = \begin{cases} 0, & |\alpha| < 1, \\ 2\pi \log|\alpha|, & |\alpha| > 1. \end{cases}$$

Note 7.4. The absolute values in the arctangent calculation are needed for negative parameters and for $|\alpha| > 1$. At $\alpha = \pm 1$ the logarithmic integral is improper at an endpoint; the differentiation argument does not apply across those parameter values.

Chapter 8

Change of variables and Jacobians

8.1 Why the determinant appears

8.1.1 The motivating polar example

Recall the one-dimensional substitution rule

$$\int_a^b f(t)\ \mathrm{d}t = \int_{u^{-1}(a)}^{u^{-1}(b)} f(u(s))u'(s)\ \mathrm{d}s$$

when the inverse and the indicated endpoints are appropriate. In several dimensions, replacing a region by its parameter domain without a volume factor does not work.

For example, let Q be the unit disk and let

$$T(r,\theta) = (r\cos\theta, r\sin\theta), \qquad A = [0,1]\times[0,2\pi].$$

Then $T(A) = Q$, but

$$\int_Q 1\ \mathrm{d}(x,y) = \pi, \qquad \int_A 1\ \mathrm{d}(r,\theta) = 2\pi.$$

The second integral is the area of the parameter rectangle. We must account for the way T changes small areas [1].

For a differentiable map $T:\mathbb{R}^n\to\mathbb{R}^n$, recall that $\mathrm{d}T_a$ is the linear map best approximating T near a:

$$T(a+h) = T(a) + T'(a)h + o(\|h\|).$$

Its matrix is the derivative matrix $T'(a)$, with rows given by the derivatives of the component functions. We call $\det T'(a)$ the Jacobian determinant.

8.1.2 Linear maps and volume

Theorem 8.1 (Linear volume scaling)**.** *If $L:\mathbb{R}^n\to\mathbb{R}^n$ is linear and B is contented, then $L(B)$ is contented and*

$$V(L(B)) = |\det L|\, V(B).$$

For a diagonal matrix $\mathrm{diag}(a_1,\dots,a_n)$, this is the familiar multiplication of side lengths by $|a_j|$, giving the factor $|a_1\cdots a_n|$. A negative determinant reverses orientation; ordinary volume is nonnegative, hence the absolute value.

Exercise 8.1 (From boxes to the linear volume theorem)**.** Complete the theorem using elementary linear maps.

Solution. For an invertible diagonal map, the image of a box is a box with volume multiplied by the product of the absolute diagonal entries. Coordinate permutations preserve box volume. An elementary shear, such as

$$S(x_1, x_2, \dots, x_n) = (x_1 + cx_2, x_2, \dots, x_n),$$

maps a box to a parallelepiped. For each fixed $(x_2, \dots, x_n)$, its first-coordinate section is an interval of exactly the original first-coordinate length. Its boundary is a finite union of pieces of affine graphs, hence negligible, and Fubini computes its volume as the original box volume.

For each elementary invertible map, approximate a contented B from inside and outside by finite box families. Their images are contented parallelepipeds. Images of disjoint interiors remain disjoint, and injectivity preserves inclusions. The inner and outer total volumes are multiplied by the elementary factor. Their gap can be made arbitrarily small, proving both contentedness and the formula for B. Gaussian elimination expresses any invertible matrix as a product of these elementary maps. Applying the preceding result successively multiplies their factors, whose product is $|\det L|$.

If L is singular, $L(B)$ is bounded and lies in a proper linear subspace. That subspace lies in a hyperplane, which can be written as the graph of a linear function after choosing a coordinate with nonzero normal component. Its bounded part is negligible by the graph result. Thus $V(L(B)) = 0 = |\det L| V(B)$.

8.2 The local volume estimate

8.2.1 Normalizing the derivative

We reduce a nonlinear map to a map close to the identity. Let Q be a cube centered at a, let T be a C^1 diffeomorphism on an open neighborhood of Q, and put

$$b = T(a), \qquad L = T'(a), \qquad F(x) = L^{-1}\big(T(a + x) - b\big).$$

If $C_r = [-r, r]^n$ and $Q = a + C_r$, then $F(0) = 0$, and the chain rule gives

$$F'(x) = T'(a)^{-1} T'(a + x).$$

This normalizes both the base point and the derivative. With $\tau_c(x) = x + c$, it is $F = L^{-1} \circ \tau_b^{-1} \circ T \circ \tau_a$.

Proposition 8.2 (Local volume distortion)**.** *Use the vector norm $\|x\|_\infty = \max_j |x_j|$ and its induced matrix norm. Suppose, for some $0 < \varepsilon < 1$,*

$$\|T'(a)^{-1} T'(x) - I\|_\infty \le \varepsilon \qquad (x \in Q).$$

Then

$$(1 - \varepsilon)^n |\det T'(a)| V(Q) \le V(T(Q)) \le (1 + \varepsilon)^n |\det T'(a)| V(Q).$$

Note 8.3. The estimate $(1 \pm \varepsilon)^n$ is stated for cubes using the induced maximum norm. For a rectangular box, use the norm scaled to its side lengths, or partition into small cubes.

Exercise 8.2 (Finishing the local volume estimate)**.** Prove the near-identity inclusions and the volume estimate.

Solution. Write $F(x) = x + H(x)$ on C_r. Integration along the segment joining two points in this convex cube gives

$$\|H(x) - H(y)\|_\infty \le \varepsilon \|x - y\|_\infty.$$

Since $H(0) = 0$, $\|H(x)\|_\infty \le \varepsilon r$. Thus $F(C_r) \subset C_{(1+\varepsilon)r}$.

For $y \in C_{(1-\varepsilon)r}$, the map $\Phi_y(x) = y - H(x)$ takes C_r into itself and has Lipschitz constant at most $\varepsilon < 1$. Starting at any $x_0 \in C_r$, define $x_{k+1} = \Phi_y(x_k)$. The successive differences are at most a geometric sequence with ratio ε, so the sequence is Cauchy. Its limit $x \in C_r$ satisfies $x = \Phi_y(x)$ by continuity, hence $F(x) = y$. Therefore

$$C_{(1-\varepsilon)r} \subset F(C_r) \subset C_{(1+\varepsilon)r}.$$

The set $F(C_r)$ is contented. Indeed F is a diffeomorphism on a neighborhood, so its boundary is $F(\partial C_r)$. On C_r the map is Lipschitz. Each face of ∂C_r can be covered by $O(\delta^{-(n-1)})$ small face boxes of diameter $O(\delta)$, whose images fit in cubes of volume $O(\delta^n)$. The total is $O(\delta) \to 0$. Thus $F(\partial C_r)$ is negligible. In dimension one, the boundary is just two points.

Taking volumes in the two inclusions gives $(1-\varepsilon)^n V(C_r) \le V(F(C_r)) \le (1+\varepsilon)^n V(C_r)$. Finally $T(Q) = b + L(F(C_r))$. Translation preserves volume, and linear scaling multiplies it by $|\det L|$, proving the proposition.

8.3 The change-of-variables formula

8.3.1 Statement and the partition calculation

Theorem 8.4 (Change of variables)**.** *Let $U, V \subset \mathbb{R}^n$ be open and $T : U \to V$ a C^1 diffeomorphism. Let $A \subset U$ be compact and contented. If f is bounded and integrable on $T(A)$ and $(f \circ T)|\det T'|$ is integrable on A, then*

$$\int_{T(A)} f(x)\, \mathrm{d}x = \int_A f(T(u))|\det T'(u)|\, \mathrm{d}u.$$

In particular, the hypotheses on the integrands hold for continuous f on $T(A)$.

Note 8.5. Polar, spherical, and squaring substitutions can have singularities or identifications on boundary sets. Apply the diffeomorphism theorem on compact subsets of their regular interiors, then pass to the limit while controlling the omitted pieces.

First suppose f is continuous. Divide the parameter region into small cells A_i and choose tags u_i. Since T is injective, their images do not overlap in their interiors. For a fine partition,

$$\begin{aligned}\int_{T(A)} f &= \sum_i \int_{T(A_i)} f \;\approx\; \sum_i f(T(u_i))V(T(A_i)) \\ &\approx \sum_i f(T(u_i))V\big(T'(u_i)(A_i - u_i)\big) \\ &= \sum_i f(T(u_i))|\det T'(u_i)|V(A_i).\end{aligned}$$

The last line is a Riemann sum for $(f \circ T)|\det T'|$. The local distortion estimate justifies the volume approximation, and uniform continuity controls the approximation of f on each image cell.

Exercise 8.3 (Making the partition argument quantitative)**.** Prove the formula for continuous f on the image of a compact box $A \subset U$.

Solution. Use a uniform grid of cubes in A. Rectangular edge cells can be avoided by first taking coordinate grids with comparable side lengths and using the corresponding scaled maximum norm; equivalently, approximate the box by inner cubes, with leftover volume tending to zero. On a fixed compact neighborhood of A, T', T'^{-1}, and $\det T'$ are bounded and uniformly continuous where appropriate. Hence, for each $\varepsilon > 0$, every sufficiently small cube Q_i with center u_i satisfies the local estimate with this same ε.

Put $J_i = |\det T'(u_i)|$. There is a constant $c_n(\varepsilon) \to 0$ such that

$$|V(T(Q_i)) - J_i V(Q_i)| \le c_n(\varepsilon) J_i V(Q_i).$$

If $M = \max|f|$ and $J_{\max} = \max_A |\det T'|$, the total error from replacing image volumes by linearized volumes is at most $M c_n(\varepsilon) J_{\max} V(A)$. Also T is Lipschitz on each small cube, with one uniform constant. Thus the diameters of their images tend uniformly to zero. Uniform continuity of f implies that the error in replacing $\int_{T(Q_i)} f$ by $f(T(u_i))V(T(Q_i))$, summed over i, is at most

$$\omega(\max_i \operatorname{diam} T(Q_i)) V(T(A)),$$

where $\omega(s) \to 0$ is a modulus of continuity. Shared image faces have zero volume by the face argument in the preceding solution. The resulting sum is a Riemann sum for the continuous function $(f \circ T)|\det T'|$. Let the mesh tend to zero and then $\varepsilon \to 0$. The two error bounds prove the equality. The omitted edge cells in an inner-cube construction have image volume bounded by a constant times their volume, by the local estimate, so their contributions vanish as well.

Exercise 8.4 (Extension to contented sets and Riemann integrands)**.** Extend the preceding continuous-on-a-box proof to compact contented A, and then prove the change-of-variables formula for bounded Riemann-integrable f on $T(A)$.

Solution. First note that a C^1 map on a neighborhood of a compact set is Lipschitz on each of finitely many small convex neighborhoods covering that set, with a common finite bound. Such maps take negligible sets to negligible sets. Indeed, finite covers by sufficiently small cubes of arbitrarily small total volume have images covered by cubes with total volume at most a fixed constant times that volume. One may obtain these cube covers from box covers by subdivision and arbitrarily small enlargements. The same observation applies to T^{-1} near the compact set $T(A)$.

Let $E \subset U$ be compact and contented. Then $\partial T(E) = T(\partial E)$ is negligible, so $T(E)$ is contented. Cover E by a fine cubical grid and retain the cubes lying wholly in E. Their finite union H satisfies $V(E \setminus H) \to 0$, because the remaining cells meet the negligible boundary. All these cells lie in a fixed compact neighborhood in U when the mesh is small. Their images have total volume bounded by a fixed constant times their original total volume, by the Lipschitz cube-cover estimate. Thus $V(T(E) \setminus T(H)) \to 0$ as well. Apply the continuous box result to $f = 1$ on each retained cube and add. With $J = |\det T'|$, boundedness of J and the two vanishing remainders give

$$V(T(E)) = \int_E J.$$

The same cube argument, using boundedness of a continuous integrand on $T(E)$, extends the continuous formula from boxes to E.

Now extend f by zero off $T(A)$. Choose target step functions $h \le f\chi_{T(A)} \le k$ with $\int(k-h) < \varepsilon$. For each closed box B appearing in either step function, set

$$E_B = A \cap T^{-1}(B).$$

This set is compact. Its boundary is contained in $\partial A \cup \big(A \cap T^{-1}(\partial B)\big)$, which is negligible: the box faces are negligible, and T^{-1} is locally Lipschitz near the relevant compact target set. Hence E_B is contented. The volume identity just proved gives

$$\int_A \chi_B(T(u))J(u)\,\mathrm{d}u = \int_{E_B} J = V(T(E_B)) = V\big(T(A)\cap B\big).$$

By linearity, the change-of-variables identity holds for h and for k restricted to $T(A)$. Moreover $J > 0$ on A, so

$$(h\circ T)J \le (f\circ T)J \le (k\circ T)J,$$

and the difference between the integrals of these integrable bounds is

$$\int_{T(A)}(k-h) \le \int(k-h) < \varepsilon.$$

Integrable lower and upper bounds of arbitrarily small integral gap imply integrability of the middle function: approximate each bound in turn by lower and upper step functions. Finally, $\int_{T(A)} f$ and $\int_A (f\circ T)J$ lie between the same two endpoint integrals with gap less than ε. Letting $\varepsilon \to 0$ proves equality. This also establishes the transformed integrability required in the theorem.

Exercise 8.5 (compact cutoffs for coordinate substitutions)**.** Justify the use of the compact change-of-variables theorem when polar, spherical, or squaring coordinates fail to be a diffeomorphism at their boundary. Assume the integrands in the resulting bounded-domain formulas are continuous and bounded.

Solution. For the unit disk, use

$$A_\delta = [\delta, 1] \times [\delta, 2\pi - \delta], \qquad 0 < \delta < 1.$$

Each cutoff is contained in the open polar chart $(0,\infty)\times(0,2\pi)$. The compact theorem applies there. The omitted part of the disk lies in the union of $[-\delta,\delta]^2$ and $[0,1]\times[-\delta,\delta]$, so its outer area is at most

$$4\delta^2 + 2\delta \longrightarrow 0.$$

If $|f| \le M$, its omitted target integral is bounded by $M(4\delta^2+2\delta)$. Integrating the polar weight r over the omitted parameter strips gives $\pi\delta^2 + \delta(1-\delta^2)$, which also tends to zero and bounds the omitted parameter integral after multiplication by M. Letting $\delta \downarrow 0$ proves the full disk formula.

For spherical coordinates on a bounded radial range, remove parameter strips near $\rho = 0$, $\phi = 0, \pi$, and the azimuthal seam. For the squaring map $(u,v) \mapsto (u^2, v^2)$ on a bounded first-quadrant region, remove strips near $u = 0$ and $v = 0$. The remaining compact contented cutoffs lie in regular one-to-one charts. All removed parameter strips have volume tending to zero. The maps are C^1 on their full compact parameter boxes and therefore Lipschitz there. The cube-cover estimate in the preceding solution bounds the outer volume of each omitted image by a fixed constant times the omitted parameter volume. The continuous Jacobians and the integrands are bounded as well. Thus both omitted integrals tend to zero. These estimates justify passage from the cutoff formulas to the full bounded-domain formulas. Singular or unbounded integrands require separate convergence estimates.

8.3.2 Polar coordinates and the disk calculation

For $T(r,\theta) = (r\cos\theta, r\sin\theta)$,

$$T'(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det T' = r.$$

For $r > 0$ and an angular interval of length less than 2π, this is an ordinary change of variables. For the full disk, the origin and a radial seam are negligible; a continuous bounded integrand is unaffected by them. Thus

$$\int_{x^2+y^2\le 1} f(x,y)\ \mathrm{d}x\ \mathrm{d}y = \int_0^{2\pi}\int_0^1 f(r\cos\theta, r\sin\theta) r\ \mathrm{d}r\ \mathrm{d}\theta.$$

In particular, the corrected area calculation is $\int_0^{2\pi}\int_0^1 r\ \mathrm{d}r\ \mathrm{d}\theta = \pi$.

Next consider $\int_Q \sin(x^2+y^2)$ over the same disk.

Exercise 8.6 (Finish the polar sine integral)**.** Complete this calculation.

Solution. Since $x^2+y^2 = r^2$,

$$\int_Q \sin(x^2+y^2)\ \mathrm{d}(x,y) = 2\pi\int_0^1 r\sin(r^2)\ \mathrm{d}r = \pi\int_0^1 \sin u\ \mathrm{d}u = \pi(1-\cos 1),$$

where $u = r^2$.

8.3.3 Spherical coordinates

Use

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

For fixed ρ and ϕ, the horizontal circle has radius $\rho\sin\phi$, so its (x,y) coordinates have the same form as polar coordinates. Here $\rho \ge 0$, $0 \le \phi \le \pi$, and θ is the azimuthal angle.

Note 8.6. The margin asks about $T : \mathbb{R}^m \to \mathbb{R}^n$ when $m \ne n$. Then T' is rectangular and has no ordinary determinant. The change-of-variables theorem here concerns equal-dimensional maps; the spherical-coordinate map is a map from three parameters to $\mathbb{R}^3$.

Exercise 8.7 (The spherical volume factor)**.** Compute the absolute Jacobian determinant for the variable order (ρ,ϕ,θ).

Solution. The derivative columns are

$$T_\rho = \begin{pmatrix} \sin\phi\cos\theta \\ \sin\phi\sin\theta \\ \cos\phi \end{pmatrix}, \quad T_\phi = \rho\begin{pmatrix} \cos\phi\cos\theta \\ \cos\phi\sin\theta \\ -\sin\phi \end{pmatrix}, \quad T_\theta = \rho\sin\phi\begin{pmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{pmatrix}.$$

They are orthogonal, with lengths $1, \rho, \rho\sin\phi$. Their ordered unit vectors form a positively oriented orthonormal basis when $0 < \phi < \pi$, so $\det T' = \rho^2\sin\phi$. In any parameter order the absolute determinant is $\rho^2\sin\phi$. The poles, origin, and angular seam are boundary exceptions to the interior coordinate chart, and have zero ordinary volume.

Let

$$R = \{(x, y, z) : x^2 + y^2 + z^2 \le 1,\ x \ge 0\}, \qquad f(x, y, z) = x.$$

Choose $-\pi/2 \le \theta \le \pi/2$, $0 \le \phi \le \pi$, and $0 \le \rho \le 1$. Then

$$\begin{aligned}\int_R x \ \mathrm{d}(x, y, z) &= \int_{-\pi/2}^{\pi/2} \int_0^{\pi} \int_0^1 (\rho \sin\phi \cos\theta)(\rho^2 \sin\phi) \ \mathrm{d}\rho \ \mathrm{d}\phi \ \mathrm{d}\theta \\ &= \left(\int_0^1 \rho^3 \ \mathrm{d}\rho\right)\left(\int_0^{\pi} \sin^2\phi \ \mathrm{d}\phi\right)\left(\int_{-\pi/2}^{\pi/2} \cos\theta \ \mathrm{d}\theta\right) \\ &= \frac{1}{4} \cdot \frac{\pi}{2} \cdot 2 = \frac{\pi}{4}.\end{aligned}$$

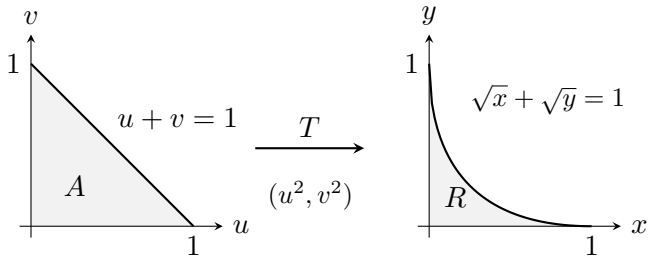
8.3.4 Straightening a square-root boundary

Let R be the first-quadrant region bounded by $x = 0$, $y = 0$, and $\sqrt{x} + \sqrt{y} = 1$. Evaluate

$$\int_R (\sqrt{x} + \sqrt{y})^{1/2} \ \mathrm{d}(x, y)$$

and introduces $u = \sqrt{x}$, $v = \sqrt{y}$, or $T(u, v) = (u^2, v^2)$. The parameter region becomes the triangle

$$A = \{(u, v) : u, v \ge 0,\ u + v \le 1\}.$$



Exercise 8.8 (Finish the square-root substitution)**.** Compute the integral, including the Jacobian and the boundary conditions.

Solution. The derivative is $T'(u, v) = \mathrm{diag}(2u, 2v)$, so the absolute determinant is $4uv$ on A. On the interior, $u, v > 0$, the map has a C^1 inverse; the two axes are negligible boundary sets. The transformed integrand is continuous even at those axes. Hence

$$I = 4 \int_0^1 \int_0^{1-u} uv(u + v)^{1/2} \ \mathrm{d}v \ \mathrm{d}u.$$

For a shorter evaluation put $s = u + v$ and retain u as the other variable. The new triangle is $0 \le s \le 1$, $0 \le u \le s$, and $v = s - u$. This linear substitution has absolute Jacobian 1. Therefore

$$\begin{aligned}I &= 4 \int_0^1 s^{1/2} \int_0^s u(s - u) \ \mathrm{d}u \ \mathrm{d}s \\ &= 4 \int_0^1 s^{1/2} \left(\frac{s^3}{2} - \frac{s^3}{3}\right) \mathrm{d}s = \frac{2}{3} \int_0^1 s^{7/2} \ \mathrm{d}s = \frac{4}{27}.\end{aligned}$$

Chapter 9

Improper integrals and unbounded regions

9.1 Definitions and the meaning of absolute convergence

9.1.1 Infinite endpoints and singular endpoints

Recall the one-variable definitions

$$\int_a^\infty f(t)\ \mathrm{d}t = \lim_{R\to\infty}\int_a^R f(t)\ \mathrm{d}t, \qquad \int_a^b f(t)\ \mathrm{d}t = \lim_{s\to b^-}\int_a^s f(t)\ \mathrm{d}t$$

when the second integral is improper because f blows up at b. The displayed limits must exist and be finite for convergence. If there are two singular endpoints, or a singularity inside an interval, split the integral and require each improper part to converge separately.

9.1.2 Compact truncations in several variables

Suppose $f : U \to \mathbb{R}$, where either f, U, or both may be unbounded. We formulate convergence by compact contented truncations: there is a number I such that for every $\varepsilon > 0$ there is a compact contented $B_\varepsilon \subset U$ for which

$$|\int_A f - I| < \varepsilon$$

for every compact contented $A \subset U$ containing B_ε. Each ordinary integral in this condition must exist. The important quantifier is *every* sufficiently large truncation, rather than only one chosen sequence. This is called absolute integrability and is equivalent, in the usual locally integrable setting, to integrability of $|f|$.

A function is locally integrable on U if its integral exists on every compact contented subset of U. In the examples below, continuity away from the singularities ensures this. When a singular point is removed from U, the compact truncations must avoid it, while approaching it in the limiting process.

Note 9.1. Absolute convergence is independent of arbitrary compact contented truncations. Convergence along a single exhaustion is a weaker condition for signed functions.

Exercise 9.1 (A practical absolute-convergence criterion)**.** Suppose U has a nested compact contented exhaustion K_j such that every compact subset of U lies in some K_j. Let f be locally integrable. Show that a finite bound

$$M = \sup\left\{\int_A |f| : A \subset U \text{ compact and contented}\right\} < \infty$$

implies that $\int_A f$ has a limit independent of the compact truncations. Deduce the comparison test: if $0 \le f \le g$, both are locally integrable, and g has finite integral, then so does f.

Solution. Given $\varepsilon > 0$, choose compact contented B with $\int_B |f| > M - \varepsilon$. If $A \supset B$ is another allowed truncation, then

$$\int_{A\setminus B} |f| = \int_A |f| - \int_B |f| < \varepsilon.$$

The difference set is contented, so its ordinary integral exists by restriction. Thus $|\int_A f - \int_B f| < \varepsilon$. More generally, for two truncations containing B, their signed integrals differ by less than 2ε, using their union. Choose a subsequence of the exhaustion containing successive such sets B. Its integrals form a Cauchy sequence, hence have a finite limit I. For any $A \supset B$, compare A to a sufficiently late member of that subsequence by taking their union; the same tail bounds give $|\int_A f - I| \le 2\varepsilon$. This is the required truncation-independent limit. For the nonnegative function $|f|$, its limit is the supremum M itself.

For comparison, $\int_A f \le \int_A g$ for every compact truncation A. A finite total integral for nonnegative g bounds all these integrals, so the same supremum criterion applies to f. For signed f, the corresponding test uses $|f| \le g$.

Exercise 9.2 (why all truncations force absolute integrability)**.** Let f be locally Riemann integrable on an open set U. Suppose its signed integrals converge to a finite number I over every sufficiently large compact contented truncation, in the sense of the preceding definition. Prove that

$$\sup_{D\subset U \text{ compact and contented}} \int_D |f| < \infty.$$

Together with the preceding criterion, this proves the equivalence with absolute integrability.

Solution. Choose a compact contented $B \subset U$ such that $|\int_A f - I| < 1$ whenever $A \supset B$ is an allowed truncation. Taking $A = B$ and $A = B \cup C$, where $C \subset U$ is compact, contented, and disjoint from B, gives

$$\left|\int_C f\right| < 2.$$

Fix another compact contented set D. Extend $f\chi_{D\setminus B}$ by zero to a containing box, and call the result g. Restriction and the contented-set calculus imply that g, $g^+ = \max(g, 0)$, and $g^- = \max(-g, 0)$ are Riemann integrable.

Take lower Darboux sums for g^+ converging to its integral. In any cell with strictly positive infimum, every point is in $D \setminus B$ and f is positive. The finite union C of these closed cells is compact, contented, and disjoint from B. Its integral of f is at least the lower sum; cells with zero infimum contribute nothing. Since $\int_C f < 2$, passage to the limit gives $\int g^+ \le 2$. Applying the same argument to $-f$ gives $\int g^- \le 2$. Hence

$$\int_D |f| \le \int_B |f| + \int g^+ + \int g^- \le \int_B |f| + 4.$$

The bound is independent of D, which proves the assertion. Cofinal compact contented exhaustions, as in the preceding criterion, are available on open subsets of Euclidean space: enumerate the closed rational boxes contained in U and take finite unions. Their interiors cover U, so compactness makes these unions cofinal.

9.2 Radial powers: the two thresholds

9.2.1 Outside the unit disk

Let $r = \sqrt{x^2 + y^2}$ and, for $p > 0$, let

$$f(x, y) = \frac{1}{(x^2 + y^2)^{p/2}} = r^{-p}.$$

On the region $r \geq 1$, polar coordinates give, for $R > 1$,

$$\int_{1 \leq r \leq R} f \, \mathrm{d}(x, y) = \int_0^{2\pi} \int_1^R r^{1-p} \, \mathrm{d}r \, \mathrm{d}\theta.$$

If $p \neq 2$, this is

$$\frac{2\pi}{2 - p}(R^{2-p} - 1).$$

For $p = 2$, it is $2\pi \log R$. Hence

$$\int_{r \geq 1} r^{-p} \, \mathrm{d}(x, y) = \begin{cases} \dfrac{2\pi}{p - 2}, & p > 2, \\ +\infty, & 0 < p \leq 2. \end{cases}$$

The density must decrease rapidly enough to offset the growing number of small area elements at large radius. In polar form the competition is explicit: the area factor is r, so the effective radial power is r^{1-p}.

9.2.2 Inside the unit disk, near the origin

For $0 < \delta < 1$,

$$\int_{\delta \leq r \leq 1} r^{-p} \, \mathrm{d}(x, y) = 2\pi \int_\delta^1 r^{1-p} \, \mathrm{d}r.$$

Thus

$$\int_{0 < r \leq 1} r^{-p} \, \mathrm{d}(x, y) = \begin{cases} \dfrac{2\pi}{2 - p}, & 0 < p < 2, \\ +\infty, & p \geq 2. \end{cases}$$

At $p = 2$, the logarithm diverges as $\delta \to 0$. The conditions at the origin and at infinity go in opposite directions. No pure power r^{-p} has a finite integral over the entire punctured plane.

Exercise 9.3 (Why radial limits suffice for these examples)**.** Explain why convergence of the radial integral gives absolute integrability over arbitrary compact truncations, rather than merely convergence along circular annuli.

Solution. The integrand is nonnegative. Every compact truncation of the exterior region lies inside some annulus $1 \le r \le R$, so its integral is bounded above by the corresponding annular integral. Conversely, the annuli themselves are allowed compact truncations. Hence the supremum of all compact-truncation integrals equals the increasing radial limit. For the punctured disk, compact truncations have positive distance from the origin and lie inside an annulus $\delta \le r \le 1$. The same argument applies, with an additional outer cutoff tending to 1 if the outer boundary is excluded from the domain. A finite supremum invokes the preceding absolute-convergence criterion; an infinite radial limit proves divergence.

9.3 The Gaussian integral

9.3.1 Squaring the one-dimensional integral

Consider the Gaussian integral

$$I = \int_{-\infty}^{\infty} e^{-x^2} \, \mathrm{d}x.$$

First, I exists and is positive. On $x \ge 1$, $e^{-x^2} \le e^{-x}$, and the latter has a finite tail integral; the negative tail is the same by symmetry. Formally, square the integral, apply Fubini, and use polar coordinates:

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} \, \mathrm{d}x \right) \left(\int_{-\infty}^{\infty} e^{-y^2} \, \mathrm{d}y \right) \\ &= \int_{\mathbb{R}^2} e^{-(x^2+y^2)} \, \mathrm{d}(x,y) \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r \, \mathrm{d}r \, \mathrm{d}\theta = \pi \int_0^{\infty} e^{-u} \, \mathrm{d}u = \pi, \end{aligned}$$

where $u = r^2$, $\mathrm{d}u = 2r \, \mathrm{d}r$. Since $I > 0$,

$$\int_{-\infty}^{\infty} e^{-x^2} \, \mathrm{d}x = \sqrt{\pi}.$$

Note 9.2. The fundamental theorem guarantees an antiderivative of the Gaussian on finite intervals. Evaluating its improper integral does not require an explicit primitive.

Exercise 9.4 (A bounded-domain justification of the Gaussian trick)**.** Justify the squaring and polar steps without assuming an improper version of Fubini in advance.

Solution. Set $I_R = \int_{-R}^{R} e^{-x^2} \, \mathrm{d}x$. Ordinary Fubini on a square gives

$$I_R^2 = \int_{[-R,R]^2} e^{-(x^2+y^2)} \, \mathrm{d}(x,y).$$

The disk of radius R is contained in this square, and the square is contained in the disk of radius $\sqrt{2}R$. Since the integrand is nonnegative, polar integration on these two finite disks yields

$$\pi(1 - e^{-R^2}) \le I_R^2 \le \pi(1 - e^{-2R^2}).$$

Both bounds tend to π. Since $I_R \to I > 0$, we obtain $I^2 = \pi$ and $I = \sqrt{\pi}$. The same bounds show that the planar nonnegative integral has finite supremum over compact truncations, so its improper interpretation is justified as well.

9.4 A finite integral with a singularity and an infinite tail

9.4.1 A radial comparison example

Consider

$$f(x,y) = \frac{1}{\sqrt{x^2+y^2}(1+x^2+y^2)} = \frac{1}{r(1+r^2)}$$

on the punctured plane. It is nonnegative and continuous away from the origin. On a compact annulus,

$$\begin{aligned}\int_{\delta\le r\le R} f \ \mathrm{d}(x,y) &= \int_0^{2\pi}\int_\delta^R \frac{r}{r(1+r^2)}\ \mathrm{d}r\ \mathrm{d}\theta \\ &= 2\pi[\arctan r]_\delta^R.\end{aligned}$$

Letting $\delta \to 0$ and $R \to \infty$ gives

$$\int_{\mathbb{R}^2\setminus\{0\}} \frac{\mathrm{d}(x,y)}{\sqrt{x^2+y^2}(1+x^2+y^2)} = \pi^2.$$

The finite radial total, together with nonnegativity, verifies absolute integrability over all compact truncations. The cancellation of the polar factor r against the factor $1/r$ makes the mild singularity at the origin integrable; at infinity the function behaves like r^{-3}, which satisfies the exterior threshold $p > 2$.

9.5 Exhaustions and conditional convergence

9.5.1 A warning about one chosen sequence

For a locally Riemann-integrable function, consider nested compact contented sets

$$K_1 \subset K_2 \subset \cdots, \qquad \bigcup_{j=1}^{\infty} K_j = \mathbb{R}^n,$$

and the limit $\lim_{j\to\infty}\int_{K_j} f$. Such a limit is meaningful for that exhaustion, but alone it does not establish absolute integrability. To approximate all compact subsets reliably, it is convenient to require the stronger exhaustion condition $\bigcup_j \operatorname{int} K_j = \mathbb{R}^n$; then compactness puts every compact subset inside some K_j. Absolute convergence makes the limit independent of the exhaustion. A conditionally convergent integral need not have this independence.

Consider $\int_0^\infty \sin x/x \ \mathrm{d}x$. At $x = 0$ the function has the removable value 1. Its usual interval truncations converge, even though its absolute integral diverges.

Exercise 9.5 (Conditional convergence and a different exhaustion)**.** Prove these two assertions and construct a nested compact exhaustion of $[0,\infty)$ along which the integrals of $\sin x/x$ tend to $+\infty$.

Solution. For $S > R \ge 1$, integration by parts gives

$$\int_R^S \frac{\sin x}{x}\ \mathrm{d}x = \left[-\frac{\cos x}{x}\right]_R^S - \int_R^S \frac{\cos x}{x^2}\ \mathrm{d}x.$$

Its absolute value is at most $1/S + 1/R + \int_R^S x^{-2}\,\mathrm{d}x \le 3/R$. This is the Cauchy criterion for the ordinary improper integral. The continuous extension at zero handles the finite initial interval.

On each interval $[k\pi, (k+1)\pi]$, with $k \ge 1$,

$$\int_{k\pi}^{(k+1)\pi} \frac{|\sin x|}{x}\,\mathrm{d}x \ge \frac{1}{(k+1)\pi}\int_{k\pi}^{(k+1)\pi} |\sin x|\,\mathrm{d}x = \frac{2}{(k+1)\pi}.$$

The harmonic series diverges, so the absolute integral is infinite. Restricting to intervals with even k proves that the positive part alone has an infinite tail integral.

Construct K_j inductively as finite unions of closed bounded intervals. Start with $K_0 = \varnothing$. Include $K_{j-1} \cup [0, j]$ in K_j, and then add finitely many positive-sine intervals beyond this bounded set until $\int_{K_j} \sin x/x\,\mathrm{d}x \ge j$. There are always enough such intervals, because the positive tail integral is infinite. Each K_j is compact and contented, $K_j \supset K_{j-1}$, and their union contains $[0, \infty)$. Yet their integrals tend to $+\infty$, whereas the interval truncations have a finite limit. This demonstrates precisely why a chosen-exhaustion limit is weaker than the all-truncations condition.

9.6 The gamma integral

Recall the improper integral

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = \int_0^1 t^{x-1}e^{-t}\,\mathrm{d}t + \int_1^\infty t^{x-1}e^{-t}\,\mathrm{d}t, \qquad x > 0.$$

At zero, $0 \le t^{x-1}e^{-t} \le t^{x-1}$ and $\int_0^1 t^{x-1}\,\mathrm{d}t = 1/x$. At infinity the exponential dominates any fixed power, so the second integral also converges. Thus Γ is well defined on $(0, \infty)$, and $\Gamma(1) = 1$.

Integration by parts, with $u = t^x$ and $\mathrm{d}v = e^{-t}\,\mathrm{d}t$, gives

$$\Gamma(x+1) = [-t^x e^{-t}]_0^\infty + x\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = x\Gamma(x).$$

In particular, $\Gamma(n+1) = n!$ for every integer $n \ge 0$.

Exercise 9.6 (justify the improper integration by parts)**.** Supply the tail estimate and the endpoint limits used above.

Solution. Choose an integer $m > x + 1$. From $e^{t/2} \ge (t/2)^m/m!$ we obtain, for $t \ge 1$,

$$t^{x-1}e^{-t} \le 2^m m!\, t^{x-1-m}e^{-t/2} \le 2^m m!\, e^{-t/2}.$$

This is integrable. The same estimate, with a sufficiently large m, shows $t^x e^{-t} \to 0$ as $t \to \infty$. Since $x > 0$, $t^x e^{-t} \to 0$ as $t \downarrow 0$. Apply ordinary integration by parts on $[\delta, R]$ and then let $\delta \downarrow 0$ and $R \to \infty$. The displayed recurrence follows. Starting with $\Gamma(1) = 1$ and applying it successively proves the factorial formula.

Chapter 10

Paths, arc length, and line integrals

10.1 Paths and polygonal approximation

10.1.1 Regular paths and piecewise smooth paths

Definition 10.1. A C^1 path in $\mathbb{R}^n$ is a continuously differentiable map $\gamma : [a, b] \to \mathbb{R}^n$, where $a < b$. It is *regular* if $\gamma'(t) \neq 0$ for every t. In this chapter, such a path is called *smooth*; infinitely many derivatives are not required.

At endpoints, derivatives are interpreted by a C^1 extension or by the corresponding one-sided derivatives. The image of a path records its points, whereas the path also records their order and multiplicity [1].

Definition 10.2. A continuous path $\gamma : [a, b] \to \mathbb{R}^n$ is *piecewise smooth* if there is a partition $a = a_0 < a_1 < \cdots < a_k = b$ such that each restriction $\gamma_i = \gamma|_{[a_{i-1}, a_i]}$ is a regular C^1 path. A path is *closed* if $\gamma(a) = \gamma(b)$. A simple arc is injective; a simple closed path is injective except for the equality of its two endpoints.

10.1.2 Length of a path

Choose a partition

$$P = \{a = t_0 < t_1 < \cdots < t_k = b\}, \qquad |P| = \max_i (t_i - t_{i-1}).$$

Join successive points $\gamma(t_{i-1})$ and $\gamma(t_i)$ by line segments. The length of this polygonal arc is

$$S(\gamma, P) = \sum_{i=1}^{k} \|\gamma(t_i) - \gamma(t_{i-1})\|.$$

We define the length by

$$S(\gamma) = \lim_{|P| \to 0} S(\gamma, P),$$

provided the limit exists. This means that for every $\varepsilon > 0$ there is $\delta > 0$ such that every partition with $|P| < \delta$ satisfies $|S(\gamma) - S(\gamma, P)| < \varepsilon$.

Theorem 10.3 (Length formula)**.** *For every C^1 path $\gamma : [a, b] \to \mathbb{R}^n$, the length exists and*

$$S(\gamma) = \int_a^b \|\gamma'(t)\| \, \mathrm{d}t.$$

Think of $\gamma(t)$ as the position of a moving particle. Then $\|\gamma'(t)\|$ is its speed, so the integral gives the total distance traveled, including any retracing of the path.

Exercise 10.1 (prove the polygonal length formula)**.** Prove the theorem directly, without using a scalar mean-value theorem for vector-valued functions.

Solution. Let

$$\eta(h) = \sup\{\|\gamma'(u) - \gamma'(v)\| : u, v \in [a, b],\ |u - v| \le h\}.$$

Uniform continuity gives $\eta(h) \to 0$ as $h \to 0$. On a subinterval of length $\Delta_i = t_i - t_{i-1}$, set $v_i = \gamma'(t_{i-1})$. The fundamental theorem of calculus and the triangle inequality give

$$\|\gamma(t_i) - \gamma(t_{i-1}) - v_i \Delta_i\| \le \eta(|P|)\Delta_i.$$

Also,

$$\left| \int_{t_{i-1}}^{t_i} \|\gamma'(t)\| \, \mathrm{d}t - \|v_i\|\Delta_i \right| \le \eta(|P|)\Delta_i.$$

Combining these bounds and summing yields

$$\left| S(\gamma, P) - \int_a^b \|\gamma'(t)\| \, \mathrm{d}t \right| \le 2(b - a)\eta(|P|) \longrightarrow 0.$$

Regularity is unnecessary here: a C^1 path may stop and still have this length formula. For a piecewise C^1 path, apply the formula on each piece and add; inserting the finitely many breakpoints into a partition does not affect the limiting result.

Example 10.4 (Coordinate notation and a graph)**.** Writing $\gamma(t) = (x_1(t), \ldots, x_n(t))$, we have

$$S(\gamma) = \int_a^b \sqrt{\left(\frac{\mathrm{d}x_1}{\mathrm{d}t}\right)^2 + \cdots + \left(\frac{\mathrm{d}x_n}{\mathrm{d}t}\right)^2} \, \mathrm{d}t.$$

If $f : [a, b] \to \mathbb{R}$ is C^1, its graph is the image of $\gamma(x) = (x, f(x))$. Since $\gamma'(x) = (1, f'(x))$, the graph has length

$$S(\gamma) = \int_a^b \sqrt{1 + f'(x)^2} \, \mathrm{d}x.$$

10.2 Equivalent parameters and arc-length parameters

10.2.1 Orientation-preserving equivalence

The length should not depend on how fast a particle traverses the same points in the same order. The following definition makes that statement precise.

Definition 10.5. Paths $\alpha : [a,b] \to \mathbb{R}^n$ and $\beta : [c,d] \to \mathbb{R}^n$ are *equivalent* if there is an onto C^1 map $\phi : [a,b] \to [c,d]$ such that

$$\alpha = \beta \circ \phi, \qquad \phi'(t) > 0 \quad (t \in [a,b]).$$

Thus $\phi(a) = c$, $\phi(b) = d$, and the orientation is preserved.

Exercise 10.2 (check equivalence and regularity)**.** Show that this is an equivalence relation, and that equivalent paths are either both regular or both nonregular.

Solution. The identity gives reflexivity. Since $\phi' > 0$, ϕ is strictly increasing and has a C^1 inverse, with

$$(\phi^{-1})'(u) = \frac{1}{\phi'(\phi^{-1}(u))} > 0.$$

The equation $\beta = \alpha \circ \phi^{-1}$ proves symmetry. If $\alpha = \beta \circ \phi$ and $\beta = \gamma \circ \psi$, then $\alpha = \gamma \circ (\psi \circ \phi)$, and the derivative of the composite is positive. This proves transitivity. Finally,

$$\alpha'(t) = \beta'(\phi(t))\phi'(t),$$

so $\alpha'(t) = 0$ exactly when $\beta'(\phi(t)) = 0$; surjectivity then proves the regularity assertion.

Theorem 10.6 (Parameter invariance of scalar path integrals)**.** *Suppose α and β are equivalent C^1 paths, and f is continuous on a set containing their common image. Then*

$$\int_a^b f(\alpha(t))\|\alpha'(t)\| \, \mathrm{d}t = \int_c^d f(\beta(u))\|\beta'(u)\| \, \mathrm{d}u.$$

In particular, $S(\alpha) = S(\beta)$.

Proof. Let $\alpha = \beta \circ \phi$ with $\phi' > 0$. By the chain rule,

$$\begin{aligned}\int_a^b f(\alpha(t))\|\alpha'(t)\| \, \mathrm{d}t &= \int_a^b f(\beta(\phi(t)))\|\beta'(\phi(t))\|\phi'(t) \, \mathrm{d}t \\ &= \int_c^d f(\beta(u))\|\beta'(u)\| \, \mathrm{d}u,\end{aligned}$$

where $u = \phi(t)$. Take $f = 1$ for the length assertion.

Note 10.7. Having the same image alone is insufficient. For example, $(\cos t, \sin t)$ on $[0, 2\pi]$ has length 2π, while the same formula on $[0, 4\pi]$ has length 4π. The second path goes around the same circle twice and is not equivalent to the first.

10.2.2 Every regular path has a unit-speed parameter

Proposition 10.8. *Every regular C^1 path $\gamma : [a,b] \to \mathbb{R}^n$ is equivalent to a regular unit-speed path $\widehat{\gamma} : [0, L] \to \mathbb{R}^n$, where $L = S(\gamma)$.*

Proof. Define the distance traveled from the initial point by

$$\sigma(t) = \int_a^t \|\gamma'(u)\| \, \mathrm{d}u.$$

Then $\sigma(a) = 0$, $\sigma(b) = L$, and $\sigma'(t) = \|\gamma'(t)\| > 0$. Consequently σ is invertible. Write $\tau = \sigma^{-1} : [0, L] \to [a, b]$; then

$$\tau'(s) = \frac{1}{\sigma'(\tau(s))} > 0.$$

Define $\widehat{\gamma}(s) = \gamma(\tau(s))$. We have $\gamma = \widehat{\gamma} \circ \sigma$, and

$$\|\widehat{\gamma}'(s)\| = \|\gamma'(\tau(s))\|\tau'(s) = \frac{\|\gamma'(\tau(s))\|}{\sigma'(\tau(s))} = 1.$$

Thus s measures distance *along the path* from its initial point, rather than Euclidean distance from that point. Substitution $s = \sigma(t)$ gives

$$\int_a^b f(\gamma(t))\|\gamma'(t)\| \, \mathrm{d}t = \int_0^L f(\widehat{\gamma}(s)) \, \mathrm{d}s.$$

This motivates the notation

$$\int_\gamma f \, \mathrm{d}s = \int_a^b f(\gamma(t))\|\gamma'(t)\| \, \mathrm{d}t, \qquad \mathrm{d}s = \|\gamma'(t)\| \, \mathrm{d}t.$$

10.3 Scalar integrals and mass

10.3.1 Density on a wire

Suppose a wire coincides with the image of a regular simple path γ and has continuous linear density ρ. The mass of a short piece is approximated by its density times its length. Thus, with tags $\xi_i \in [t_{i-1}, t_i]$, the polygonal approximation is

$$m(\gamma, \rho, P) = \sum_{i=1}^{k} \rho(\gamma(\xi_i))\|\gamma(t_i) - \gamma(t_{i-1})\|.$$

Its limit is

$$m = \int_\gamma \rho \, \mathrm{d}s = \int_a^b \rho(\gamma(t))\|\gamma'(t)\| \, \mathrm{d}t.$$

For a physical density assume $\rho \geq 0$. A general continuous f still defines a scalar integral but may have either sign.

Note 10.9. A path integral counts every traversal. For the mass of an ordinary wire, use a parameterization that traverses the wire once, or account for multiplicity explicitly.

Example 10.10 (Mass of the helix)**.** Let

$$\gamma(t) = (\cos t, \sin t, t), \qquad 0 \leq t \leq 4\pi, \qquad \rho(x, y, z) = (x^2 + y^2)z.$$

Here

$$\gamma'(t) = (-\sin t, \cos t, 1), \qquad \|\gamma'(t)\| = \sqrt{2}, \qquad \rho(\gamma(t)) = t.$$

Consequently,

$$m = \int_\gamma \rho \, \mathrm{d}s = \sqrt{2} \int_0^{4\pi} t \, \mathrm{d}t = \sqrt{2} \left[\frac{t^2}{2}\right]_0^{4\pi} = 8\sqrt{2}\,\pi^2.$$

The length is $4\pi\sqrt{2}$. An arc-length parametrization is $\widehat{\gamma}(s) = (\cos(s/\sqrt{2}), \sin(s/\sqrt{2}), s/\sqrt{2})$ for $0 \leq s \leq 4\pi\sqrt{2}$.

10.4 Work and integrals of one-forms

10.4.1 Work of a force field

Let $F : U \to \mathbb{R}^n$ be a continuous force field and let $\gamma : [a, b] \to U$ be a piecewise C^1 path describing a particle's motion. If F is constant, the work is $F \cdot (\gamma(b) - \gamma(a))$. For a varying field, use the sum

$$W(F, \gamma, P) = \sum_{i=1}^{k} F(\gamma(\xi_i)) \cdot (\gamma(t_i) - \gamma(t_{i-1})).$$

In the limit this gives, with the integral summed over the C^1 pieces,

$$W = \int_a^b F(\gamma(t)) \cdot \gamma'(t) \, \mathrm{d}t.$$

For a regular path put

$$T(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}.$$

Then

$$\begin{aligned} W &= \int_a^b F(\gamma(t)) \cdot T(t) \|\gamma'(t)\| \, \mathrm{d}t \\ &= \int_\gamma F \cdot T \, \mathrm{d}s = \int_\gamma F \cdot \mathrm{d}\mathbf{r}. \end{aligned}$$

Thus work is the scalar arc-length integral of the tangential component of the force. It is sensitive to the direction of motion.

Writing $F = (F_1, \ldots, F_n)$ and $\gamma(t) = (x_1(t), \ldots, x_n(t))$, we also have

$$\begin{aligned} W &= \int_a^b \sum_{i=1}^{n} F_i(\gamma(t)) x_i'(t) \, \mathrm{d}t \\ &= \int_\gamma F_1 \, \mathrm{d}x_1 + \cdots + F_n \, \mathrm{d}x_n. \end{aligned}$$

For example,

$$\int_\gamma f \, \mathrm{d}x_i = \int_a^b f(\gamma(t)) \gamma_i'(t) \, \mathrm{d}t.$$

This is different from $\int_\gamma f \, \mathrm{d}s$.

10.4.2 A one-form assigns a linear functional at each point

Definition 10.11. A differential one-form ω on an open set $U \subseteq \mathbb{R}^n$ assigns to each $x \in U$ a linear map $\omega_x : \mathbb{R}^n \to \mathbb{R}$. Let $\mathrm{d}x_i$ denote the coordinate functional $\mathrm{d}x_i(v) = v_i$. There are unique functions $a_1, \ldots, a_n$ such that

$$\omega_x = a_1(x) \, \mathrm{d}x_1 + \cdots + a_n(x) \, \mathrm{d}x_n.$$

The form is continuous, or C^1, when its coefficient functions have that regularity.

Exercise 10.3 (existence and uniqueness of the coefficients)**.** Prove the coordinate representation of a one-form.

Solution. Let e_i be the ith standard basis vector and set $a_i(x) = \omega_x(e_i)$. Since $v = \sum_i v_i e_i$, linearity gives

$$\omega_x(v) = \sum_i v_i \omega_x(e_i) = \sum_i a_i(x)\,\mathrm{d}x_i(v).$$

Evaluating any proposed representation on e_i recovers $a_i(x)$, so the coefficients are unique.

Definition 10.12. For a continuous one-form ω and a C^1 path γ in U, we define

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\gamma'(t))\ \mathrm{d}t = \int_a^b \sum_i a_i(\gamma(t))\gamma_i'(t)\ \mathrm{d}t.$$

For a piecewise C^1 path, add the integrals over its finitely many pieces. This definition does not require regularity.

We simply substitute $x_i = \gamma_i(t)$ and $\mathrm{d}x_i = \gamma_i'(t)\,\mathrm{d}t$, then integrate from a to b.

Exercise 10.4 (reversal, concatenation, and parameter invariance)**.** Prove that an increasing change of parameter preserves a one-form integral, reversing a path changes its sign, and concatenation adds integrals. Compare reversal with scalar arc-length integration.

Solution. If $\alpha = \beta \circ \phi$ and $\phi' > 0$, then linearity and the chain rule give

$$\omega_{\alpha(t)}(\alpha'(t)) = \omega_{\beta(\phi(t))}(\beta'(\phi(t)))\phi'(t).$$

Substitution $u = \phi(t)$ proves invariance. Define $\overline{\gamma}(t) = \gamma(a+b-t)$. Its derivative is $-\gamma'(a+b-t)$, so the same substitution gives

$$\int_{\overline{\gamma}} \omega = -\int_\gamma \omega.$$

For the scalar integral, taking the norm removes this minus sign: $\int_{\overline{\gamma}} f\ \mathrm{d}s = \int_\gamma f\ \mathrm{d}s$. If α ends where β begins, parametrize their concatenation on adjacent intervals. The definition of the piecewise integral gives $\int_{\alpha*\beta}\omega = \int_\alpha \omega + \int_\beta \omega$. An increasing affine rescaling of either interval does not affect this identity by the first calculation.

Exercise 10.5 (complete the two displayed coordinate integrals)**.** Compare $\int_\gamma xy^2z\ \mathrm{d}y$ and $\int_\gamma xy^2z\ \mathrm{d}s$ using the helix from Example 10.10 and compute both.

Solution. Along this helix, $xy^2z = t\cos t\sin^2 t$, $\mathrm{d}y = \cos t\,\mathrm{d}t$, and $\mathrm{d}s = \sqrt{2}\,\mathrm{d}t$. Hence

$$\begin{aligned}\int_\gamma xy^2z\ \mathrm{d}y &= \int_0^{4\pi} t\cos^2 t\sin^2 t\ \mathrm{d}t \\ &= \frac{1}{8}\int_0^{4\pi} t(1-\cos 4t)\ \mathrm{d}t = \pi^2.\end{aligned}$$

Indeed $\int t\cos 4t\ \mathrm{d}t = t\sin 4t/4 + \cos 4t/16$, whose endpoint difference is zero. For the second integral, integration by parts gives

$$\begin{aligned}\int_\gamma xy^2z\ \mathrm{d}s &= \sqrt{2}\int_0^{4\pi} t\cos t\sin^2 t\ \mathrm{d}t \\ &= \frac{\sqrt{2}}{3}[t\sin^3 t]_0^{4\pi} - \frac{\sqrt{2}}{3}\int_0^{4\pi}\sin^3 t\ \mathrm{d}t = 0.\end{aligned}$$

The last integrand has zero integral over complete periods. Here xy^2z is a signed function, not a nonnegative mass density.

10.4.3 Orientation and the arc-length form on a curve

Suppose a regular simple path parametrizes an arc C. Choosing one of its two directions chooses an orientation. At a noncorner point $x = \gamma(t)$, put

$$T(x) = \frac{\gamma'(t)}{\|\gamma'(t)\|}.$$

An equivalent parametrization gives the same T, since its positive parameter derivative cancels in the quotient. For a simple closed curve, use the analogous definition with the endpoints identified. At corners the tangent is understood separately on the adjacent pieces. Oppositely oriented parametrizations give $-T$.

Define the orientation-dependent arc-length one-form along C by

$$(\mathrm{d}s_C)_x(v) = T(x) \cdot v.$$

For a parametrization compatible with the chosen orientation,

$$(\mathrm{d}s_C)_{\gamma(t)}(\gamma'(t)) = \|\gamma'(t)\|, \qquad \int_\gamma f \; \mathrm{d}s_C = \int_\gamma f \; \mathrm{d}s.$$

In particular, $S(\gamma) = \int_\gamma \mathrm{d}s_C$ and $\int_\gamma F \cdot \mathrm{d}\mathbf{r} = \int_\gamma F \cdot T \; \mathrm{d}s$.

Note 10.13. The scalar element $\mathrm{d}s$ is independent of orientation. The one-form $\mathrm{d}s_C$ depends on the chosen oriented curve. Reversing a parameterization with $\mathrm{d}s_C$ fixed reverses its integral; reversing the chosen orientation also replaces $\mathrm{d}s_C$ by $-\mathrm{d}s_C$.

Chapter 11

Conservative fields and local potentials

11.1 The fundamental theorem for line integrals

11.1.1 Differentials and endpoint evaluation

Recall that if $f : U \to \mathbb{R}$ is C^1, its differential at x is

$$(\mathrm{d}f)_x(v) = Df(x)v, \qquad \mathrm{d}f = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\, \mathrm{d}x_i.$$

Theorem 11.1 (Fundamental theorem for line integrals)**.** *If $f \in C^1(U)$ and $\gamma : [a, b] \to U$ is piecewise C^1, then*

$$\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a)).$$

Proof. For a C^1 piece define $g(t) = f(\gamma(t))$. The chain rule gives $g'(t) = Df(\gamma(t))\gamma'(t)$, so

$$\begin{aligned}\int_\gamma \mathrm{d}f &= \int_a^b (\mathrm{d}f)_{\gamma(t)}(\gamma'(t))\, \mathrm{d}t \\ &= \int_a^b g'(t)\, \mathrm{d}t = g(b) - g(a) = f(\gamma(b)) - f(\gamma(a)).\end{aligned}$$

For several pieces, sum these equalities. The intermediate endpoint values cancel.

Definition 11.2. A one-form ω is *exact* on U if $\omega = \mathrm{d}f$ for a single-valued C^1 function $f : U \to \mathbb{R}$. Such an f is a *potential*. A continuous vector field F is *conservative* if $F = \nabla f$; equivalently, $F_1\, \mathrm{d}x_1 + \cdots + F_n\, \mathrm{d}x_n = \mathrm{d}f$.

The theorem says that work in a conservative field depends only on the endpoints. Every closed-path integral of an exact form is zero.

11.1.2 Finding a potential by integrating one coefficient

Example 11.3. Evaluate the integral along $\gamma(t) = (t, t^2, t^3)$, $0 \le t \le 1$, of

$$\omega = (e^{yz} - 2xy)\, \mathrm{d}x + (xze^{yz} - x^2)\, \mathrm{d}y + xye^{yz}\, \mathrm{d}z.$$

Suppose $\omega = \mathrm{d}f$. The first coefficient gives

$$\frac{\partial f}{\partial x} = e^{yz} - 2xy, \qquad f(x,y,z) = xe^{yz} - x^2y + c(y,z).$$

Differentiating with respect to the remaining variables gives

$$f_y = xze^{yz} - x^2 + c_y, \qquad f_z = xye^{yz} + c_z.$$

Comparison with the given coefficients requires $c_y = c_z = 0$. We may take $c = 0$, obtaining

$$f(x,y,z) = xe^{yz} - x^2y.$$

This verifies exactness, rather than merely guessing it. Since $\gamma(0) = (0,0,0)$ and $\gamma(1) = (1,1,1)$,

$$\int_\gamma \omega = f(1,1,1) - f(0,0,0) = e - 1.$$

11.2 Path independence and the construction of potentials

11.2.1 Exactness and closed paths

Theorem 11.4. *Let $U \subseteq \mathbb{R}^n$ be open and path connected, and let ω be a continuous one-form on U. The following are equivalent:*

(a) ω is exact on U.
(b) For any $a, b \in U$, the integral from a to b is independent of the piecewise C^1 path chosen in U.
(c) $\int_\gamma \omega = 0$ for every closed piecewise C^1 path in U.

Exercise 11.1 (complete the path-independence argument)**.** Prove the theorem, including the differentiability of the function defined by integrating from a fixed point.

Solution. The fundamental theorem proves (a)⇒(b) and (a)⇒(c). If α, β have the same endpoints, $\alpha * \overline{\beta}$ is closed and

$$\int_{\alpha * \overline{\beta}} \omega = \int_\alpha \omega - \int_\beta \omega.$$

Thus (c) implies (b). Conversely, (b) applied to a closed path and the constant path gives (c).

For (b)⇒(a), fix $p \in U$ and define

$$f(x) = \int_{p \text{ to } x} \omega.$$

There is a piecewise linear path between these points: the set of points reachable from p by such paths is open in U, because a small ball allows one more line segment; its complement is open for the same reason. Connectedness of U makes this reachable set all of U. Hypothesis (b) makes f well defined.

Write $\omega = \sum_i a_i \, \mathrm{d}x_i$. For fixed x, choose a ball $B(x,r) \subseteq U$. For $|h| < r$, use a path to x followed by the coordinate segment from x to $x + he_i$. Then

$$f(x + he_i) - f(x) = \int_0^h a_i(x + te_i) \, \mathrm{d}t.$$

Dividing by h and using continuity gives $\partial_i f(x) = a_i(x)$. All these partial derivatives are continuous, so $f \in C^1(U)$ and $\mathrm{d}f = \omega$.

Note 11.5. If U has several connected components, the construction is made separately on each component. Two potentials on a connected open set differ by a constant: the differential of their difference vanishes, so the fundamental theorem along a connecting polygonal path makes the difference the same at both endpoints.

11.2.2 Closed forms and the missing domain hypothesis

For a C^1 one-form in the plane,

$$\omega = P\,\mathrm{d}x + Q\,\mathrm{d}y, \qquad \mathrm{d}\omega = (Q_x - P_y)\,\mathrm{d}x \wedge \mathrm{d}y.$$

The form is *closed* when $Q_x = P_y$ throughout its domain. If $\omega = \mathrm{d}f$ and ω is C^1, then $f_x = P$ and $f_y = Q$ are C^1, so f is C^2 and equality of mixed partial derivatives gives

$$\mathrm{d}\omega = (f_{yx} - f_{xy})\,\mathrm{d}x \wedge \mathrm{d}y = 0.$$

Thus every C^1 exact form is closed. The converse needs a hypothesis on the domain.

Theorem 11.6 (Potentials on the plane)**.** *For a C^1 one-form $\omega = P\,\mathrm{d}x + Q\,\mathrm{d}y$ on all of $\mathbb{R}^2$, the following statements are equivalent:*

(a) There is f on $\mathbb{R}^2$ such that $\omega = \mathrm{d}f$.
(b) $\mathrm{d}\omega = 0$, that is, $Q_x = P_y$.
(c) Integrals between two points are independent of the piecewise smooth path chosen.

Exercise 11.2 (Construct the coordinate potential)**.** Assuming $Q_x = P_y$ on $\mathbb{R}^2$, prove the theorem by completing the horizontal-then-vertical construction.

Solution. Use the broken line from $(0,0)$ to $(x,0)$ and then to (x,y) and set

$$f(x,y) = \int_0^x P(t,0)\,\mathrm{d}t + \int_0^y Q(x,t)\,\mathrm{d}t.$$

The fundamental theorem immediately gives $f_y = Q(x,y)$. Since $Q \in C^1$, differentiation under this finite integral gives

$$\begin{aligned} f_x(x,y) &= P(x,0) + \int_0^y Q_x(x,t)\,\mathrm{d}t \\ &= P(x,0) + \int_0^y P_y(x,t)\,\mathrm{d}t = P(x,y). \end{aligned}$$

These formulas remain valid when x or y is negative, using oriented one-variable integrals. Therefore $\mathrm{d}f = \omega$. Together with the mixed-partial calculation and Theorem 11.4, this proves all three equivalences.

Note 11.7. A closed loop may intersect itself or retrace segments, so it need not be the boundary of a single simple region. The potential theorem on $\mathbb{R}^2$ must not be extended unchanged to arbitrary punctured domains.

11.2.3 Local potentials on convex balls

The preceding construction uses horizontal and vertical segments that remain in $\mathbb{R}^2$. On a general open set, a ball supplies a smaller region in which a potential can be constructed without leaving the domain.

Exercise 11.3 (construct a potential on a convex set)**.** Let $V \subseteq \mathbb{R}^n$ be open and convex, let $\omega = \sum_i a_i \, \mathrm{d}x_i$ with $a_i \in C^1(V)$, and suppose

$$\partial_j a_i = \partial_i a_j \qquad (1 \le i, j \le n).$$

Fix $c \in V$. Show that

$$f(x) = \int_0^1 \sum_{i=1}^{n} a_i(c + t(x-c))(x_i - c_i) \, \mathrm{d}t$$

is a potential. Deduce local exactness of every closed C^1 one-form on an open subset of $\mathbb{R}^2$.

Solution. Convexity keeps the whole segment $c + t(x-c)$ inside V. For fixed x, compactness of this segment and openness of V allow x to vary in a sufficiently small neighborhood while all these segments remain in V. The integrand and its first x-derivatives are then continuous on the corresponding compact parameter sets, which justifies differentiation under the integral. Write $z(t) = c + t(x-c)$. For the jth partial derivative,

$$\begin{aligned}
\partial_j f(x) &= \int_0^1 \left[a_j(z(t)) + t \sum_i (x_i - c_i) \partial_j a_i(z(t)) \right] \mathrm{d}t \\
&= \int_0^1 \left[a_j(z(t)) + t \sum_i (x_i - c_i) \partial_i a_j(z(t)) \right] \mathrm{d}t \\
&= \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} \big[t a_j(z(t)) \big] \, \mathrm{d}t = a_j(x).
\end{aligned}$$

Thus $\mathrm{d}f = \omega$. In the plane, closedness is exactly the nontrivial symmetry condition $Q_x = P_y$. Every point of an open set has a convex ball contained in that set, so a closed C^1 form has a potential on each such ball. Potentials on overlapping balls need only differ by a constant on each connected component of their overlap; they need not combine to a single global potential.

11.3 The angular form: locally exact but not globally exact

11.3.1 The angular form

On $U = \mathbb{R}^2 \setminus \{(0,0)\}$, consider

$$\omega = \frac{-y \, \mathrm{d}x + x \, \mathrm{d}y}{x^2 + y^2}.$$

This is a smooth one-form on its domain. Along the counterclockwise unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$,

$$\begin{aligned}
\int_\gamma \omega &= \int_0^{2\pi} \frac{-\sin t(-\sin t) + \cos t \cos t}{\cos^2 t + \sin^2 t} \, \mathrm{d}t \\
&= \int_0^{2\pi} 1 \, \mathrm{d}t = 2\pi.
\end{aligned}$$

If $\omega = \mathrm{d}f$ on all of U, the same closed-path integral would be zero. Hence ω has no global single-valued potential on the punctured plane.

Nevertheless, set $P = -y/(x^2+y^2)$ and $Q = x/(x^2+y^2)$. Then

$$Q_x = \frac{y^2-x^2}{(x^2+y^2)^2}, \qquad P_y = \frac{y^2-x^2}{(x^2+y^2)^2},$$

so $\mathrm{d}\omega = 0$ at every point of U. This is the basic example showing that closedness alone does not imply global exactness.

11.3.2 The local angle calculation

Derive the angular form from $\tan\theta = y/x$. On a region where $x \neq 0$, take a local choice $\theta = \arctan(y/x)$, possibly shifted by a constant multiple of π. Differentiation gives

$$\mathrm{d}\theta = \frac{1}{1+(y/x)^2}\,\mathrm{d}\left(\frac{y}{x}\right) = \frac{x^2}{x^2+y^2}\left(-\frac{y}{x^2}\,\mathrm{d}x + \frac{1}{x}\,\mathrm{d}y\right) = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2+y^2} = \omega.$$

The constant adjustment selects an angle branch but does not change its differential.

Exercise 11.4 (finish the local angle and radius comparison)**.** Find a potential near a point where $x = 0$ and $y \neq 0$. Also compute both $\int_\gamma \omega$ and $\int_\gamma \mathrm{d}s$ for $\gamma(t) = (R\cos t, R\sin t)$, $0 \le t \le 2\pi$, $R > 0$.

Solution. In a neighborhood on which $y \neq 0$, take $\theta = \arctan(-x/y)$. Since

$$\mathrm{d}(-x/y) = -\frac{1}{y}\,\mathrm{d}x + \frac{x}{y^2}\,\mathrm{d}y,$$

we obtain

$$\mathrm{d}\theta = \frac{y^2}{x^2+y^2}\left(-\frac{1}{y}\,\mathrm{d}x + \frac{x}{y^2}\,\mathrm{d}y\right) = \omega.$$

An added constant makes this agree with the desired local angle. For the radius-R circle, the numerator after substitution is $R^2\,\mathrm{d}t$, the denominator is R^2, and the speed is R. Therefore

$$\int_\gamma \omega = 2\pi, \qquad \int_\gamma \mathrm{d}s = 2\pi R.$$

Reversing the path changes the first value to -2π and preserves the second. No global angle branch on the entire punctured plane can have this differential, because its closed-circle increment would have to be both 0 and 2π.

Chapter 12

Green's theorem and oriented regions

12.1 Forms in the plane and boundary orientation

12.1.1 Zero-, one-, and two-forms

A zero-form is just a function. A one-form in the plane has the form $\omega = P\,\mathrm{d}x + Q\,\mathrm{d}y$. A two-form has the form

$$\alpha = A(x,y)\ \mathrm{d}x \wedge \mathrm{d}y.$$

Here $\mathrm{d}x \wedge \mathrm{d}y$ is the alternating area form: on vectors v, w it has value $v_1 w_2 - v_2 w_1$. In particular, $\mathrm{d}y \wedge \mathrm{d}x = -\,\mathrm{d}x \wedge \mathrm{d}y$ and $\mathrm{d}x \wedge \mathrm{d}x = \mathrm{d}y \wedge \mathrm{d}y = 0$. With the usual orientation of the plane, we define

$$\int_D A\ \mathrm{d}x \wedge \mathrm{d}y = \iint_D A(x,y)\ \mathrm{d}x\ \mathrm{d}y.$$

For a C^1 one-form we define

$$\mathrm{d}\omega = (Q_x - P_y)\ \mathrm{d}x \wedge \mathrm{d}y.$$

We write $\mathrm{d}x \wedge \mathrm{d}y$ for the oriented alternating form, distinguishing it from unsigned area notation.

Recall the one-variable fundamental theorem, $\int_a^b f'(t)\,\mathrm{d}t = f(b) - f(a)$. Green's theorem has the corresponding boundary form

$$\int_D \mathrm{d}\omega = \int_{\partial D} \omega.$$

This is the two-dimensional instance of Stokes' formula $\int_V \mathrm{d}\alpha = \int_{\partial V} \alpha$: a derivative in the interior is paired with an oriented integral on the boundary.

12.1.2 What positive orientation means

A planar region here is the closure $D = \overline{\Omega}$ of a bounded connected open set, with boundary a finite disjoint union of simple closed piecewise smooth curves. The positive boundary direction is the one for which the region is on our left as we traverse it. Thus the outer boundary is counterclockwise, while the boundary of each hole is clockwise. Integrating over ∂D means summing these oriented boundary integrals.

For the proof, we use regions admitting a finite compatible cellulation: they are finite unions of oriented two-cells, with disjoint interiors, and their shared boundary arcs can be subdivided into finitely many matching edges. The definition of a two-cell is given below. All boundaries used in this proof are piecewise C^2.

12.2 Green's theorem on the unit square

12.2.1 The four boundary paths

Let $I^2 = [0,1]^2$, with counterclockwise boundary paths, all on $0 \le t \le 1$,

$$\gamma_1(t) = (t,0), \quad \gamma_2(t) = (1,t), \quad \gamma_3(t) = (1-t,1), \quad \gamma_4(t) = (0,1-t).$$

Lemma 12.1 (Green's theorem for the unit square)**.** *If P, Q are C^1 on an open neighborhood of I^2, then*

$$\iint_{I^2} (Q_x - P_y)\, \mathrm{d}x\, \mathrm{d}y = \int_{\partial I^2} P\, \mathrm{d}x + Q\, \mathrm{d}y.$$

Proof. Fubini's theorem and the one-variable fundamental theorem give

$$\begin{aligned}\iint_{I^2} (Q_x - P_y)\, \mathrm{d}x\, \mathrm{d}y &= \int_0^1 [Q(1,y) - Q(0,y)]\, \mathrm{d}y \\ &\quad - \int_0^1 [P(x,1) - P(x,0)]\, \mathrm{d}x.\end{aligned}$$

On the bottom and right edges,

$$\int_{\gamma_1} \omega = \int_0^1 P(t,0)\, \mathrm{d}t, \qquad \int_{\gamma_2} \omega = \int_0^1 Q(1,t)\, \mathrm{d}t.$$

On the top and left edges the direction is reversed:

$$\int_{\gamma_3} \omega = -\int_0^1 P(1-t,1)\, \mathrm{d}t = -\int_0^1 P(x,1)\, \mathrm{d}x,$$

$$\int_{\gamma_4} \omega = -\int_0^1 Q(0,1-t)\, \mathrm{d}t = -\int_0^1 Q(0,y)\, \mathrm{d}y.$$

Their sum is exactly the preceding double integral.

12.3 Pullback and the passage to two-cells

12.3.1 Pullback formulas

Suppose $F : U \to \mathbb{R}^2$ is C^1, where U is open in the (u,v)-plane, and write $F(u,v) = (x(u,v), y(u,v))$. The pullback turns a form in the target (x,y)-plane into a form on the parameter (u,v)-plane. For a zero-form φ,

$$F^*\varphi = \varphi \circ F.$$

For coordinate one-forms,

$$F^*(\mathrm{d}x) = x_u\, \mathrm{d}u + x_v\, \mathrm{d}v, \qquad F^*(\mathrm{d}y) = y_u\, \mathrm{d}u + y_v\, \mathrm{d}v.$$

Consequently,

$$\begin{aligned}F^*(P\, \mathrm{d}x + Q\, \mathrm{d}y) &= (P \circ F)(x_u\, \mathrm{d}u + x_v\, \mathrm{d}v) + (Q \circ F)(y_u\, \mathrm{d}u + y_v\, \mathrm{d}v) \\ &= [(P \circ F)x_u + (Q \circ F)y_u]\, \mathrm{d}u \\ &\quad + [(P \circ F)x_v + (Q \circ F)y_v]\, \mathrm{d}v.\end{aligned}$$

For a two-form,

$$\begin{aligned} F^*(A\,\mathrm{d}x \wedge \mathrm{d}y) &= (A \circ F)(x_u\,\mathrm{d}u + x_v\,\mathrm{d}v) \wedge (y_u\,\mathrm{d}u + y_v\,\mathrm{d}v) \\ &= (A \circ F)(x_u y_v - x_v y_u)\,\mathrm{d}u \wedge \mathrm{d}v \\ &= (A \circ F) \det DF\ \mathrm{d}u \wedge \mathrm{d}v. \end{aligned}$$

The determinant, rather than its absolute value, records orientation.

Example 12.2 (Pullbacks in degrees zero, one, and two)**.** Let

$$F(u,v) = (2u - v, 3u + 2v), \qquad \det DF = 7.$$

For the zero-form $\varphi(x,y) = x^2 - y^2$,

$$F^*\varphi = (2u - v)^2 - (3u + 2v)^2 = -5u^2 - 16uv - 3v^2.$$

For $\omega = -y\,\mathrm{d}x + 2x\,\mathrm{d}y$, use $F^*\,\mathrm{d}x = 2\,\mathrm{d}u - \mathrm{d}v$ and $F^*\,\mathrm{d}y = 3\,\mathrm{d}u + 2\,\mathrm{d}v$. Therefore

$$\begin{aligned} F^*\omega &= -(3u + 2v)(2\,\mathrm{d}u - \mathrm{d}v) + 2(2u - v)(3\,\mathrm{d}u + 2\,\mathrm{d}v) \\ &= (6u - 10v)\,\mathrm{d}u + (11u - 2v)\,\mathrm{d}v. \end{aligned}$$

For $\alpha = e^{x+y}\,\mathrm{d}x \wedge \mathrm{d}y$,

$$F^*\alpha = e^{5u+v}(2\,\mathrm{d}u - \mathrm{d}v) \wedge (3\,\mathrm{d}u + 2\,\mathrm{d}v) = 7e^{5u+v}\,\mathrm{d}u \wedge \mathrm{d}v.$$

12.3.2 An oriented two-cell

Definition 12.3. An oriented two-cell is $D = F(I^2)$, where F is an injective C^2 map on an open neighborhood of I^2 and $\det DF > 0$ there. Its boundary orientation is transported from the counterclockwise orientation of ∂I^2 by F.

The inverse function theorem makes F a diffeomorphism onto its image. Positivity of the determinant ensures that the transported orientation is the ordinary positive orientation of the planar cell.

Lemma 12.4 (The three pullback identities)**.** *Let $D = F(I^2)$ be such a cell. For a C^1 one-form ω and a continuous two-form α on a neighborhood of D,*

$$\begin{aligned} \int_{\partial D} \omega &= \int_{\partial I^2} F^*\omega, \\ \int_D \alpha &= \int_{I^2} F^*\alpha, \\ \mathrm{d}(F^*\omega) &= F^*(\mathrm{d}\omega). \end{aligned}$$

Exercise 12.1 (prove the pullback identities)**.** Supply the boundary chain-rule calculation, change-of-variables argument, and coefficient calculation for these identities.

Solution. For an edge path η of the square,

$$(F^*\omega)_{\eta(t)}(\eta'(t)) = \omega_{F(\eta(t))}(DF(\eta(t))\eta'(t)) = \omega_{(F\circ\eta)(t)}((F \circ \eta)'(t)).$$

Integrating and summing over the four edges proves the first identity. If $\alpha = A\,\mathrm{d}x \wedge \mathrm{d}y$, change of variables gives

$$\int_D \alpha = \iint_{I^2} (A \circ F)|\det DF|\ \mathrm{d}u\ \mathrm{d}v = \int_{I^2} F^*\alpha,$$

since $\det DF > 0$.

For the third identity, abbreviate

$$B = (P \circ F)x_u + (Q \circ F)y_u, \qquad C = (P \circ F)x_v + (Q \circ F)y_v.$$

Then $\mathrm{d}(F^*\omega) = (C_u - B_v)\,\mathrm{d}u \wedge \mathrm{d}v$. In expanding $C_u - B_v$, the terms containing P_x or Q_y cancel. The terms containing P and Q without a derivative cancel because $x_{vu} = x_{uv}$ and $y_{vu} = y_{uv}$. The remaining terms are

$$\begin{aligned} C_u - B_v &= (P_y \circ F)(y_u x_v - y_v x_u) + (Q_x \circ F)(x_u y_v - x_v y_u) \\ &= ((Q_x - P_y) \circ F) \det DF. \end{aligned}$$

This is precisely the coefficient of $F^*(\mathrm{d}\omega)$.

Lemma 12.5 (Green's theorem for a two-cell)**.** *If D is an oriented two-cell and ω is C^1 on an open neighborhood of D, then $\int_D \mathrm{d}\omega = \int_{\partial D} \omega$.*

Proof. Apply the three identities and the square theorem in order:

$$\int_D \mathrm{d}\omega = \int_{I^2} F^*(\mathrm{d}\omega) = \int_{I^2} \mathrm{d}(F^*\omega) = \int_{\partial I^2} F^*\omega = \int_{\partial D} \omega.$$

12.4 Global gluing and the role of holes

12.4.1 Cancellation of interior edges

Theorem 12.6 (Green's theorem for a finitely cellulated region)**.** *Let D be a planar region with a finite compatible decomposition into the oriented two-cells $D_1, \ldots, D_m$. Suppose $\omega = P\,\mathrm{d}x + Q\,\mathrm{d}y$ is C^1 on an open neighborhood of D. With the positive orientation on every boundary component,*

$$\iint_D (Q_x - P_y)\ \mathrm{d}x\ \mathrm{d}y = \int_{\partial D} P\ \mathrm{d}x + Q\ \mathrm{d}y.$$

Proof. Each boundary arc is C^1 on finitely many pieces and has planar area zero. The cells have disjoint interiors, so additivity gives

$$\int_D \mathrm{d}\omega = \sum_{j=1}^{m} \int_{D_j} \mathrm{d}\omega = \sum_{j=1}^{m} \int_{\partial D_j} \omega.$$

A shared edge appears twice. Its positive direction for one cell is the negative direction for its neighbor, because each cell must lie to the left of its own boundary traversal. The two integrals therefore cancel. Finite compatibility lets us make this cancellation edge by edge after subdivision. Only the outer edges and the edges around holes remain, with the region on the left. They form exactly the positively oriented ∂D.

Exercise 12.2 (cellulations for the circular examples)**.** Show that a closed disk, an annulus, and an ellipse have finite cellulations of the kind used in this proof.

Solution. For the disk of radius $R > 0$, choose $0 < a < R/\sqrt{2}$ and take the central square $[-a, a]^2$ as one cell. The region to its right inside the disk is parametrized by

$$\theta(v) = -\frac{\pi}{4} + \frac{\pi}{2}v, \qquad r(u, v) = (1 - u)a \sec\theta(v) + uR,$$

$$F(u, v) = (r(u, v)\cos\theta(v), r(u, v)\sin\theta(v)), \qquad (u, v) \in I^2.$$

Its inner edge is $x = a$, $-a \le y \le a$, and its outer edge is the corresponding circular arc. The other two edges are radial segments. Moreover,

$$\det DF = r\,(R - a\sec\theta)\,\frac{\pi}{2} > 0.$$

Both r and $R - a\sec\theta$ stay positive on a neighborhood of the square. Polar coordinates and strict increase of r with u show injectivity on a sufficiently small such neighborhood. Rotating this cell by $\pi/2$, π, and $3\pi/2$ gives the remaining three cells. Together with the central square, they cover the disk and match along their edges.

For $0 < a < b$, subdivide $a \le r \le b$ into four angular sectors. For $j = 0, 1, 2, 3$, use

$$r = a + (b - a)u, \quad \theta = j\pi/2 + (\pi/2)v, \quad F_j(u, v) = (r\cos\theta, r\sin\theta).$$

Each determinant is $r(b-a)\pi/2 > 0$, and each sector map extends injectively to a neighborhood of its square. These four cells form the annulus. Finally, the positive linear map $(x, y) \mapsto (Ax, By)$, with $A, B > 0$, takes a cellulated unit disk to the ellipse $x^2/A^2 + y^2/B^2 \le 1$ and preserves compatibility and positive Jacobians.

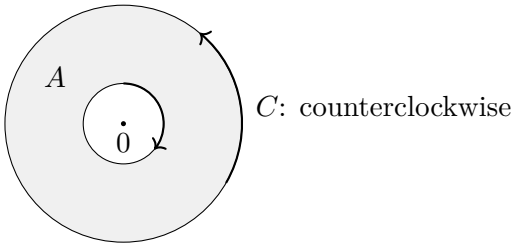
12.4.2 The angular form and an excised origin

Suppose C is a positively oriented simple closed piecewise smooth curve enclosing the origin, and let D denote its enclosed region. Choose $\varepsilon > 0$ so that the closed disk B_ε lies in the interior of D. Assume the punctured region $A = D \setminus \operatorname{int} B_\varepsilon$ has a finite cellulation as above. The angular form is C^1 on a neighborhood of A and $\mathrm{d}\omega = 0$ there. Hence Green's theorem gives

$$0 = \int_A \mathrm{d}\omega = \int_C \omega - \int_{C_\varepsilon} \omega,$$

where $C_\varepsilon(t) = (\varepsilon\cos t, \varepsilon\sin t)$ is written *counterclockwise*. The minus sign occurs because the inner boundary of A is actually clockwise. Since $\int_{C_\varepsilon} \omega = 2\pi$, we obtain

$$\int_C \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2} = 2\pi.$$



inner boundary: clockwise

The region stays on the left along both arrows. The small hole is essential: applying Green's theorem directly to D would require ω to be defined and C^1 at the origin, which it is not.

Note 12.7. A circle traversed k times counterclockwise has angular integral $2\pi k$. The value 2π for a simple positively oriented curve assumes one traversal.

12.5 Computations from Green's theorem

12.5.1 A zero circulation and an area formula

Example 12.8. For $\omega = 2xy\,\mathrm{d}x + x^2\,\mathrm{d}y$,

$$Q_x - P_y = 2x - 2x = 0.$$

Thus $\int_{\partial D}\omega = 0$ for every admissible region D. In fact $\omega = \mathrm{d}(x^2y)$, so its integral is zero on every closed piecewise C^1 path, with no simplicity assumption needed.

If D has positive boundary orientation, choose

$$\omega = -\tfrac{1}{2}y\,\mathrm{d}x + \tfrac{1}{2}x\,\mathrm{d}y, \qquad \mathrm{d}\omega = \mathrm{d}x \wedge \mathrm{d}y.$$

Green's theorem gives the area formula

$$\mathrm{Area}(D) = \frac{1}{2}\int_{\partial D}(x\,\mathrm{d}y - y\,\mathrm{d}x).$$

The choices $\omega = x\,\mathrm{d}y$ and $\omega = -y\,\mathrm{d}x$ likewise give

$$\mathrm{Area}(D) = \int_{\partial D} x\,\mathrm{d}y = -\int_{\partial D} y\,\mathrm{d}x.$$

For a region with holes, all boundary components must be included with their positive directions.

Example 12.9 (Area of an ellipse)**.** Let $a, b > 0$ and $D = \{(x, y) : x^2/a^2 + y^2/b^2 \le 1\}$. Its positive boundary parametrization is

$$x = a\cos t, \quad y = b\sin t, \qquad 0 \le t \le 2\pi.$$

Therefore

$$\begin{aligned}\mathrm{Area}(D) &= \frac{1}{2}\int_0^{2\pi}\left[(a\cos t)(b\cos t) - (b\sin t)(-a\sin t)\right]\,\mathrm{d}t \\ &= \frac{ab}{2}\int_0^{2\pi}(\cos^2 t + \sin^2 t)\,\mathrm{d}t = \pi ab.\end{aligned}$$

The determinant of the scaling from the unit disk is $ab > 0$, consistent with the direction of this parametrization.

12.5.2 Combining two double integrals

The same page contains a separate change-of-order calculation. Its two displayed pieces are

$$I = \int_0^{1/2} \int_0^1 (\arctan x)^2 \, \mathrm{d}x \, \mathrm{d}y + \int_{1/2}^1 \int_0^{\sqrt{1/y-1}} (\arctan x)^2 \, \mathrm{d}x \, \mathrm{d}y.$$

The curved boundary can be written in either form

$$x = \sqrt{\frac{1}{y} - 1} \quad \Longleftrightarrow \quad y = \frac{1}{1+x^2}, \qquad x \geq 0.$$

Exercise 12.3 (complete the change of order and cancellation)**.** Describe the combined region, reverse the order of integration, and evaluate the result.

Solution. The first integral covers the rectangle $0 \leq x \leq 1$, $0 \leq y \leq 1/2$. The second covers the region above that rectangle and below $y = 1/(1+x^2)$, since the curve runs from $(0,1)$ to $(1,1/2)$. Their union is

$$D = \{(x,y) : 0 \leq x \leq 1, \ 0 \leq y \leq 1/(1+x^2)\}.$$

The integrand is continuous and the pieces have disjoint interiors, so Fubini's theorem gives

$$\begin{aligned} I &= \int_0^1 \int_0^{1/(1+x^2)} (\arctan x)^2 \, \mathrm{d}y \, \mathrm{d}x \\ &= \int_0^1 \frac{(\arctan x)^2}{1+x^2} \, \mathrm{d}x. \end{aligned}$$

With $u = \arctan x$, $\mathrm{d}u = \mathrm{d}x/(1+x^2)$, the endpoints are 0 and $\pi/4$. Thus

$$I = \int_0^{\pi/4} u^2 \, \mathrm{d}u = \frac{1}{3}\left(\frac{\pi}{4}\right)^3 = \frac{\pi^3}{192}.$$

The second piece alone is

$$\int_0^1 \left(\frac{1}{1+x^2} - \frac{1}{2}\right) (\arctan x)^2 \, \mathrm{d}x.$$

The first piece is $\frac{1}{2}\int_0^1 (\arctan x)^2 \, \mathrm{d}x$. Those two $1/2$ terms cancel *between the two pieces*; the subtracted term cannot simply be discarded in the second integral alone. This accounts for the otherwise unexplained cancellation.

Chapter 13

Differential forms and the exterior derivative

13.1 What is a differential form?

13.1.1 The parameterization problem

Recall the expression

$$\int_0^1 t^2 \, \mathrm{d}t.$$

We integrate the function t^2 over $[0, 1]$, but the expression being integrated includes $\mathrm{d}t$. This is the simplest example of a differential form. In higher dimensions, a differential k-form is the kind of object we integrate over an oriented k-dimensional manifold [2, Chs. 3–5].

Suppose $f(x, y) = y^2$, and let C be the top half of the unit circle in $\mathbb{R}^2$. Compare

$$\phi_1(a) = (a, \sqrt{1 - a^2}), \qquad \phi_2(t) = (\cos t, \sin t).$$

Simply integrating $f \circ \phi_1$ and $f \circ \phi_2$ with respect to their parameters does not give a parameter-independent construction. The small pieces of the curve have different lengths in the two parameterizations. If a parameter interval has length Δa, its image has approximate length

$$L_i \simeq \|\phi'(a_i)\| \Delta a.$$

Thus the Riemann-sum expression $\sum_i f(\phi(a_i)) L_i$ leads to

$$\int_C f \, \mathrm{d}s = \int f(\phi(a)) \|\phi'(a)\| \, \mathrm{d}a.$$

At $p \in C$, the tangent space T_pC consists of tangent vectors; for a regular parameterization, it is spanned by $\phi'(a)$ when $\phi(a) = p$.

Note 13.1. The map $v \mapsto f(p)\|v\|$ is generally not linear. Thus $f \, \mathrm{d}s$ is a scalar length density, not a differential one-form. A one-form is linear on tangent vectors and changes sign when the tangent reverses. A k-form is alternating and multilinear in k vectors. The graph parameterization ϕ_1 is regular only for $-1 < a < 1$, so its endpoint integrals are interpreted as limits.

Exercise 13.1 (finish the semicircle comparison)**.** Compute the two naive parameter integrals and the scalar integral $\int_C y^2 \, \mathrm{d}s$. Explain the distinction from integrating $y^2 \, \mathrm{d}x$.

Solution. The naive integrals are

$$\int_{-1}^{1}(1-a^2)\,\mathrm{d}a=\frac{4}{3},\qquad \int_0^{\pi}\sin^2 t\,\mathrm{d}t=\frac{\pi}{2}.$$

For the graph parameterization,

$$\phi_1'(a)=\left(1,-\frac{a}{\sqrt{1-a^2}}\right),\qquad \|\phi_1'(a)\|=\frac{1}{\sqrt{1-a^2}}.$$

It follows that

$$\int_C y^2\,\mathrm{d}s=\int_{-1}^{1}\sqrt{1-a^2}\,\mathrm{d}a=\frac{\pi}{2}.$$

For ϕ_2, the speed is one, so the same scalar integral is $\int_0^{\pi}\sin^2 t\,\mathrm{d}t=\pi/2$.

The one-form $y^2\,\mathrm{d}x$ has a different evaluation rule:

$$(y^2\,\mathrm{d}x)_p(v)=y(p)^2v_x.$$

Along ϕ_1, with a increasing, its integral is $4/3$. Along ϕ_2, with t increasing, it is

$$\int_0^{\pi}\sin^2 t(-\sin t)\,\mathrm{d}t=-\frac{4}{3}.$$

These parameterizations traverse the semicircle in opposite directions. Reversing one makes the one-form integrals agree. Scalar length integrals agree without an orientation choice.

13.2 Alternating multilinear algebra

13.2.1 Multilinearity and alternation

Definition 13.2. A function $M:(\mathbb{R}^n)^k\to\mathbb{R}$ is *k-multilinear* if it is linear in each argument separately. Thus, with the other arguments fixed,

$$\begin{aligned}M(v_1,\dots,sv_j+tw_j,\dots,v_k)&=sM(v_1,\dots,v_j,\dots,v_k)\\&\quad+tM(v_1,\dots,w_j,\dots,v_k).\end{aligned}$$

It is *alternating* if its value is zero whenever two arguments are equal.

Example 13.3. The function

$$M((x_1,y_1),(x_2,y_2))=x_1y_2$$

is bilinear: holding the second argument fixed gives a multiple of the first coordinate of the first argument, and holding the first argument fixed gives a multiple of the second coordinate of the second argument. It is not alternating; $M((1,1),(1,1))=1$.

The function $H((x_1,y_1),(x_2,y_2))=x_1x_2y_2$ is not bilinear. Scaling its second argument by s multiplies the result by s^2, not s, when $x_1x_2y_2\neq 0$.

Exercise 13.2 (swapping arguments)**.** Show that an alternating multilinear function changes sign when two argument positions are interchanged.

Solution. Fix all other positions. Alternation and multilinearity give

$$0 = M(\dots, u+v, \dots, u+v, \dots) = M(\dots, u, \dots, v, \dots) + M(\dots, v, \dots, u, \dots),$$

because the two terms with repeated u or repeated v vanish. Therefore the two displayed terms have opposite signs. A permutation is a product of transpositions, so permuting all arguments multiplies the value by the sign of the permutation.

Proposition 13.4. *If M is alternating and k-multilinear and $v_1, \dots, v_k$ are linearly dependent, then $M(v_1, \dots, v_k) = 0$.*

Proof. After relabeling, one vector can be expressed as a linear combination of the others:

$$v_1 = c_2 v_2 + \cdots + c_k v_k.$$

Expand in the first position. Each summand has a repeated vector and is zero. If one vector is zero, multilinearity already gives zero.

Corollary 13.5. *If $k > n$, the only alternating k-multilinear function on $\mathbb{R}^n$ is the zero function.*

Proof. Every list of more than n vectors in $\mathbb{R}^n$ is linearly dependent.

13.2.2 Coordinate covectors and the basis of k-forms

Recall that $\mathrm{d}x_i : \mathbb{R}^n \to \mathbb{R}$ picks out the ith coordinate of a column vector:

$$\mathrm{d}x_i(v) = v_i.$$

For a k-tuple $I = (i_1, \dots, i_k)$ of coordinate indices, define

$$\mathrm{d}x_I(v_1, \dots, v_k) = \det \begin{pmatrix} (v_1)_{i_1} & \cdots & (v_k)_{i_1} \\ \vdots & & \vdots \\ (v_1)_{i_k} & \cdots & (v_k)_{i_k} \end{pmatrix}.$$

This is alternating and multilinear because a determinant is alternating and multilinear in its columns. We write

$$\mathrm{d}x_I = \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}.$$

For example, when $n = 3$ and $I = (1, 3)$,

$$\mathrm{d}x_1 \wedge \mathrm{d}x_3((x_1, y_1, z_1), (x_2, y_2, z_2)) = x_1 z_2 - x_2 z_1.$$

Interchanging coordinate rows changes the sign. In particular,

$$\mathrm{d}x_1 \wedge \mathrm{d}x_3 = -\,\mathrm{d}x_3 \wedge \mathrm{d}x_1, \qquad \mathrm{d}x_i \wedge \mathrm{d}x_i = 0.$$

A coordinate wedge is zero whenever an index repeats.

Theorem 13.6 (Basis theorem for alternating forms)**.** *Let $0 \le k \le n$. Every alternating k-multilinear function M on $\mathbb{R}^n$ has a unique expression*

$$M = \sum_{i_1 < \cdots < i_k} a_I \, \mathrm{d}x_I, \qquad a_I = M(e_{i_1}, \dots, e_{i_k}).$$

Consequently, the vector space of these functions, denoted $\Lambda^k(\mathbb{R}^n)$, has dimension $\binom{n}{k}$. For $k = 0$, it is $\mathbb{R}$.

Exercise 13.3 (prove the basis theorem)**.** Prove the coordinate expansion and the uniqueness of its coefficients.

Solution. Write $v_j = \sum_{r=1}^{n} (v_j)_r e_r$. Multilinearity gives

$$M(v_1, \dots, v_k) = \sum_{r_1, \dots, r_k} (v_1)_{r_1} \cdots (v_k)_{r_k} M(e_{r_1}, \dots, e_{r_k}).$$

Terms with repeated indices vanish. Group the remaining terms by the increasing rearrangement $I = (i_1 < \cdots < i_k)$ of their indices. The coefficient of $a_I = M(e_{i_1}, \dots, e_{i_k})$ becomes

$$\sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) (v_1)_{i_{\sigma(1)}} \cdots (v_k)_{i_{\sigma(k)}} = \mathrm{d}x_I(v_1, \dots, v_k),$$

by the determinant expansion. This proves existence. On an increasing list $(e_{j_1}, \dots, e_{j_k})$, the form $\mathrm{d}x_I$ evaluates to 1 if $I = J$ and 0 otherwise. Evaluating any proposed expansion on this list proves uniqueness and linear independence. There are $\binom{n}{k}$ increasing lists.

In $\mathbb{R}^3$, the alternating bilinear functions are precisely

$$a\ \mathrm{d}x \wedge \mathrm{d}y + b\ \mathrm{d}x \wedge \mathrm{d}z + c\ \mathrm{d}y \wedge \mathrm{d}z.$$

The coefficients are numbers here. For differential forms they become functions of the base point.

13.3 Forms with variable coefficients and wedge products

13.3.1 Differential forms on an open set

Definition 13.7. Let $U \subset \mathbb{R}^n$ be open. A differential k-form α on U assigns to each $p \in U$ an alternating k-multilinear function α_p. In coordinates,

$$\alpha = \sum_{i_1 < \cdots < i_k} a_I(x)\ \mathrm{d}x_I.$$

It is continuous, C^1, or C^r when all coefficient functions a_I have that regularity. The notation $\Omega^k(U)$ denotes the space of smooth k-forms, with weaker regularity stated when needed. A zero-form is a scalar function.

For example, every two-form on an open subset of $\mathbb{R}^3$ has the form

$$\alpha = a_{12}(x, y, z)\ \mathrm{d}x \wedge \mathrm{d}y + a_{13}(x, y, z)\ \mathrm{d}x \wedge \mathrm{d}z + a_{23}(x, y, z)\ \mathrm{d}y \wedge \mathrm{d}z.$$

Thus a differential form includes both a base point and a rule for evaluating tangent vectors at that point.

13.3.2 Multiplying forms

We define the wedge product first on coordinate wedges by concatenating their factors, and then extend by distributivity:

$$\left(\sum_I a_I \, \mathrm{d}x_I\right) \wedge \left(\sum_J b_J \, \mathrm{d}x_J\right) = \sum_{I,J} a_I b_J \, \mathrm{d}x_I \wedge \mathrm{d}x_J.$$

Repeated coordinate factors make a term zero; otherwise reorder into increasing order, keeping the sign of every interchange. The product of a k-form and an ℓ-form is a $(k+\ell)$-form. Scalar coefficients multiply in the usual way, but the order of one-form factors matters.

Example 13.8 (Two one-forms in three dimensions)**.** Let

$$\alpha = a_1 \, \mathrm{d}x + a_2 \, \mathrm{d}y + a_3 \, \mathrm{d}z, \qquad \beta = b_1 \, \mathrm{d}x + b_2 \, \mathrm{d}y + b_3 \, \mathrm{d}z.$$

Expanding the nine products, deleting repeated factors, and collecting gives

$$\begin{aligned} \alpha \wedge \beta = {} & (a_1 b_2 - a_2 b_1) \, \mathrm{d}x \wedge \mathrm{d}y \\ & + (a_2 b_3 - a_3 b_2) \, \mathrm{d}y \wedge \mathrm{d}z \\ & + (a_3 b_1 - a_1 b_3) \, \mathrm{d}z \wedge \mathrm{d}x. \end{aligned}$$

The last display uses the cyclic basis $\mathrm{d}x \wedge \mathrm{d}y$, $\mathrm{d}y \wedge \mathrm{d}z$, $\mathrm{d}z \wedge \mathrm{d}x$. Its coefficients are the third, first, and second components of $a \times b$, respectively. In the increasing basis, the coefficient of $\mathrm{d}x \wedge \mathrm{d}z$ is $a_1 b_3 - a_3 b_1$.

Example 13.9 (A one-form times a two-form)**.** For another calculation, let

$$\omega = P \, \mathrm{d}x + Q \, \mathrm{d}y + R \, \mathrm{d}z, \qquad \alpha = A \, \mathrm{d}x \wedge \mathrm{d}y + B \, \mathrm{d}z \wedge \mathrm{d}x + C \, \mathrm{d}y \wedge \mathrm{d}z.$$

Only three products survive:

$$\begin{aligned} \omega \wedge \alpha &= PC \, \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + QB \, \mathrm{d}y \wedge \mathrm{d}z \wedge \mathrm{d}x + RA \, \mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y \\ &= (PC + QB + RA) \, \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z. \end{aligned}$$

Both cyclic permutations have positive sign because each requires two interchanges.

Exercise 13.4 (the general sign rule)**.** Show that the wedge product is associative and that a k-form α and an ℓ-form β satisfy

$$\alpha \wedge \beta = (-1)^{k\ell} \beta \wedge \alpha.$$

Deduce that $\alpha \wedge \alpha = 0$ whenever k is odd.

Solution. It is enough to check coordinate wedges and then use distributivity. Associativity follows because either parenthesization concatenates the same ordered list. If the two lists share an index, both products are zero. Otherwise moving the k factors of the first list past the ℓ factors of the second takes $k\ell$ interchanges, giving the sign $(-1)^{k\ell}$. When k is odd and $\beta = \alpha$, the identity says $\alpha \wedge \alpha = -\alpha \wedge \alpha$; over $\mathbb{R}$ this forces zero. For even degrees the conclusion need not hold: on $\mathbb{R}^4$,

$$(\mathrm{d}x_1 \wedge \mathrm{d}x_2 + \mathrm{d}x_3 \wedge \mathrm{d}x_4)^2 = 2 \, \mathrm{d}x_1 \wedge \mathrm{d}x_2 \wedge \mathrm{d}x_3 \wedge \mathrm{d}x_4.$$

Here the square means wedge product with itself.

13.3.3 The Jacobian appears in a wedge product

Suppose $T(u,v) = (x(u,v), y(u,v))$ is C^1. Recall

$$\mathrm{d}x = x_u\ \mathrm{d}u + x_v\ \mathrm{d}v, \qquad \mathrm{d}y = y_u\ \mathrm{d}u + y_v\ \mathrm{d}v.$$

Using alternation,

$$\begin{aligned}\mathrm{d}x \wedge \mathrm{d}y &= x_u y_v\ \mathrm{d}u \wedge \mathrm{d}v + x_v y_u\ \mathrm{d}v \wedge \mathrm{d}u \\ &= (x_u y_v - x_v y_u)\ \mathrm{d}u \wedge \mathrm{d}v = \det DT\ \mathrm{d}u \wedge \mathrm{d}v.\end{aligned}$$

This determinant is also written $\partial(x,y)/\partial(u,v)$. In n dimensions the corresponding identity is

$$\mathrm{d}T_1 \wedge \cdots \wedge \mathrm{d}T_n = \det DT\ \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_n.$$

The determinant expansion follows by selecting one term from each differential; repeated parameter indices vanish, and the surviving terms are the signed permutations.

Recall the wedge-order identity

$$\int_R x^2\ \mathrm{d}y \wedge \mathrm{d}x = -\int_R x^2\ \mathrm{d}x \wedge \mathrm{d}y$$

for a fixed orientation of R.

Note 13.10. The sign from interchanging differential-form factors is distinct from changing the order of scalar iterated integration in Fubini's theorem.

Exercise 13.5 (finish the polar-coordinate differentials)**.** For $x = r\cos\theta$ and $y = r\sin\theta$, compute $\mathrm{d}x$, $\mathrm{d}y$, and $\mathrm{d}x \wedge \mathrm{d}y$.

Solution. We have

$$\mathrm{d}x = \cos\theta\ \mathrm{d}r - r\sin\theta\ \mathrm{d}\theta, \qquad \mathrm{d}y = \sin\theta\ \mathrm{d}r + r\cos\theta\ \mathrm{d}\theta.$$

Therefore

$$\begin{aligned}\mathrm{d}x \wedge \mathrm{d}y &= r\cos^2\theta\ \mathrm{d}r \wedge \mathrm{d}\theta - r\sin^2\theta\ \mathrm{d}\theta \wedge \mathrm{d}r \\ &= r\ \mathrm{d}r \wedge \mathrm{d}\theta.\end{aligned}$$

The sign depends on the order (r,θ); writing (θ,r) reverses it. For $r > 0$ this map has positive Jacobian r. At $r = 0$ the derivative loses rank, so it is not a regular surface coordinate chart there.

13.4 Exterior differentiation

13.4.1 Definition and coordinate calculations

Recall that the differential of a C^1 scalar function is

$$\mathrm{d}f = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}\ \mathrm{d}x_j.$$

Definition 13.11. If $\alpha = \sum_I a_I \, \mathrm{d}x_I$ is a C^1 k-form, its exterior derivative is the continuous $(k+1)$-form

$$\mathrm{d}\alpha = \sum_I \mathrm{d}a_I \wedge \mathrm{d}x_I = \sum_I \sum_{j=1}^{n} \frac{\partial a_I}{\partial x_j} \, \mathrm{d}x_j \wedge \mathrm{d}x_I.$$

In particular $\mathrm{d}(\mathrm{d}x_i) = 0$. Differentiation is linear: $\mathrm{d}(c\alpha + \beta) = c \, \mathrm{d}\alpha + \mathrm{d}\beta$ for constant c.

Example 13.12 (Curl)**.** For $\omega = P \, \mathrm{d}x + Q \, \mathrm{d}y + R \, \mathrm{d}z$,

$$\begin{aligned} \mathrm{d}\omega &= \mathrm{d}P \wedge \mathrm{d}x + \mathrm{d}Q \wedge \mathrm{d}y + \mathrm{d}R \wedge \mathrm{d}z \\ &= (Q_x - P_y) \, \mathrm{d}x \wedge \mathrm{d}y + (R_y - Q_z) \, \mathrm{d}y \wedge \mathrm{d}z + (P_z - R_x) \, \mathrm{d}z \wedge \mathrm{d}x. \end{aligned}$$

For instance $P_y \, \mathrm{d}y \wedge \mathrm{d}x = -P_y \, \mathrm{d}x \wedge \mathrm{d}y$, while $P_x \, \mathrm{d}x \wedge \mathrm{d}x = 0$. Under the identification

$$(A, B, C) \longleftrightarrow A \, \mathrm{d}y \wedge \mathrm{d}z + B \, \mathrm{d}z \wedge \mathrm{d}x + C \, \mathrm{d}x \wedge \mathrm{d}y,$$

$\mathrm{d}\omega$ corresponds to

$$\operatorname{curl}(P, Q, R) = (R_y - Q_z, \; P_z - R_x, \; Q_x - P_y).$$

Example 13.13 (Divergence)**.** For $\alpha = A \, \mathrm{d}y \wedge \mathrm{d}z + B \, \mathrm{d}z \wedge \mathrm{d}x + C \, \mathrm{d}x \wedge \mathrm{d}y$,

$$\begin{aligned} \mathrm{d}\alpha &= \mathrm{d}A \wedge \mathrm{d}y \wedge \mathrm{d}z + \mathrm{d}B \wedge \mathrm{d}z \wedge \mathrm{d}x + \mathrm{d}C \wedge \mathrm{d}x \wedge \mathrm{d}y \\ &= A_x \, \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + B_y \, \mathrm{d}y \wedge \mathrm{d}z \wedge \mathrm{d}x + C_z \, \mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y \\ &= (A_x + B_y + C_z) \, \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z. \end{aligned}$$

All other terms have repeated factors. This is the divergence of (A, B, C) multiplied by the standard volume form.

Exercise 13.6 (the product rule for exterior differentiation)**.** For a C^1 k-form α and a C^1 form β, prove

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^k \alpha \wedge \mathrm{d}\beta.$$

Solution. By linearity, take $\alpha = a \, \mathrm{d}x_I$ and $\beta = b \, \mathrm{d}x_J$, with I of length k. The ordinary product rule gives

$$\mathrm{d}(\alpha \wedge \beta) = b \, \mathrm{d}a \wedge \mathrm{d}x_I \wedge \mathrm{d}x_J + a \, \mathrm{d}b \wedge \mathrm{d}x_I \wedge \mathrm{d}x_J.$$

The first term is $\mathrm{d}\alpha \wedge \beta$. Moving the one-form $\mathrm{d}b$ past the k factors in $\mathrm{d}x_I$ introduces $(-1)^k$, yielding the second term in the claimed formula. Summing proves the result.

13.4.2 Why $\mathrm{d}^2 = 0$

Proposition 13.14. *If α is a C^2 differential form on an open subset of $\mathbb{R}^n$, then*

$$\mathrm{d}(\mathrm{d}\alpha) = 0.$$

Proof. By linearity it is enough to take $\alpha = f \, \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}$. Then

$$\mathrm{d}(\mathrm{d}\alpha) = \sum_{r=1}^{n} \sum_{s=1}^{n} \frac{\partial^2 f}{\partial x_s \partial x_r} \, \mathrm{d}x_s \wedge \mathrm{d}x_r \wedge \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}.$$

If $s = r$, the wedge has a repeated factor and is zero. If $s \neq r$, pair the (r, s) term with the (s, r) term. Their coefficients agree because f is C^2, and their wedge factors have opposite signs. Thus they cancel. Every summand belongs to one of these cases.

Note 13.15. The classical calculation of d^2 requires C^2 coefficients. For a C^1 form, the second classical derivative need not exist. Forms of degree greater than the ambient dimension vanish algebraically.

Exercise 13.7 (the vector-calculus identities)**.** Translate $\mathrm{d}^2 = 0$ into identities for gradient, curl, and divergence in $\mathbb{R}^3$, with sufficient differentiability assumptions.

Solution. For $f \in C^2$, identify $\mathrm{d}f = f_x\,\mathrm{d}x + f_y\,\mathrm{d}y + f_z\,\mathrm{d}z$ with ∇f. The identity $\mathrm{d}(\mathrm{d}f) = 0$ gives

$$\operatorname{curl}(\nabla f) = 0.$$

For a C^2 vector field $F = (P, Q, R)$, identify it first with the one-form $P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z$. Its exterior derivative is the flux two-form of $\operatorname{curl} F$. Differentiating again gives

$$\operatorname{div}(\operatorname{curl} F) = 0.$$

For example, the coefficient in the second calculation is

$$\partial_x(R_y - Q_z) + \partial_y(P_z - R_x) + \partial_z(Q_x - P_y) = 0,$$

by equality of the corresponding mixed partial derivatives. No converse is asserted here: knowing a derivative vanishes does not by itself produce a globally defined potential.

Chapter 14

Pullbacks, surface integrals, and orientation

14.1 Pulling a form back to parameter space

14.1.1 The pullback operation

Suppose $\Phi : U \subset \mathbb{R}^k \to V \subset \mathbb{R}^n$ is C^1. The substitution rule is

$$x_i = \Phi_i(u_1, \ldots, u_k), \qquad \mathrm{d}x_i = \sum_{j=1}^{k} \frac{\partial \Phi_i}{\partial u_j}\, \mathrm{d}u_j.$$

Coefficients must also be evaluated at $\Phi(u)$. This operation is the pullback [2].

Definition 14.1. For an ℓ-form α on V, define the ℓ-form $\Phi^*\alpha$ on U by

$$(\Phi^*\alpha)_u(v_1, \ldots, v_\ell) = \alpha_{\Phi(u)}(D\Phi(u)v_1, \ldots, D\Phi(u)v_\ell).$$

For a scalar function f, $\Phi^* f = f \circ \Phi$.

Thus

$$\Phi^*(f\, \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_\ell}) = (f \circ \Phi)\, \mathrm{d}\Phi_{i_1} \wedge \cdots \wedge \mathrm{d}\Phi_{i_\ell}.$$

The pullback is defined for any C^1 map; injectivity and maximal rank are additional conditions for a regular parameterization, not conditions for this algebraic definition.

Example 14.2 (The coefficient is a minor)**.** If $\Phi(u, v) = (\Phi_1, \Phi_2, \Phi_3)$ and $\alpha = f\, \mathrm{d}y \wedge \mathrm{d}z$, we have

$$\begin{aligned} \Phi^*\alpha &= (f \circ \Phi)\, \mathrm{d}\Phi_2 \wedge \mathrm{d}\Phi_3 \\ &= (f \circ \Phi) \left(\frac{\partial \Phi_2}{\partial u}\frac{\partial \Phi_3}{\partial v} - \frac{\partial \Phi_2}{\partial v}\frac{\partial \Phi_3}{\partial u} \right) \mathrm{d}u \wedge \mathrm{d}v. \end{aligned}$$

This is precisely $\alpha_{\Phi(u,v)}(\Phi_u, \Phi_v)\, \mathrm{d}u \wedge \mathrm{d}v$.

Exercise 14.1 (the pullback rules used in integration)**.** Verify composition, wedge compatibility, and compatibility with exterior differentiation:

$$(\Phi \circ \Psi)^* = \Psi^*\Phi^*, \qquad \Phi^*(\alpha \wedge \beta) = \Phi^*\alpha \wedge \Phi^*\beta, \qquad \mathrm{d}(\Phi^*\alpha) = \Phi^*(\mathrm{d}\alpha).$$

Use C^1 maps for the first two identities and a C^2 map and C^1 form for the last.

Solution. The first identity follows from $D(\Phi \circ \Psi) = D\Phi\, D\Psi$ in the defining evaluation on vectors; on coefficients it is ordinary composition. For the second, pull back coordinate differentials by $\mathrm{d}x_i \mapsto \mathrm{d}\Phi_i$. Both sides concatenate the same differential factors, with the same coefficients evaluated at Φ. Distributivity proves it for all forms.

For the third it is enough to take $\alpha = f\ \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_\ell}$. Since Φ is C^2, $\mathrm{d}(\mathrm{d}\Phi_i) = 0$. Apply the exterior product rule and the ordinary chain rule:

$$\begin{aligned}\mathrm{d}(\Phi^*\alpha) &= \mathrm{d}(f \circ \Phi) \wedge \mathrm{d}\Phi_{i_1} \wedge \cdots \wedge \mathrm{d}\Phi_{i_\ell} \\ &= \sum_j (f_{x_j} \circ \Phi)\ \mathrm{d}\Phi_j \wedge \mathrm{d}\Phi_{i_1} \wedge \cdots \wedge \mathrm{d}\Phi_{i_\ell} \\ &= \Phi^*(\mathrm{d}\alpha).\end{aligned}$$

The omitted product-rule terms are zero individually because they contain $\mathrm{d}(\mathrm{d}\Phi_i)$. Thus no additional smoothness of the coefficient f beyond C^1 is needed.

14.2 Integration over a parameterized patch

14.2.1 Cells, patches, and the definition of the integral

Definition 14.3. A regular C^1 k-cell in $\mathbb{R}^n$ is the image of $I^k = [0,1]^k$ under a one-to-one C^1 map $\Phi : U \to \mathbb{R}^n$, where $U \subset \mathbb{R}^k$ is open, contains I^k, and $D\Phi$ has rank k on I^k. The restriction of Φ to I^k is a parameterization of the cell.

More generally, consider a k-dimensional surface patch $\Phi : Q \to \mathbb{R}^n$, with Q a k-dimensional interval. Unless explicitly allowing a singular parameterization, we take the map to have rank k. Injectivity is needed when the goal is to count each point of the geometric patch once.

Definition 14.4. Let $Q \subset \mathbb{R}^k$ be a compact rectangle with its standard coordinate orientation, let Φ be C^1 on a neighborhood of Q, and let α be a continuous k-form near $\Phi(Q)$. Define

$$\begin{aligned}\int_\Phi \alpha &= \int_Q \alpha_{\Phi(u)}(D\Phi(u)e_1, \ldots, D\Phi(u)e_k)\ \mathrm{d}u_1 \cdots \mathrm{d}u_k \\ &= \int_Q \Phi^*\alpha.\end{aligned}$$

In the second expression, if $\Phi^*\alpha = g\ \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_k$, the integral means the ordinary integral of g over Q.

The integrand is a continuous real-valued function. There is no extra area factor after evaluating the form on all the parameter derivatives; that evaluation already contains the relevant determinants.

For $k = 1$, a curve $c : [a,b] \to \mathbb{R}^n$ and a one-form $\omega = \sum_i a_i\ \mathrm{d}x_i$ give the familiar formula

$$\int_c \omega = \int_a^b \omega_{c(t)}(c'(t))\ \mathrm{d}t = \int_a^b \sum_i a_i(c(t))c_i'(t)\ \mathrm{d}t.$$

14.2.2 A polynomial patch

Example 14.5. Let $Q = [0,1] \times [0,1]$ and

$$\Phi(u,v) = (u+v, u-v, uv), \qquad \alpha = x\, \mathrm{d}y \wedge \mathrm{d}z + y\, \mathrm{d}x \wedge \mathrm{d}z.$$

Substitute

$$\mathrm{d}x = \mathrm{d}u + \mathrm{d}v, \qquad \mathrm{d}y = \mathrm{d}u - \mathrm{d}v, \qquad \mathrm{d}z = v\, \mathrm{d}u + u\, \mathrm{d}v.$$

Then

$$\begin{aligned} \Phi^*\alpha &= (u+v)(\mathrm{d}u - \mathrm{d}v) \wedge (v\, \mathrm{d}u + u\, \mathrm{d}v) \\ &\quad + (u-v)(\mathrm{d}u + \mathrm{d}v) \wedge (v\, \mathrm{d}u + u\, \mathrm{d}v) \\ &= \left((u+v)^2 + (u-v)^2\right)\, \mathrm{d}u \wedge \mathrm{d}v \\ &= 2(u^2+v^2)\, \mathrm{d}u \wedge \mathrm{d}v. \end{aligned}$$

Exercise 14.2 (finish the patch integral in two ways)**.** Evaluate $\int_\Phi \alpha$, and check the pullback using the defining evaluation on Φ_u, Φ_v.

Solution. The first method integrates the calculated coefficient:

$$\int_\Phi \alpha = 2\int_0^1 \int_0^1 (u^2+v^2)\, \mathrm{d}u\, \mathrm{d}v = 2\left(\frac{1}{3} + \frac{1}{3}\right) = \frac{4}{3}.$$

For the second method,

$$\Phi_u = (1,1,v), \qquad \Phi_v = (1,-1,u).$$

Thus

$$(\mathrm{d}y \wedge \mathrm{d}z)(\Phi_u, \Phi_v) = u+v, \qquad (\mathrm{d}x \wedge \mathrm{d}z)(\Phi_u, \Phi_v) = u - v.$$

Multiplying by $x = u+v$ and $y = u-v$ gives the same sum of squares. Moreover,

$$\Phi_u \times \Phi_v = (u+v, v-u, -2) \neq 0,$$

so the patch is regular even at $(0,0)$. The first two coordinates determine $u = (x+y)/2$ and $v = (x-y)/2$, so it is also injective.

14.3 Orientation and independence of parameterization

14.3.1 Which changes of parameters preserve an integral?

Parameterization invariance must account for orientation. Two regular one-to-one parameterizations of the same oriented surface give the same integral when they preserve its orientation, and opposite integrals when their orientations disagree.

Definition 14.6. For $k \geq 1$, an orientation of a k-dimensional tangent space is a choice of equivalence class of ordered bases, where two bases are equivalent when their change-of-basis matrix has positive determinant. An oriented manifold has such choices varying consistently from chart to chart. A regular parameterization is positive when its ordered parameter derivatives represent the chosen orientation. In dimension zero, an orientation assigns a sign to each point; integration over a finite oriented set of points is the corresponding signed sum.

Changing to a basis with negative determinant reverses the orientation. A consistent orientation need not exist on every manifold. On an oriented surface in $\mathbb{R}^3$, the ordered pair (Φ_u, Φ_v) corresponds to the normal direction $\Phi_u \times \Phi_v$.

Exercise 14.3 (prove parameterization invariance with the sign)**.** Let $h : \widetilde{Q} \to Q$ be a C^1 diffeomorphism of connected parameter domains, extending with the needed regularity to their boundaries, and let $\widetilde{\Phi} = \Phi \circ h$. Show

$$\int_{\widetilde{\Phi}} \alpha = \begin{cases} \int_\Phi \alpha, & \det Dh > 0, \\ -\int_\Phi \alpha, & \det Dh < 0. \end{cases}$$

Solution. Write $\Phi^*\alpha = g(u)\ \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_k$. Composition of pullbacks and the determinant formula give

$$\widetilde{\Phi}^*\alpha = g(h(v)) \det Dh(v)\ \mathrm{d}v_1 \wedge \cdots \wedge \mathrm{d}v_k.$$

Because h is a diffeomorphism, its determinant never vanishes. Continuity and connectedness make its sign constant. The scalar change-of-variables theorem uses $|\det Dh|$; factoring out the sign proves both cases. For disconnected domains the argument applies on each component.

Note 14.7. Parameterization invariance up to sign assumes regular one-to-one parameterizations related by a change of coordinates. Repeated coverings count multiplicity. Integrals over an oriented manifold combine local parameterizations by subdivision or a partition of unity, so overlaps are counted correctly.

14.3.2 Area and flux are different integrals

For a regular surface patch $\Phi(u, v)$ in $\mathbb{R}^3$, the scalar area element is

$$\mathrm{d}S = \|\Phi_u \times \Phi_v\|\ \mathrm{d}u\ \mathrm{d}v.$$

For an oriented patch, set

$$n = \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}.$$

A two-form

$$\alpha = A\ \mathrm{d}y \wedge \mathrm{d}z + B\ \mathrm{d}z \wedge \mathrm{d}x + C\ \mathrm{d}x \wedge \mathrm{d}y$$

corresponds to $F = (A, B, C)$, and its pullback satisfies

$$\alpha(\Phi_u, \Phi_v) = F(\Phi) \cdot (\Phi_u \times \Phi_v).$$

Thus $\int_\Phi \alpha$ is the flux integral $\int_S F \cdot n\ \mathrm{d}S$ with the indicated orientation. In contrast, a scalar integral $\int_S f\ \mathrm{d}S$ uses the magnitude of the cross product and does not change when the orientation reverses.

Exercise 14.4 (justify the area and flux factors)**.** Explain these formulas from the infinitesimal parallelogram determined by Φ_u and Φ_v. Give the corresponding area factor for a regular k-patch.

Solution. The parallelogram spanned by $a, b \in \mathbb{R}^3$ has area $\|a \times b\|$. Its squared area is

$$\|a\|^2\|b\|^2 - (a \cdot b)^2 = \det \begin{pmatrix} a \cdot a & a \cdot b \\ b \cdot a & b \cdot b \end{pmatrix}.$$

Linearizing Φ sends a small parameter rectangle to the parallelogram spanned by $\Phi_u\,\Delta u$ and $\Phi_v\,\Delta v$. Dividing its area by $\Delta u\Delta v$ yields $\|\Phi_u \times \Phi_v\|$. The three components of $\Phi_u \times \Phi_v$ are exactly the three minors evaluating $\mathrm{d}y \wedge \mathrm{d}z$, $\mathrm{d}z \wedge \mathrm{d}x$, and $\mathrm{d}x \wedge \mathrm{d}y$. This proves the flux identity directly by expansion.

For a k-patch, the corresponding parallelepiped volume is

$$J_k\Phi = \sqrt{\det((D\Phi)^T D\Phi)}.$$

Indeed, choose an orthonormal basis of the image of $D\Phi$, and write $D\Phi = ER$, where $E^TE = I_k$ and R is a nonsingular $k \times k$ matrix. In that orthonormal image space the volume is $|\det R|$, while $(D\Phi)^T D\Phi = R^TR$ has determinant $(\det R)^2$. For $k = 2, n = 3$ this is the preceding cross-product factor.

14.3.3 The upper-hemisphere example

Example 14.8. Let $\alpha = z\,\mathrm{d}x \wedge \mathrm{d}y$ on $\mathbb{R}^3$, and let S_+ be the unit upper hemisphere. Parameterize it by

$$x = \sin\phi\cos\theta, \qquad y = \sin\phi\sin\theta, \qquad z = \cos\phi,$$

with $0 \le \phi \le \pi/2$ and $0 \le \theta \le 2\pi$. We use parameter order (ϕ, θ).

Exercise 14.5 (complete the hemisphere differentials and integral)**.** Compute the coordinate differentials and evaluate the integral with the outward orientation.

Solution. We have

$$\begin{aligned}
\mathrm{d}x &= \cos\phi\cos\theta\ \mathrm{d}\phi - \sin\phi\sin\theta\ \mathrm{d}\theta,\\
\mathrm{d}y &= \cos\phi\sin\theta\ \mathrm{d}\phi + \sin\phi\cos\theta\ \mathrm{d}\theta,\\
\mathrm{d}z &= -\sin\phi\ \mathrm{d}\phi.
\end{aligned}$$

Hence

$$\mathrm{d}x \wedge \mathrm{d}y = \cos\phi\sin\phi\ \mathrm{d}\phi \wedge \mathrm{d}\theta, \qquad \Phi^*\alpha = \cos^2\phi\sin\phi\ \mathrm{d}\phi \wedge \mathrm{d}\theta.$$

For $0 < \phi < \pi/2$,

$$\Phi_\phi \times \Phi_\theta = \sin\phi(\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi) = \sin\phi\,\Phi,$$

which points outward. Therefore

$$\int_{S_+} z\ \mathrm{d}x \wedge \mathrm{d}y = \int_0^{2\pi}\int_0^{\pi/2} \cos^2\phi\sin\phi\ \mathrm{d}\phi\ \mathrm{d}\theta = 2\pi\left[-\frac{\cos^3\phi}{3}\right]_0^{\pi/2} = \frac{2\pi}{3}.$$

The formula degenerates at the pole and identifies the meridian edges $\theta = 0, 2\pi$; it is not a single injective regular cell on the closed rectangle. One may restrict to interior angular patches and take limits. The omitted pole and meridian have zero surface area, and the continuous coefficient above has a well-defined limiting integral. This justifies using the usual spherical-coordinate integral for the whole hemisphere.

Chapter 15

Boundary orientation and generalized Stokes

15.1 Flux and the boundary convention

15.1.1 The planar divergence calculation

Suppose $D \subset \mathbb{R}^2$ is a bounded region with piecewise C^1 boundary, positively oriented so that D lies to the left as its boundary is traversed. Let $F = (F_1, F_2)$ be C^1 on a neighborhood of $\overline{D}$. Recall

$$\operatorname{div} F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}.$$

For a regular boundary curve $c(t) = (x(t), y(t))$, the unit tangent and outward unit normal are

$$T = \frac{(x', y')}{\|c'\|}, \qquad N = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} T = \frac{(y', -x')}{\|c'\|}.$$

The normal is a clockwise quarter-turn of the positively directed tangent. Therefore the flux through the boundary is

$$\begin{aligned} \int_{\partial D} F \cdot N \, \mathrm{d}s &= \int (F_1(c(t))y'(t) - F_2(c(t))x'(t)) \, \mathrm{d}t \\ &= \int_{\partial D} F_1 \, \mathrm{d}y - F_2 \, \mathrm{d}x \\ &= \int_D \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) \mathrm{d}x \, \mathrm{d}y. \end{aligned}$$

The last equality is Green's theorem applied to $-F_2 \, \mathrm{d}x + F_1 \, \mathrm{d}y$. It measures the total outward flux.

Note 15.1. On an inner boundary component the positive direction is clockwise. The same normal formula points into the hole, outward from the region.

15.1.2 Manifolds with boundary

Definition 15.2. For $k \geq 1$, a C^2 embedded k-manifold with boundary is locally parameterized by a homeomorphism $\Phi : O \to M \cap W$, where W is ambient open and O is relatively open in the half-space

$$H^k = \{u \in \mathbb{R}^k : u_k \geq 0\}.$$

The chart has a C^2 extension across $u_k = 0$ with derivative of rank k, and the changes of coordinates are compatible C^2 maps. Boundary points correspond to $u_k = 0$; interior points correspond to $u_k > 0$. The tangent space, including at a boundary point $p = \Phi(u)$, is

$$T_pM = D\Phi(u)(\mathbb{R}^k).$$

At a boundary point,

$$T_p\partial M = D\Phi(u)(\mathbb{R}^{k-1} \times \{0\}).$$

A transverse tangent vector is outward-pointing when its preimage under $D\Phi(u)$ has negative last coordinate. These definitions agree on compatible chart overlaps.

For example, $T_0[0,1] = \mathbb{R}$. Velocities of two-sided curves remaining in $[0,1]$ through 0 are all zero, so that earlier velocity description for manifolds without boundary does not define the full tangent space at a boundary point. The half-space chart gives the needed tangent space and outward direction.

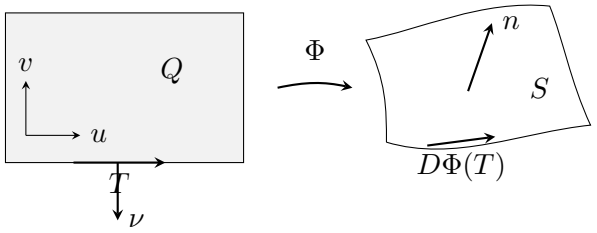
15.1.3 Outward normal first

Definition 15.3. Let M be an oriented k-manifold with boundary, $k \geq 2$. The induced orientation on ∂M is determined by the following rule: an ordered basis $(v_1, \ldots, v_{k-1})$ of $T_p\partial M$ is positive if

$$(\nu, v_1, \ldots, v_{k-1})$$

is positive in T_pM, where ν is an outward-pointing vector tangent to M and transverse to ∂M. This is the *outward-normal-first* convention. When $k = 1$, the induced boundary signs are positive at a terminal endpoint and negative at an initial endpoint, relative to the chosen orientation of the interval.

Here "outward" concerns the boundary inside M. For a surface in $\mathbb{R}^3$, ν is an outward conormal tangent to the surface; it differs from the ambient surface normal n. If T is the induced boundary tangent, the convention says that $\nu \times T$ points along n.



On the bottom edge of a positively oriented (u, v) rectangle, $\nu = -e_v$ and $T = e_u$. The ordered pair $(-e_v, e_u)$ has positive determinant, so the boundary goes to the right there and counterclockwise around the rectangle. A positive surface parameterization carries this boundary direction to the surface.

Exercise 15.1 (endpoint and face signs)**.** Determine the induced orientations of an interval and of the faces of the oriented cube $[0,1]^k$.

Solution. An interval $[a, b]$ oriented by increasing coordinate has boundary $\{b\} - \{a\}$. At b the outward vector is positive, and at a it is negative. An oriented zero-dimensional integral therefore sums the endpoint values with signs $+1$ and -1.

On the face $u_i = 1$ of a cube the outward vector is e_i. Prepending e_i to the ordered list

$$(e_1, \dots, \widehat{e_i}, \dots, e_k)$$

has determinant $(-1)^{i-1}$. Thus the induced face orientation is $(-1)^{i-1}$ times the listed coordinate orientation. On $u_i = 0$, the outward vector is $-e_i$, and the sign is $(-1)^i$. The hat means that the indicated vector is omitted. In two dimensions these signs give counterclockwise traversal of the outer boundary.

15.2 Stokes' theorem

15.2.1 Statement, hypotheses, and the bounding issue

Theorem 15.4 (Generalized Stokes)**.** *Let M be a compact oriented smooth k-manifold with boundary, $k \geq 1$, embedded in $\mathbb{R}^n$. Let ω be a C^1 $(k-1)$-form on an open neighborhood of M. With the induced outward-normal-first orientation on ∂M,*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

It suffices for the manifold and its charts to be C^2. If $\partial M = \varnothing$, the right-hand side is zero. Piecewise smooth domains with compatible face orientations are also covered by applying the theorem to pieces; shared faces cancel [2, Chs. 6–7].

The integral on M is of a k-form, and the integral on its boundary is of a $(k-1)$-form. For a ball with its boundary sphere, the ball is the domain of integration on the left and its sphere is the boundary on the right. The degree of the form must match the dimension in each integral.

Note 15.5. Having a boundary and being the boundary of another manifold are distinct properties. Stokes' theorem on M does not require that M bound a larger manifold. For the unit sphere, the unit ball supplies an explicit bounding manifold.

15.2.2 The fundamental local calculation

Exercise 15.2 (prove Stokes on a rectangle)**.** Let $Q = \prod_{i=1}^{k} [\ell_i, r_i]$ have its standard orientation. Write a C^1 $(k-1)$-form as

$$\eta = \sum_{i=1}^{k} (-1)^{i-1} a_i(u)\, \mathrm{d}u_1 \wedge \cdots \wedge \widehat{\mathrm{d}u_i} \wedge \cdots \wedge \mathrm{d}u_k.$$

Prove $\int_Q \mathrm{d}\eta = \int_{\partial Q} \eta$ using the fundamental theorem of calculus and the face signs above.

Solution. Only differentiation with respect to the missing coordinate survives:

$$\mathrm{d}\eta = \sum_{i=1}^{k} \partial_i a_i\, \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_k.$$

Indeed, moving $\mathrm{d}u_i$ from the front to its ith place introduces $(-1)^{i-1}$, canceling the prefactor. Integrating each term first in u_i gives

$$\int_Q \mathrm{d}\eta = \sum_{i=1}^{k} \int_{Q_i} \big(a_i(u_1, \dots, r_i, \dots, u_k) - a_i(u_1, \dots, \ell_i, \dots, u_k)\big)\, \mathrm{d}u_{\widehat{i}},$$

where Q_i is the product of the other coordinate intervals and $\mathrm{d}u_{\widehat{i}}$ is their ordinary volume measure. On a face with u_i fixed, every summand of η except the ith pulls back to zero, because it contains $\mathrm{d}u_i$. On the upper face, the coefficient sign $(-1)^{i-1}$ and the induced orientation sign $(-1)^{i-1}$ multiply to $+1$. On the lower face the product is $(-1)^{i-1}(-1)^i = -1$. Thus the boundary integral is exactly the same sum of upper-minus-lower coefficient integrals. Fubini and the one-dimensional fundamental theorem justify the preceding integrations for C^1 coefficients on a compact rectangle.

15.2.3 A complete local-to-global proof

Exercise 15.3 (prove generalized Stokes)**.** Prove the theorem. Reduce to compactly supported forms in interior or boundary coordinates and justify that these local reductions can be summed.

Solution. Use manifold charts with their orientation signs $\sigma = +1$ or -1 recorded. At an interior point they have domains open in $\mathbb{R}^k$; at a boundary point use domains relatively open in

$$H = \{u_k \geq 0\} \subset \mathbb{R}^k.$$

Charts are C^2, so pullback commutes with exterior differentiation for C^1 forms. Both the manifold integral and its induced boundary integral are σ times their standard-coordinate integrals. We first prove the coordinate identity, then multiply both sides by σ.

First consider an interior chart and a $(k-1)$-form η compactly supported inside its parameter domain. Extend its coefficient functions by zero to $\mathbb{R}^k$; this is a C^1 extension because the support is separated from the edges of that domain. Write

$$\eta = \sum_i (-1)^{i-1} b_i \, \mathrm{d}u_1 \wedge \cdots \wedge \widehat{\mathrm{d}u_i} \wedge \cdots \wedge \mathrm{d}u_k.$$

Then $\mathrm{d}\eta = (\sum_i \partial_i b_i) \, \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_k$. Each derivative has zero integral on $\mathbb{R}^k$: integrate first in its differentiated coordinate over an interval outside the compact support, obtaining zero endpoint values. There is no manifold boundary in an interior chart, so the desired local identity follows.

Next consider a boundary chart and a compactly supported form relative to its domain in H. Extend by zero within H, away from the chart domain, keeping its possible values on $u_k = 0$. For $i < k$ the same endpoint calculation gives zero. For $i = k$,

$$\int_H \partial_k b_k \, \mathrm{d}u = -\int_{\mathbb{R}^{k-1}} b_k(u_1, \ldots, u_{k-1}, 0) \, \mathrm{d}u_1 \cdots \mathrm{d}u_{k-1}.$$

The outward vector of H on its boundary is $-e_k$. The orientation it induces on the coordinate frame $(e_1, \ldots, e_{k-1})$ has sign $(-1)^k$. The restriction of η to this boundary is

$$(-1)^{k-1} b_k(u', 0) \, \mathrm{d}u_1 \wedge \cdots \wedge \mathrm{d}u_{k-1}.$$

The coefficient and orientation signs multiply to $(-1)^{k-1}(-1)^k = -1$, giving exactly the negative integral above. Multiply the coordinate identity by σ to obtain Stokes in the given orientation. For $k = 1$, the half-line's boundary point is negative. At a terminal endpoint, $\Phi(u) = b - u$ has $\sigma = -1$, turning $-\omega(b)$ into $+\omega(b)$. A positive chart at the initial endpoint gives $-\omega(a)$. Thus the signed calculation covers both endpoints.

We now justify localization rather than simply assuming that overlapping chart integrals may be added. Take finitely many relatively open chart neighborhoods W_j covering M, using compactness.

Refine them by finitely many ambient balls B_j such that the closures of the doubled balls meet M only inside an assigned chart and the smaller balls still cover M. This is possible by first taking sufficiently small balls at each point and then taking a finite subcover. Choose smooth ambient bump functions $\rho_j \geq 0$ supported in the doubled balls, positive on the corresponding smaller balls. For example, translated and rescaled versions of

$$\rho(x) = \begin{cases} \exp(-1/(1-\|x\|^2)), & \|x\| < 1, \\ 0, & \|x\| \geq 1 \end{cases}$$

provide such functions. Their sum $s = \sum_j \rho_j$ is positive on a neighborhood of M. There define $\lambda_j = \rho_j/s$. On M,

$$\sum_j \lambda_j = 1, \qquad \sum_j \mathrm{d}\lambda_j = 0,$$

and each $\lambda_j|_M$ has compact support inside its assigned chart. The same holds after restriction to ∂M.

Apply the proven chart calculation to the pullback of $\lambda_j\omega$. Pullback compatibility gives

$$\int_M \mathrm{d}(\lambda_j\omega) = \int_{\partial M} \lambda_j\omega.$$

These integrals have support in a single chart, where their definitions are the parameter integrals already used. Sum over j. The ordinary exterior product rule gives

$$\sum_j \mathrm{d}(\lambda_j\omega) = \sum_j \mathrm{d}\lambda_j \wedge \omega + \sum_j \lambda_j \, \mathrm{d}\omega = \mathrm{d}\omega,$$
$$\sum_j (\lambda_j\omega)|_{\partial M} = \omega|_{\partial M}.$$

Finite additivity and the change-of-parameter rule, with the recorded chart orientation signs, now yield $\int_M \mathrm{d}\omega = \int_{\partial M} \omega$.

For completeness, let θ be a top-degree form on an oriented manifold N. Its integral defined by these local weights does not depend on the chosen weights. If $\{\mu_\ell\}$ is another finite such family, insert $1 = \sum_\ell \mu_\ell$ into every weighted integral of θ. The resulting terms $\lambda_j\mu_\ell\theta$ are supported in chart overlaps; change of coordinates gives the same integral in either chart. Summing in the other order recovers the construction using μ_ℓ. Apply this argument on $N = M$ to $\theta = \mathrm{d}\omega$, and on $N = \partial M$ to the restriction of ω. Thus both integrals in Stokes' theorem are independent of the chosen cover and weights.

15.3 Cells, chains, and cancellation of boundaries

Recall that a parameterization records more than its image. In particular, it records orientation and multiplicity. To add parameterized pieces, we use chains [2, Sec. 6.1].

Definition 15.6. A smooth parameterized k-cell in a smooth manifold M is a smooth map $\sigma : [0,1]^k \to M$, smooth up to the faces. A k-chain is a finite formal sum

$$C = \sum_{j=1}^{N} n_j\sigma_j, \qquad n_j \in \mathbb{Z}.$$

The cube has its standard ordered-coordinate orientation. A minus sign reverses the sign of integration. A 0-cell is a point, and $\int_p f = f(p)$ for a 0-form f. Integration is extended by linearity:

$$\int_C \omega = \sum_j n_j \int_{[0,1]^k} \sigma_j^* \omega.$$

The parameter maps need not be injective or regular. Two maps with the same image are not thereby the same cell.

Let $\delta_i^\epsilon : [0,1]^{k-1} \to [0,1]^k$ insert $\epsilon \in \{0,1\}$ in the ith position, leaving the remaining coordinates in their original order. Define

$$\partial\sigma = \sum_{i=1}^{k} (-1)^{i-1} (\sigma \circ \delta_i^1 - \sigma \circ \delta_i^0), \qquad \partial C = \sum_j n_j \partial\sigma_j.$$

The boundary of a 0-chain is zero. For a path, $\partial\sigma = \sigma(1) - \sigma(0)$. For a square, the four terms give counterclockwise traversal of its boundary.

Exercise 15.4 (the boundary of a boundary)**.** Prove that $\partial^2 C = 0$. Deduce the chain version of Stokes' theorem,

$$\int_{\partial C} \omega = \int_C \mathrm{d}\omega,$$

for smooth k-chains and C^1 $(k-1)$-forms defined near their images.

Solution. By linearity, it suffices to consider one k-cell. A codimension-two face fixes two coordinates, say $i < j$, each to a specified endpoint. If coordinate i is fixed first, the original coordinate j has index $j-1$ in the remaining face. Its two orientation factors therefore have product

$$(-1)^{i-1}(-1)^{j-2} = (-1)^{i+j-3}.$$

If coordinate j is fixed first, coordinate i retains its index, giving

$$(-1)^{j-1}(-1)^{i-1} = (-1)^{i+j-2}.$$

The endpoint factors are the same in both orders, and the resulting parameter maps agree. Thus each codimension-two face occurs twice with opposite signs. This proves $\partial^2\sigma = 0$, including the immediate cases $k = 0, 1$.

Apply the rectangle version of Stokes to $\sigma^*\omega$ on $[0,1]^k$. Its boundary terms are exactly the terms in $\partial\sigma$, and naturality gives $\mathrm{d}(\sigma^*\omega) = \sigma^*(\mathrm{d}\omega)$. Hence

$$\int_{\partial\sigma} \omega = \int_{[0,1]^k} \mathrm{d}(\sigma^*\omega) = \int_\sigma \mathrm{d}\omega.$$

No inverse parameter map is used, so regularity and injectivity are unnecessary. Sum over the finitely many cells to obtain the chain formula.

Definition 15.7. A chain C is a *cycle* if $\partial C = 0$. It is a *boundary* if $C = \partial B$ for a chain B one dimension higher. Every boundary is a cycle, since $\partial^2 = 0$. The converse depends on the space in which the chains are allowed to lie.

15.4 Classical consequences and the final sphere example

15.4.1 The common formula behind the classical theorems

Exercise 15.5 (recover the familiar integral theorems)**.** Derive the fundamental theorem for curves, Green's theorem, the classical surface Stokes theorem, and the divergence theorem from generalized Stokes. State the orientation in each case.

Solution. For an oriented interval and a C^1 function, Stokes gives $\int_a^b \mathrm{d}f = f(b) - f(a)$. Pulling back a C^1 function along a C^1 curve c gives

$$\int_c \mathrm{d}f = f(c(b)) - f(c(a)),$$

by the ordinary chain rule. The endpoints are signed terminal minus initial.

For a planar region D oriented by $\mathrm{d}x \wedge \mathrm{d}y$ and $\omega = P\ \mathrm{d}x + Q\ \mathrm{d}y$,

$$\int_{\partial D} P\ \mathrm{d}x + Q\ \mathrm{d}y = \int_D (Q_x - P_y)\ \mathrm{d}x\ \mathrm{d}y.$$

Its induced boundary is counterclockwise on the outer boundary and clockwise around holes. Taking $P = -F_2$, $Q = F_1$ gives the planar flux formula above.

For an oriented compact surface $S \subset \mathbb{R}^3$ and the one-form corresponding to a C^1 field F,

$$\int_{\partial S} F \cdot T\ \mathrm{d}s = \int_S (\operatorname{curl} F) \cdot n\ \mathrm{d}S.$$

This follows from the coordinate formula for $\mathrm{d}\omega$ and the flux identification. The induced tangent satisfies $\nu \times T = n$, with ν the outward unit conormal.

For a compact smooth region $B \subset \mathbb{R}^3$ with standard volume orientation and a C^1 field $F = (A, B_2, C)$, use the two-form

$$\alpha = A\ \mathrm{d}y \wedge \mathrm{d}z + B_2\ \mathrm{d}z \wedge \mathrm{d}x + C\ \mathrm{d}x \wedge \mathrm{d}y.$$

The notation B_2 keeps the second field component distinct from the region B. Then

$$\int_{\partial B} F \cdot n\ \mathrm{d}S = \int_B \operatorname{div} F\ \mathrm{d}x\ \mathrm{d}y\ \mathrm{d}z,$$

where n is the outward normal. The two-form derivative calculation establishes the integrand, and outward-normal-first establishes the sign. Smoothness can be replaced by the piecewise regular hypotheses under which the pieces and their boundary integrals are well defined.

15.4.2 Maxwell's equations in differential-form notation

Exercise 15.6 (four vector equations and two form equations)**.** Let E, B, J be smooth time-dependent vector fields on an open subset of $\mathbb{R}_t \times \mathbb{R}^3$, and let ρ be a smooth scalar function. Use Gaussian units with the speed of light set equal to 1, as in Bachman [2, Sec. 6.4]. Write

$$\mathrm{vol}_3 = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z, \qquad \mathcal{S}(V) = V_1\ \mathrm{d}y \wedge \mathrm{d}z + V_2\ \mathrm{d}z \wedge \mathrm{d}x + V_3\ \mathrm{d}x \wedge \mathrm{d}y,$$

and $e = E_1\,\mathrm{d}x + E_2\,\mathrm{d}y + E_3\,\mathrm{d}z$, $b = B_1\,\mathrm{d}x + B_2\,\mathrm{d}y + B_3\,\mathrm{d}z$. Define the spacetime forms

$$\mathcal{F} = \mathcal{S}(B) + e \wedge \mathrm{d}t, \qquad \mathcal{G} = \mathcal{S}(E) + \mathrm{d}t \wedge b, \qquad \mathcal{J} = \rho\,\mathrm{vol}_3 - \mathrm{d}t \wedge \mathcal{S}(J).$$

Show that $\mathrm{d}\mathcal{F} = 0$ and $\mathrm{d}\mathcal{G} = 4\pi\mathcal{J}$ are equivalent to

$$\operatorname{div} B = 0, \qquad \partial_t B + \operatorname{curl} E = 0, \qquad \operatorname{div} E = 4\pi\rho, \qquad \partial_t E - \operatorname{curl} B = -4\pi J.$$

Deduce charge conservation, $\partial_t \rho + \operatorname{div} J = 0$.

Solution. Let d_s denote exterior differentiation in the three spatial variables. The previously established curl and divergence identities give

$$\mathrm{d}_s e = \mathcal{S}(\operatorname{curl} E), \qquad \mathrm{d}_s b = \mathcal{S}(\operatorname{curl} B), \qquad \mathrm{d}_s \mathcal{S}(V) = (\operatorname{div} V)\operatorname{vol}_3.$$

Separating the time derivative, and noting that moving $\mathrm{d}t$ past a two-form gives no sign change, yields

$$\begin{aligned} \mathrm{d}\mathcal{F} &= (\operatorname{div} B)\operatorname{vol}_3 + \mathrm{d}t \wedge \mathcal{S}(\partial_t B + \operatorname{curl} E), \\ \mathrm{d}\mathcal{G} &= (\operatorname{div} E)\operatorname{vol}_3 + \mathrm{d}t \wedge \mathcal{S}(\partial_t E - \operatorname{curl} B). \end{aligned}$$

For the second identity, the minus sign comes from $\mathrm{d}(\mathrm{d}t \wedge b) = -\,\mathrm{d}t \wedge \mathrm{d}b$. The four coordinate three-forms are independent, so comparing coefficients gives exactly the four vector equations, in both directions.

Finally, $\mathrm{d}^2\mathcal{G} = 0$ implies $\mathrm{d}\mathcal{J} = 0$. Direct differentiation gives

$$\mathrm{d}\mathcal{J} = (\partial_t \rho + \operatorname{div} J)\ \mathrm{d}t \wedge \operatorname{vol}_3,$$

which proves the conservation law. The normalizations and signs depend on the stated units and on placing $\mathrm{d}t$ first in the spacetime volume form.

15.4.3 Closed forms and an explicit bounding ball

Example 15.8 (A closed two-form on the sphere)**.** Let S be the unit sphere in $\mathbb{R}^3$, and let

$$\begin{aligned} \omega = {} & \arctan(1+x^2)\ \mathrm{d}x \wedge \mathrm{d}y \\ & + \sin\bigl(\log(1+y^2+z^2)\bigr)\ \mathrm{d}y \wedge \mathrm{d}z \\ & + e^{z^2-x^2}\ \mathrm{d}z \wedge \mathrm{d}x. \end{aligned}$$

Evaluate $\displaystyle\int_S \omega$.

Exercise 15.7 (solve the final sphere example)**.** Evaluate the two-form integral by differentiating the coefficients complementary to their wedge factors, and check every hypothesis required for using the ball.

Solution. Write

$$A = \sin\bigl(\log(1+y^2+z^2)\bigr), \qquad B = e^{z^2-x^2}, \qquad C = \arctan(1+x^2),$$

so $\omega = A\ \mathrm{d}y \wedge \mathrm{d}z + B\ \mathrm{d}z \wedge \mathrm{d}x + C\ \mathrm{d}x \wedge \mathrm{d}y$. The complementary derivatives are

$$A_x = 0, \qquad B_y = 0, \qquad C_z = 0.$$

Therefore

$$\mathrm{d}\omega = (A_x + B_y + C_z)\ \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z = 0.$$

One must not differentiate C with respect to x for this purpose: such a term contains $\mathrm{d}x \wedge \mathrm{d}x$ and vanishes. The same observation applies to derivatives of A in y, z and of B in x, z.

Let $\mathcal{B} = \{(x, y, z) : x^2 + y^2 + z^2 \le 1\}$. It is a compact smooth three-manifold with boundary S. Orient it by $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$; its induced sphere orientation is outward. All three coefficients are smooth on all of $\mathbb{R}^3$: in particular, $1 + y^2 + z^2 \ge 1$, so the logarithm has no singularity anywhere in the ball or near it. Stokes gives

$$\int_S \omega = \int_{\mathcal{B}} \mathrm{d}\omega = 0.$$

Thus ω is closed, meaning that its exterior derivative vanishes. This is a calculation of the two-form itself, not of a scalar surface area integral.

Exercise 15.8 (why the neighborhood hypothesis matters)**.** For comparison, consider on $\mathbb{R}^3 \setminus \{0\}$ the two-form

$$\beta = \frac{x\ \mathrm{d}y \wedge \mathrm{d}z + y\ \mathrm{d}z \wedge \mathrm{d}x + z\ \mathrm{d}x \wedge \mathrm{d}y}{(x^2 + y^2 + z^2)^{3/2}}.$$

Show that $\mathrm{d}\beta = 0$ on its domain but its integral over the outward unit sphere is 4π. Explain why this does not contradict Stokes or the preceding example.

Solution. Put $r = \sqrt{x^2 + y^2 + z^2}$. The corresponding field is $F = (x, y, z)/r^3$. Away from the origin,

$$\operatorname{div} F = 3r^{-3} - 3(x^2 + y^2 + z^2)r^{-5} = 0,$$

so $\mathrm{d}\beta = 0$. On the unit sphere, $F = (x, y, z) = n$, hence $F \cdot n = 1$ and

$$\int_S \beta = \int_S 1\ \mathrm{d}S = \int_0^{2\pi} \int_0^{\pi} \sin\phi\ \mathrm{d}\phi\ \mathrm{d}\theta = 4\pi.$$

The form is singular at the origin, so Stokes does not apply to the whole ball. On the shell $\varepsilon \le r \le 1$, it gives cancellation of the two sphere fluxes, whose induced orientations are opposite under radial identification. The preceding form has no such singularity.

15.5 Closed forms, exact forms, and cohomology

The angular form and the punctured-ball example have the same feature: a form can be closed without having a global primitive. The distinction is recorded by de Rham cohomology [2, Sec. 7.7].

15.5.1 Local primitives in every positive degree

Definition 15.9. A smooth k-form ω is *closed* if $\mathrm{d}\omega = 0$. For $k \ge 1$, it is *exact* if $\omega = \mathrm{d}\eta$ for a smooth $(k-1)$-form η. Since $\mathrm{d}^2 = 0$, every exact form is closed.

For the following algebraic discussion, all forms are smooth. This avoids losing a derivative when applying d repeatedly. The earlier integral theorems retain their stated C^1 hypotheses.

Exercise 15.9 (Poincaré lemma on a star-shaped open set)**.** Let $U \subset \mathbb{R}^n$ be open and star-shaped about 0: $tx \in U$ for $x \in U$ and $0 \le t \le 1$. Prove that every smooth closed k-form on U, $k \ge 1$, is exact. Show that a primitive is

$$\eta_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \omega_{tx}(x, v_1, \dots, v_{k-1})\, \mathrm{d}t.$$

For $k = 1$, the right side is a scalar function. Compare it with the earlier construction of a potential on a convex set.

Solution. Set $F(t, x) = tx$. Separate its pullback into terms without and with $\mathrm{d}t$:

$$F^*\omega = \alpha(t, x) + \mathrm{d}t \wedge \beta(t, x).$$

Here α is a k-form and β a $(k-1)$-form in the x variables. At fixed t, $\alpha = F_t^*\omega$, and evaluation on $\partial_t, v_1, \dots, v_{k-1}$ gives

$$\beta(t, x)(v_1, \dots, v_{k-1}) = t^{k-1} \omega_{tx}(x, v_1, \dots, v_{k-1}).$$

Writing d_x for differentiation in x, closedness and naturality imply

$$0 = \mathrm{d}(F^*\omega) = \mathrm{d}_x\alpha + \mathrm{d}t \wedge (\partial_t\alpha - \mathrm{d}_x\beta).$$

Consequently $\partial_t\alpha = \mathrm{d}_x\beta$. The proposed $\eta = \int_0^1 \beta\, \mathrm{d}t$ is smooth: near any fixed x, the radial segments stay in a compact subset of U, so derivatives can be passed through the finite integral. Therefore

$$\mathrm{d}\eta = \int_0^1 \mathrm{d}_x\beta\, \mathrm{d}t = \alpha(1, x) - \alpha(0, x) = \omega.$$

The last equality uses $F_1 = \mathrm{id}$ and the fact that the pullback of a positive-degree form by the constant map F_0 is zero. For a different star center, translate the coordinates first. Every point has a ball inside its open domain, so closed positive-degree forms have local primitives. This does not assert a global primitive on an arbitrary open set.

15.5.2 Quotienting out exact forms

A k-form on a smooth manifold M assigns an alternating k-linear function on T_pM to each point p. It is smooth if its pullback to every smooth coordinate chart has smooth coefficients. Write $\Omega^k(M)$ for the vector space of these forms. Define exterior differentiation in charts; the pullback identity for d makes these definitions agree on overlaps. Restricting an ambient form to tangent vectors gives such an intrinsic form, and exterior differentiation commutes with this restriction. Set

$$Z^k(M) = \ker\bigl(\mathrm{d} : \Omega^k(M) \to \Omega^{k+1}(M)\bigr), \qquad B^k(M) = \mathrm{im}\bigl(\mathrm{d} : \Omega^{k-1}(M) \to \Omega^k(M)\bigr).$$

For $k = 0$, set $B^0(M) = \{0\}$. Since $B^k(M) \subset Z^k(M)$, we can define

$$H^k_{\mathrm{dR}}(M) = Z^k(M)/B^k(M).$$

Thus two closed k-forms represent the same class exactly when their difference is exact. Every exact form represents zero.

Exercise 15.10 (degree zero and periods)**.** Show that $H^0_{\mathrm{dR}}(M) = \mathbb{R}$ when M is nonempty and connected. For $k \ge 1$, show that the integral of a closed k-form over a smooth k-cycle depends only on its cohomology class. Show also that closed forms integrate to zero over boundaries.

Solution. If $\mathrm{d}f = 0$, every coordinate partial derivative of f is zero. On a sufficiently small coordinate ball, the fundamental theorem along line segments makes f constant. Thus f is locally constant, and connectedness makes it constant on all of M. Conversely every constant is closed, and there are no nonzero exact 0-forms under the stated convention. This proves the degree-zero assertion.

Let C be a cycle and let $\widetilde{\omega} = \omega + \mathrm{d}\eta$. Chain Stokes gives

$$\int_C \widetilde{\omega} - \int_C \omega = \int_C \mathrm{d}\eta = \int_{\partial C} \eta = 0.$$

If $C = \partial B$ and ω is closed, then

$$\int_C \omega = \int_B \mathrm{d}\omega = 0.$$

More generally, cycles differing by a boundary have equal integrals of every closed form. In particular, one nonzero period proves that a closed form is not exact and that its integration cycle is not a boundary in the given manifold. No converse about all cycles is needed for this conclusion.

15.5.3 The punctured plane

On $U = \mathbb{R}^2 \setminus \{0\}$, let

$$\vartheta = \frac{-y\ \mathrm{d}x + x\ \mathrm{d}y}{x^2 + y^2}.$$

We already computed $\mathrm{d}\vartheta = 0$ and $\int_{S^1} \vartheta = 2\pi$ with the counterclockwise orientation. Thus $[\vartheta] \neq 0$ in $H^1_{\mathrm{dR}}(U)$.

Exercise 15.11 (classify closed one-forms on the punctured plane)**.** Prove that every smooth closed one-form ω on U can be written

$$\omega = c\vartheta + \mathrm{d}f, \qquad c = \frac{1}{2\pi}\int_{S^1} \omega,$$

where $f : U \to \mathbb{R}$ is smooth. Deduce $H^1_{\mathrm{dR}}(U) \cong \mathbb{R}$.

Solution. Use the polar map $p(r,t) = (r\cos t, r\sin t)$ on $(0,\infty)\times\mathbb{R}$. Write

$$p^*\omega = a(r,t)\ \mathrm{d}r + b(r,t)\ \mathrm{d}t.$$

Both coefficients are 2π-periodic in t, and closedness says $\partial_r b = \partial_t a$. It follows that

$$\frac{\mathrm{d}}{\mathrm{d}r}\int_0^{2\pi} b(r,t)\ \mathrm{d}t = \int_0^{2\pi} \partial_t a(r,t)\ \mathrm{d}t = 0.$$

Thus this integral is always $2\pi c$, where c is determined by the unit circle. Define

$$g(r,t) = \int_1^r a(s,0)\ \mathrm{d}s + \int_0^t \big(b(r,u) - c\big)\ \mathrm{d}u.$$

Then

$$g_t = b - c, \qquad g_r = a(r,0) + \int_0^t \partial_r b(r,u)\ \mathrm{d}u = a(r,t).$$

The zero integral of $b - c$ over each full period gives $g(r,t+2\pi) = g(r,t)$. Hence g descends to a smooth, single-valued function f on U. Since $p^*\vartheta = \mathrm{d}t$, we have

$$p^*(\omega - c\vartheta) = \mathrm{d}g = p^*(\mathrm{d}f).$$

The map p is onto and is a local diffeomorphism, so equality of these pullbacks implies $\omega - c\vartheta = \mathrm{d}f$. Finally the period map $[\omega] \mapsto (2\pi)^{-1}\int_{S^1}\omega$ is linear, vanishes precisely on the zero class by the construction, and takes $[\vartheta]$ to 1. It is therefore an isomorphism onto $\mathbb{R}$.

The same local calculus applies on the plane and the punctured plane, but their global primitives differ. On a star-shaped open set the Poincaré lemma gives $H^k_{\mathrm{dR}} = 0$ for $k \geq 1$; on the punctured plane the angular class survives.

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